

3D Gaussian Splatting: A Comprehensive Survey of Representations, Optimization, Rendering, and Applications

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1 Introduction and Foundational Concepts

1.1 Evolution of 3D Scene Representations

The evolution of 3D scene representation has been driven by a persistent pursuit of two often-competing goals: photorealistic fidelity and real-time computational efficiency. The earliest attempts at representing 3D scenes for novel view synthesis relied on point-based rendering, where scenes were depicted as collections of discrete, unconnected 3D points. While these point clouds could be rendered quickly by projecting points onto an image plane and applying splatting—a technique where each point is expanded into a small footprint, such as a circle or ellipse, to fill gaps—they suffered from significant limitations. Point-based methods struggled to accurately capture complex surface properties like specular highlights, transparency, and fine-grained texture, and they lacked a compact, continuous representation of geometry. The visual output was often sparse and noisy, with noticeable holes and aliasing artifacts, particularly when viewing scenes at close range or from novel angles. This sparsity and lack of a coherent surface model made high-fidelity image synthesis highly challenging, and despite decades of research, traditional point-based rendering remained a niche technique, largely supplanted by mesh-based graphics pipelines for most applications.

A paradigm shift occurred with the introduction of Neural Radiance Fields (NeRF). This approach represented a fundamental departure from prior methods by using implicit, coordinate-based neural networks to encode a scene. Instead of storing discrete primitives, NeRF represented a scene as a continuous volumetric function where a multi-layer perceptron (MLP) maps a 5D input—a spatial coordinate and a viewing direction—to an emitted color and volume density. Rendering an image requires ray marching through the volume, sampling points along each camera ray, and accumulating the color and density along these samples to produce the final pixel value. This volumetric rendering approach proved revolutionary: it enabled the synthesis of novel views with an unprecedented level of photorealism, accurately capturing complex occlusions, reflections, and view-dependent effects. This success established NeRF-based implicit representations as the de facto standard for high-quality 3D scene reconstruction [1, 2]. However, the power of this implicit representation came at a steep computational price. The neural network encoding meant that every scene had to be optimized from scratch, leading to slow training times. More critically, rendering required evaluating the MLP hundreds of times per pixel along a ray, making the process extremely slow and preventing real-time or interactive applications. The implicit and continuous nature of the representation also made it difficult to edit or manipulate the reconstructed scene. Unlike an explicit point cloud or mesh, there were no discrete primitives to grab, move, or deform, hindering its use in content creation, VR/AR, and robotics where interactivity is paramount [3].

The field was therefore at an impasse: implicit representations like NeRF offered unmatched quality but were computationally intractable for real-time use, while explicit representations like point clouds were fast but lacked the expressiveness and fidelity needed for photorealistic rendering [4]. The introduction of 3D Gaussian Splatting (3DGS) was a watershed moment that elegantly resolved this tension and sparked a rapid transformation of the landscape of 3D scene representations [5, 6]. 3DGS pioneered a novel, explicit representation that leverages millions of learnable 3D Gaussian primitives. Each primitive is defined by a full 3D covariance matrix (encoding its anisotropic scale and rotation), a position, an opacity value, and a set of spherical harmonic coefficients for view-dependent color. This explicit parameterization offers a key advantage over the implicit neural network of NeRF: the scene is directly defined by its primitives, making it inherently editable and suitable for manipulation [1].

The magic of 3DGS lies not just in its explicit representation but in its highly efficient, differentiable rasterization pipeline, which was a game-changer for real-time performance. This

rendering mechanism projects these 3D Gaussians onto the 2D image plane, sorts them by depth, and efficiently composites them using alpha-blending, a process that is both extremely fast on modern GPUs and fully differentiable [2]. This hybrid approach combines the strengths of volumetric rendering’s image formation model with the computational efficiency of primitive-based rasterization, achieving a synergistic balance that was previously thought impossible. This departure from the ray-marching paradigm of NeRF toward a rasterization-based, point-centric approach allowed 3DGS to deliver real-time, high-quality novel view synthesis, while also facilitating faster optimization times and the explicit editability that was missing from implicit models. It effectively combined the high visual fidelity associated with NeRF with the interactive speeds and control of point-based graphics [1, 7]. The introduction of 3DGS thus marked a pivotal shift from the deep, expensive implicit representations back towards the explicit, point-based philosophy, but now armed with the powerful, differentiable tools and optimization strategies developed during the era of neural rendering [5]. This convergence created a new, powerful, and versatile framework that has since spurred a vast and rapidly expanding body of research dedicated to enhancing, applying, and extending Gaussian-based representations [8].

The foundational departure of 3D Gaussian Splatting (3DGS) from preceding neural rendering techniques lies in its explicit, parametric representation of a scene as a collection of 3D Gaussian primitives. This paradigm shift from continuous implicit functions to a discrete set of optimizable primitives is the bedrock of its real-time rendering capability and inherent editability. A rigorous understanding of this representation requires a detailed examination of the mathematical formulation of these primitives, their parameterization, and the mechanisms through which they encode both geometry and appearance.

At its core, a single 3D Gaussian primitive is defined as a probability distribution in 3D space, mathematically expressed as a Gaussian function. The fundamental parameters for each primitive include a positional mean, which essentially acts as the center of the primitive in 3D space, and a covariance matrix, which defines the spatial extent, orientation, and shape of the primitive. The covariance matrix is the key to the representation’s flexibility; it is not restricted to be isotropic, allowing the Gaussians to stretch and rotate to conform to the underlying geometry of the scene. This anisotropic nature is a crucial advantage, as it allows for the efficient representation of surfaces with a minimal number of primitives. Anisotropic ellipsoids can be flattened to represent near-planar surfaces or elongated to cover thin, edge-like structures with far greater fidelity than isotropic spheres, which would require significant over-proliferation to achieve similar coverage.

The covariance matrix is typically decomposed into a rotation matrix and a scaling matrix to ensure it remains positive semi-definite during optimization. This decomposition effectively parameterizes the primitives with a quaternion for rotation and a 3D vector for scale, making the optimization process more stable and interpretable. The shape of the resulting Gaussian ellipsoid has direct implications for geometry reconstruction. As identified in several analyses, an unbounded optimization can lead to Gaussians converging into highly anisotropic, “needle-like” shapes with an effective rank of one, which causes artifacts in the reconstructed geometry and normals [9]. This observation has motivated the introduction of regularization techniques that constrain the shape of the Gaussians to be more planar or spherical, improving surface alignment and reducing artifacts [9]. Similarly, geometric losses based on eigenvalue-derived 3D shape features, such as “planarity” or “eigenentropy” of Gaussian neighborhoods, have been proposed to guide the optimization toward more geometrically accurate and artifact-reduced reconstructions [10, 11].

Accompanying the geometry parameters are two critical attributes for appearance and visibility. Each Gaussian possesses an opacity value, which controls its contribution to the final rendered image. This value is used in alpha-blending to composite the colors from different primitives along a view ray. The color of each Gaussian is not a single static value but is modeled as a function of viewing

direction to capture view-dependent effects like specular highlights. This is achieved through the use of spherical harmonic (SH) coefficients, a compact basis for representing functions on a sphere. While low-order SH is efficient, it has limitations. It struggles to represent high-frequency, anisotropic components of the appearance, which is particularly problematic for scenes with complex lighting or specular surfaces [12, 13]. Furthermore, the SH representation is fundamentally asymmetric in its relationship to the geometric rotation of the Gaussian, which complicates the creation of equivariant processing architectures [14]. This has led to numerous extensions that replace or augment SH with alternative appearance models, such as Spherical Gaussians for better specularly and efficiency [12, 15] or general feature vectors that are decoded by a neural network [16].

The standard 3DGS representation assumes that a Gaussian has a uniform color and opacity distribution within its ellipsoidal bound, scaled by the Gaussian’s falloff. This limits the expressivity of a single primitive, as its color is effectively a solid color modulated by the Gaussian’s radial decay [17]. To overcome this limitation, a significant body of work has focused on increasing the expressive power of individual primitives by decoupling their visual appearance from their geometric extent. One approach is to augment each Gaussian with texture maps, allowing for spatially varying color and opacity across the primitive’s surface [17, 18]. This “textured Gaussian” concept can significantly improve the representation of fine details and complex textures while maintaining a compact set of primitives, but it simultaneously introduces significant memory and parameterization challenges [19, 20].

A significant and conceptually important variation of the Gaussian primitive is the 2D Gaussian surfel. By setting the z-scale of a 3D Gaussian to zero, the primitive is flattened into a 2D ellipse, which aligns it with the concept of a surface element or “surfel” [21]. This distinction is crucial for surface-oriented modeling, as 3D Gaussian volumes are inherently volumetric and have an opaque, blob-like nature, making accurate surface extraction and alignment difficult [22]. In contrast, 2D surfels have a clear normal direction (aligned with the local z-axis) and are naturally better suited for tasks requiring geometric precision, such as surface reconstruction and mesh extraction [21, 23]. Gaussian surfels also serve as a base representation for hybrid models that integrate them with 3D volumetric Gaussians to represent both coarse-scale surfaces and fine-scale volumetric details [24]. The explicit and structured nature of the Gaussian primitives is the ultimate source of the representation’s strength. Unlike the latent, high-dimensional parameters of a NeRF’s neural network, the properties of a Gaussian—its position, shape, orientation, opacity, and color—are directly interpretable. This makes the representation easily editable, as an artist or an algorithm can directly manipulate the object. This explicit parameterization is also what enables the use of uncertainty estimation in a more physically grounded way, by propagating the covariance of the parameters through the rendering Jacobian [25]. Ultimately, the mathematical formulation of these primitives is not just a rendering detail but the core enabling factor for the versatility, efficiency, and scalability that define the 3DGS paradigm across its many applications.

1.2 Mathematical Foundations of 3D Gaussian Primitives

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1.3 Differentiable Rasterization and Alpha-Blending Pipeline

Building upon the explicit and parameterizable primitive representation detailed previously, the rendering pipeline of 3D Gaussian Splatting (3DGS) constitutes the crucial operational bridge between this mathematical scene description and the final output image. This pipeline represents a fundamental departure from the ray-marching paradigm of neural radiance fields, replacing per-pixel volumetric integration with a highly efficient, GPU-friendly rasterization process. At its core, the pipeline transforms the continuous volumetric representation of Gaussian primitives into discrete 2D image contributions through a sequence of projection, sorting, and compositing operations. Each 3D Gaussian is first projected onto the 2D image plane, where it becomes a 2D splat whose shape and extent are determined by the viewing transformation and the Gaussian’s covariance matrix. The projected splats are then sorted by their depth relative to the camera, and finally, the sorted splats are composited front-to-back using alpha-blending to produce the final pixel color. This process, while conceptually simple, involves several critical design choices that balance computational efficiency with physical accuracy.

The tile-based rasterization strategy is a cornerstone of the 3DGS rendering pipeline’s efficiency. Rather than processing each pixel independently, the image is subdivided into a grid of non-overlapping tiles, and for each tile, the set of Gaussians whose projected footprints overlap the tile is identified. This hierarchical culling significantly reduces the number of Gaussians that need to be considered per tile, as only a small fraction of the total scene primitives typically contribute to any given region of the image. The per-tile Gaussian lists are then sorted by depth, and the splats are evaluated in a front-to-back order for each pixel within the tile. This approach not only reduces the computational cost of sorting and blending but also improves memory locality and parallelism,

as each tile can be processed independently by separate threads or thread groups. Subsequent optimizations have further refined this pipeline, such as hierarchical tiled rendering that decouples coarse partitioning from fine rasterization to balance the costs of binning and primitive evaluation [26].

The differentiability of the rasterization process is a key enabler of end-to-end optimization. Every operation in the pipeline—from the projection of 3D Gaussians to 2D splats, through the depth-based sorting, to the alpha-blending compositing—is designed to support gradient flow. This allows the Gaussian parameters (position, covariance, opacity, and spherical harmonic coefficients) to be updated via backpropagation based on photometric loss functions comparing rendered images to ground truth. The explicit nature of the representation means that gradients are propagated directly to individual primitives, enabling precise and localized adjustments. However, the requirement for differentiability also imposes constraints on the implementation. For example, the sorting operation must be handled carefully, as it is inherently non-smooth. Recent works have explored differentiable hardware rasterization for 3DGS, achieving faster backward passes by leveraging programmable blending and hybrid gradient reduction strategies to overcome the memory and performance limitations of tile-based software rasterization [27].

The 3DGS rendering pipeline makes several approximations relative to the full volumetric rendering equation. The most significant of these is the treatment of the Gaussian’s 3D volumetric density as a 2D splat with a single depth value per pixel. This simplification, while enabling efficient rasterization, leads to artifacts such as popping when the viewpoint changes, as the depth ordering of Gaussians can become inconsistent across frames. To address this issue, hierarchical rasterization approaches have been proposed that systematically resort and cull splats based on per-pixel depth computation, effectively eliminating popping artifacts while maintaining real-time performance [28]. Additionally, the alpha-blending process in standard 3DGS computes alpha and transmittance as scalar quantities per pixel, which can lead to erosion and dilation artifacts at unseen sampling rates. Replacing this with spatially varying alpha and transmittance distributions, as done in Gaussian Blending, enables more accurate composition of background splats at various sampling rates [29].

Despite these approximations, the practical effectiveness of 3DGS is largely attributed to the sheer number of optimized Gaussians. A key insight is that the large, over-parameterized set of primitives can compensate for the errors introduced by the approximate rendering model. In-depth analyses have shown that while more accurate volumetric rendering can indeed improve quality when the number of primitives is low, the power of efficient optimization with thousands of Gaussians allows 3DGS to outperform principled volumetric rendering despite its approximations [7]. This finding suggests that the optimization process effectively adapts the Gaussian distribution to the idiosyncrasies of the rasterization model, embedding the approximation errors into the learned representation.

In parallel with rasterization-based approaches, alternative rendering paradigms have been explored. Ray tracing for 3D Gaussians has been proposed as a means to overcome the limitations of rasterization, such as rigid pinhole camera assumptions and inaccurate handling of reflections and refractions. Early ray tracing approaches sort all intersecting Gaussians along each ray, which is computationally expensive. However, stochastic ray tracing methods that randomly accept and shade a single intersection per ray have been shown to achieve unbiased results with low variance, providing a compelling alternative for both standard and relightable 3DGS scenes [30]. Fast ray tracers like GRay have demonstrated that dense initialization, which slows down rasterization, can actually accelerate ray tracing, leading to performance competitive with splatting-based renderers [31].

The sorting operation in the rasterization pipeline is a significant bottleneck, particularly on resource-constrained platforms. To address this, order-independent transparency (OIT) methods

have been developed. One approach uses a dual-hierarchy framework with cell proxies to manage local Gaussians, combined with a physically-inspired weighted sum rendering technique that eliminates popping and transparency artifacts without costly sequential alpha-blending [32]. Similarly, the weighted sum rendering approach approximates alpha blending with weighted sums, removing the need for sorting entirely and simplifying implementation while achieving competitive image quality [33]. Moment-based methods offer another route, characterizing the density distribution along each camera ray with statistical moments to reconstruct a continuous transmittance function, thereby bridging the gap between rasterization and physical accuracy without the need for ray tracing or per-pixel sorting [34].

Furthermore, efforts to improve the physical accuracy of the splatting process itself have gained traction. Volumetrically consistent rasterization directly integrates 3D Gaussians to compute transmittance analytically, producing more physically-accurate alpha values than the standard 3DGS formulation. This approach more closely follows the volume rendering equation while retaining the speed benefits of rasterization, achieving higher accuracy for opaque surfaces with fewer points [35]. The approximations in the standard pipeline have also been analyzed from a theoretical standpoint, leading to improvements such as non-exponential transmittance models that reduce overdraw and accelerate rendering in complex scenes [36]. These developments collectively illustrate an ongoing trajectory toward reconciling the efficiency of rasterization with the fidelity of physically-based rendering, thereby expanding the applicability of 3DGS across diverse domains. This rendering efficiency, in turn, is the foundational enabler of the real-time capabilities, rapid optimization convergence, and practical editability that define the 3DGS paradigm and its downstream applications, as discussed in the following subsection.

1.4 Comparative Advantages and Core Properties of 3DGS

The emergence of 3D Gaussian Splatting (3DGS) represents a paradigm shift in 3D scene representation, not merely as an incremental improvement but as a fundamental reimagining of the trade-offs that have historically constrained neural rendering. While the preceding discussion detailed the internal mechanics of its rendering pipeline—the projection, sorting, and compositing operations that define its operational efficiency—this subsection shifts focus outward to examine how the distinctive properties of 3DGS—real-time rendering, rapid optimization, inherent editability, compatibility with established graphics pipelines, and a balanced pursuit of visual fidelity and geometric accuracy—collectively position it as a transformative technology bridging neural rendering and traditional computer graphics [5, 2]. Understanding these core advantages is essential, as they directly determine the breadth of downstream applications explored in the following subsection.

Real-Time Rendering via GPU-Friendly Rasterization. The most celebrated advantage of 3DGS is its capability for real-time, high-fidelity rendering, a feat largely unattainable by its primary predecessor, the Neural Radiance Field (NeRF), which relies on computationally intensive ray marching. Unlike NeRF’s implicit volumetric rendering, 3DGS leverages an explicit, point-based representation that is projected onto the 2D image plane and composited via alpha-blending, a process that maps exceptionally well onto highly optimized GPU rasterization pipelines [37]. This fundamental architectural shift replaces slow, per-pixel ray casting with fast, parallelizable splatting operations. The practical impact is profound: scenes composed of millions of Gaussian primitives can be rendered at interactive frame rates on commodity hardware. This relentless focus on performance confirms that real-time capability is not just a feature but the foundational property that makes 3DGS a practical tool for interactive applications, complementing the tile-based culling and sorting strategies detailed previously. The efficiency of this pipeline has spurred significant research dedicated to further acceleration, underscoring the importance of the real-time capability.

For instance, hardware-software co-design approaches enhance existing GPU triangle rasterizers to support 3DGS operations, achieving substantial speedups and energy reductions for edge deployment [38]. Similarly, architectural innovations have leveraged Tensor Cores for the rasterization stage, demonstrating a $1.65\times$ end-to-end performance improvement [39]. System-level optimizations, such as inter-frame caching and tile scheduling, have targeted redundancy in large-scale scenes, yielding up to a $10\times$ speedup over the original algorithm [40]. Even with the added complexity of dynamic scenes, the field is working towards maintaining this real-time performance, with platforms like Visionary aiming to provide web-native, real-time rendering of dynamic Gaussian content [41].

Rapid Optimization and Convergence. Just as the rendering pipeline is optimized for real-time performance, the explicit nature of the 3DGS representation confers a second major advantage: significantly faster training times compared to implicit models. NeRF and its variants often require hours or even days of optimization to converge for a single scene. In contrast, 3DGS can achieve high-quality reconstructions in a matter of minutes, a direct consequence of the differentiable rasterization pipeline described earlier that allows for efficient gradient flow to individual primitives. This speedup is attributed to the direct, differentiable rasterization pipeline that allows for efficient gradient flow, bypassing the need for costly volume rendering integrals. The optimization process itself has become a fertile area of research, with efforts focused on refining the densification and pruning strategies to improve both speed and quality [42]. The importance of efficient optimization is highlighted by work like Mini-Splatting2, which introduces a structure-aware distribution and region-prioritized optimization to achieve up to $3\times$ faster convergence while using fewer Gaussians [43]. Furthermore, the initial point cloud quality is critical; methods like GaussianPro have borrowed concepts from multi-view stereo to progressively propagate Gaussians, improving optimization in texture-less regions where SfM initialization is poor [44]. This rapid optimization cycle is not merely a convenience; it enables new workflows such as interactive reconstruction and iterative scene refinement, which are essential for applications in robotics and content creation. The development of specialized hardware further underscores this focus, with prototypes like Vorion being designed to accelerate not only rendering but also the training iterations themselves [45].

Inherent Editability and Manipulability. The explicit and discrete nature of 3DGS, where each Gaussian is an independently parameterized primitive as described in the pipeline section, provides an unparalleled level of editability, a stark contrast to the monolithic and opaque nature of implicit neural representations. Each Gaussian primitive is an independent, parameterized entity with explicit properties—position, covariance, opacity, and color—which can be individually manipulated, added, or removed. This property empowers a wide range of downstream tasks, including object-level editing, semantic manipulation, and physical simulation [1, 8]. For instance, the representation’s structure facilitates semantic understanding and editing, allowing for text-guided or segment-based manipulations of the scene [8]. This editability is also crucial for scene completion and generation, where training-free plugins can generate and register new Gaussian primitives to fill in missing information from sparse inputs [46]. The ability to treat Gaussians as discrete building blocks makes 3DGS a powerful medium for creative and interactive applications, from virtual content creation to advanced scene understanding and reasoning.

Compatibility with Mesh-Based Graphics Pipelines. While 3DGS is an explicit primitive, its compatibility with traditional mesh-based rendering pipelines has been a subject of active research and development. The standard graphics pipeline used in game engines and VR headset ecosystems is deeply entrenched in triangle meshes. To bridge this gap between the splatting-based approach detailed previously and conventional graphics hardware, several approaches have been explored. One strategy involves hybrid representations that integrate 3D Gaussians with textured meshes, using meshes for texture-rich flat areas and Gaussians for intricate geometries, thereby improving rendering efficiency and compatibility [47]. Another ambitious direction is the development of differentiable

renderers that directly optimize opaque triangle primitives as a replacement for Gaussians, aiming to produce outputs that are immediately usable in standard graphics engines without post-processing [48]. By demonstrating that mesh-like primitives can be optimized effectively, these works pave the way for 3DGS-inspired techniques to penetrate the vast ecosystem of existing mesh-based software and hardware, which is a critical factor for industrial adoption [37].

Balancing Visual Fidelity and Geometric Accuracy. 3DGS offers a unique equilibrium between photorealistic appearance and the fidelity of the underlying 3D geometry, a balance that is directly influenced by the approximate nature of its rendering model discussed earlier. While early reconstructions were put to question regarding geometric accuracy, a large body of subsequent work has focused on enhancing this aspect without sacrificing the rendering speed enabled by efficient splatting. Techniques range from incorporating normal and depth priors to regularize the Gaussians towards surface alignment [49], to hybrid representations that integrate signed distance fields (SDFs) for more globally consistent geometry. However, this balance is not trivial; for instance, while 2D Gaussian representations enforce multi-view consistency and better surface alignment, they can compromise texture details [50]. This tension is well-recognized, and hybrid methods that combine the strengths of different representations are becoming increasingly common [51]. The 3DGS community is actively working to ensure that “photorealism” and “correct geometry” are not seen as opposing goals but as complementary ones, a trend that is critical for applications like robotic navigation and digital twins that demand both a visually compelling and geometrically reliable representation [4].

In summary, the convergence of real-time rendering, rapid training, explicit editability, and growing compatibility with standard graphics pipelines makes 3DGS a uniquely versatile tool. The internal pipeline efficiencies that enable real-time performance, coupled with the explicit primitive structure that permits rapid optimization and direct manipulation, collectively ensure that 3DGS is not merely a rendering technique but a comprehensive 3D data structure. This balanced approach positions it to serve as the backbone for a new generation of applications. From enabling photorealistic digital twins for autonomous driving [52] and supporting memory-efficient real-time SLAM systems [53] to powering immersive VR experiences and generative world models [4, 41], the core properties of 3DGS establish it as a foundational technology. As research continues to address its limitations in areas such as robustness and scalability, its role as a bridge between the worlds of neural rendering and practical computer graphics is set to become even more central [1, 54].

2 Core Representation and Rendering Techniques

2.1 Alternative Primitive Designs and Kernel Extensions

The foundational 3D Gaussian Splatting (3DGS) framework relies on symmetric, isotropic Gaussian primitives that encode a single view-dependent color and a global opacity. While this representation achieves impressive results, its inherent limitations—particularly its low-pass nature and the coupling of geometry with appearance—often necessitate a large number of primitives to faithfully capture sharp boundaries, high-frequency textures, and complex surface details. To overcome these constraints, a rich body of research has explored alternative primitive designs and kernel extensions. These innovations generally aim to enhance the expressivity of individual primitives, thereby improving representation compactness, rendering sharpness, and fidelity at challenging regions such as object silhouettes and high-frequency structures. This subsection surveys these advances, which form the representational foundation upon which subsequent techniques for scale-aware regularization, frequency-domain supervision, and multi-view consistency are built.

One prominent direction replaces the standard 3D Gaussian volume with **2D Gaussian surfels**.

By setting the z-scale of a 3D Gaussian to zero, these primitives become flat, planar ellipses that align well with object surfaces. This design has been shown to improve both reconstruction quality and surface alignment, as it provides clearer geometric guidance during optimization. The use of 2D Gaussian surfels within the 3DGS framework has enabled high-fidelity geometric reconstruction alongside novel view synthesis [21]. Furthermore, this planar representation serves as a foundation for integrating per-primitive textures, effectively decoupling geometry and appearance, which proves beneficial for scenes with simple geometry but complex textures [55]. This early separation of geometry and appearance is a recurrent theme that resonates with the multi-view consistency and transparency challenges addressed later in this survey.

Another key advancement involves **asymmetric and skewed kernels** to better capture discontinuities. The standard Gaussian’s symmetric shape is ill-suited for representing one-sided surfaces or sharp edges. To address this, researchers have introduced the Azzalini Skew-Normal distribution as a more general primitive. By incorporating a learnable skewness parameter, these skew-normal kernels can continuously interpolate between symmetric Gaussians and half-Gaussian-like shapes, enabling flexible modeling of both sharp boundaries and interior regions while maintaining analytical tractability for rasterization [56]. Similarly, the 3D Skew Gaussian Splatting (3DSGS) framework extends the standard primitive to a skew-normal counterpart, enhancing structural fidelity and compactness, particularly in regions with intricate details [57]. Beyond these, the concept of deformable kernels has been explored to overcome the radial symmetry constraint. For instance, Deformable Radial Kernels (DRK) use learnable radial bases with adjustable angles and scales, allowing for precise control over edge sharpness and boundary curvature and leading to higher representation efficiency [58]. The Deformable Beta Splatting (DBS) method replaces Gaussian kernels with deformable Beta kernels that offer bounded support and adaptive frequency control, capturing fine geometric details with greater fidelity and improved memory efficiency [59]. These kernel-level modifications directly address the boundary sharpness issues that underpin the aliasing artifacts and high-frequency detail losses discussed in the subsequent subsection.

To enhance the fidelity of each primitive, another strand of research focuses on **textured and anisotropic kernels**. The core idea is to augment each Gaussian with a spatially varying color or opacity model. Innovatively, this allows a single primitive to represent complex visual phenomena that would otherwise require a cluster of standard Gaussians. The SVGS method implements this by using spatially varying functions, such as bilinear interpolation, and demonstrates that even a single 2D Gaussian surfel can effectively capture varying colors across its extent [60]. Similarly, GStex textures each 2D Gaussian primitive, making appearance representation agnostic to geometry complexity and enabling improved view synthesis when the number of primitives is reduced [55]. The “Textured Gaussians” approach generalizes the Gaussian appearance representation by augmenting each Gaussian with alpha, RGB, or RGBA texture maps, and finds that alpha-only and RGB texture maps significantly boost expressivity [17]. Further advancing this direction, adaptive anisotropic textured Gaussians (A²TG) dynamically adjust texture resolution and aspect ratio based on gradient guidance, leading to more efficient memory allocation while maintaining high rendering fidelity [20]. Content-aware texturing methods further refine this by adapting texture resolution during optimization, ensuring a principled trade-off between quality and the number of primitives [61]. By equipping primitives with internal spatial variation, these methods implicitly provide a form of scale-adaptive representation, complementing the explicit multi-scale supervision strategies described in the following subsection.

A significant limitation of standard 3DGS is its low-pass nature, which struggles to represent high-frequency details. To address this, a class of methods substitutes the Gaussian kernel with **Fourier-encoded or Gabor-based primitives**. The 3D Gabor kernel, a combination of a Gaussian term and wave functions, is introduced to directly represent spatial high-frequency signals,

such as fine stripes, which would normally require a large number of skinny Gaussian kernels [62]. Extending this, 3D Gabor Splatting (3DGabSplat) leverages a filter bank of multiple directional 3D Gabor kernels at different frequencies, developing an efficient CUDA-based rasterizer to handle the additional frequency components and a frequency-adaptive mechanism for joint optimization [63]. Neural Gabor Splatting takes a hybrid approach, augmenting each Gaussian primitive with a lightweight multi-layer perceptron (MLP) to model a wide range of color variations within a single primitive, coupled with a frequency-aware densification strategy to control primitive numbers [64]. Fourier Splatting represents another form of scalable primitives. This method parameterizes planar surfels with Fourier descriptors, naming the new primitive “Fourier-encoded”, which allows a single model to be rendered at varying levels of detail by simply truncating different numbers of Fourier coefficients at runtime [65]. The “Brane” primitive, introduced in Resonant Brane Splatting, augments the standard Gaussian envelope with internal Gaussian-Hermite modes, assigning distinct color coefficients to each mode. This enables a single Brane to represent spatially varying colors and high-frequency local contrast, improving the speed-quality trade-off in super-resolution tasks [66]. The frequency-aware design of these primitives naturally complements the frequency-domain supervision losses explored in the next subsection, where explicit control over spectral content is used to regularize optimization.

The pursuit of higher expressivity also extends to **kernel extensions with higher-order or non-Gaussian forms**. Drawing inspiration from quantum physics, the Gaussian-Hermite kernel has been proposed as a new primitive, offering improved performance in both geometry reconstruction and novel view synthesis due to its enhanced ability to model sharp edges and anisotropies [67]. Similarly, 3D Linear Splatting (3DLS) replaces Gaussian kernels with linear kernels that possess sharp, hard boundaries, resolving the blurring artifacts caused by the soft boundaries of Gaussian ellipsoids and achieving sharper results in high-frequency regions [68]. These kernel-level innovations directly target the sampling and reconstruction fidelity issues that are central to the anti-aliasing and detail-preservation challenges.

Many of these alternative primitive designs also incorporate enhancements to the rasterization or rendering pipeline to ensure they are compatible with real-time performance and remain differentiable. For example, the depth-peeling strategy is used with Gaussian-enhanced surfels (GES) to establish accurate per-pixel ordering, eliminating popping artifacts and enabling correct transmittance modulation in a sort-free rendering pipeline [69]. Another design, dual-opacity primitives, decouples a Gaussian’s visual existence from its instance occupancy, which proves crucial for object-aware segmentation tasks [70]. The CT-based factorized representation of Gaussian-Enhanced Surfels (GES) supports semi-transparent rendering and high-performance, sort-free rendering, successfully avoiding popping artifacts [24]. A factorization into geometric occupancy and optical opacity, as in RT-Splatting, allows a single set of primitives to handle both high-frequency specular reflections and clear transmission on semi-transparent surfaces, mitigating the common pitfalls of blurry reflections or overly occluded transmissions [71]. These pipeline-level improvements directly address the transparency ordering and multi-view consistency issues that are further elaborated in the subsequent subsection, where advanced transparency modeling is discussed in the context of aliasing and detail preservation.

Overall, these diverse kernel designs substantially expand the representational capacity of Gaussian primitives. By moving beyond simple symmetric Gaussian functions, these methods address the critical issues of boundary sharpness, high-frequency detail modeling, and appearance-geometry decoupling, leading to more compact scene representations and improved fidelity at object boundaries and in high-frequency regions. The continuous evolution of these primitive designs represents a fundamental avenue for advancing the efficiency and quality of radiance field rendering, and it provides the enriched representational toolkit upon which the anti-aliasing, frequency-domain

supervision, and multi-view consistency techniques of the subsequent subsection are built.

2.2 Anti-Aliasing, Multi-View Consistency, and High-Frequency Detail Modeling

The pursuit of high-fidelity novel view synthesis with 3D Gaussian Splatting (3DGS) is fundamentally challenged by two intertwined issues: aliasing artifacts arising from the discrete sampling of continuous scenes and the accurate reconstruction of high-frequency details, which are often sacrificed in favor of smooth, low-frequency approximations. Furthermore, ensuring that the reconstructed scene remains consistent across multiple viewpoints—a property that is not inherently guaranteed by per-pixel photometric losses—is critical to avoid flickering and ghosting artifacts. The primitive designs discussed in the previous subsection—such as Gabor-based kernels, skewed distributions, and textured surfels—expand the representational capacity of individual Gaussians, yet they do not by themselves resolve the scale-dependent and frequency-dependent pitfalls of optimization. To fully leverage these enhanced primitives, the optimization process itself must be regularized to respect scale, frequency, and multi-view coherence. This subsection surveys techniques designed to address these challenges, focusing on scale-aware regularization, multi-scale supervision, frequency-domain losses, and advanced transparency modeling.

Anti-Aliasing and Scale-Aware Regularization. Aliasing in 3DGS manifests as flickering textures, jagged edges, and popping artifacts, particularly when rendering at scales different from those observed during training. The root cause is the mismatch between the size of the Gaussian primitives and the pixel footprint at a given viewpoint. A direct inspiration comes from classical graphics, where mipmaps pre-filter textures to provide scale-appropriate representations. Analogously, Mipmap-GS [72] introduces a scale-consistency guidance loss and creates pseudo ground-truth images for target scales, effectively injecting scale information into the Gaussians. This allows the primitives to self-adjust their properties (color, shape, size, and position) for arbitrary zoom levels, leading to substantial PSNR improvements for both zoom-in and zoom-out scenarios compared to the baseline 3DGS model. This concept is further extended by works that use pyramidal multi-scale supervision to enforce coherence across different viewing distances. For large-scale urban environments, PrismGS [73] proposes a physically-grounded regularization framework that combines pyramidal multi-scale supervision with an explicit size regularizer. The multi-scale supervision compels the model to learn an inherently anti-aliased representation that remains coherent across scales, while the size regularization imposes a physical lower bound on Gaussian dimensions to prevent the formation of degenerate, view-dependent primitives, thereby reducing jagged edges and stabilizing optimization in demanding 4K rendering scenarios. At a more fundamental level, the choice of optimization objective itself can influence aliasing. A multi-sample anti-aliasing (MSAA) approach integrates adaptive blending of quadruple subsamples to compute pixel colors, effectively reducing aliasing in high-frequency regions. This is coupled with dual geometric constraints—an adaptive weighting strategy that prioritizes under-reconstructed regions and gradient differential constraints that enforce geometric regularization at object boundaries [74]. This targeted approach ensures that computational resources are allocated to critical regions requiring refinement while maintaining global consistency. These scale-aware strategies resonate with the textured and anisotropic primitives introduced earlier, as spatial internal variation within a primitive inherently provides a form of scale adaptation, complementing the explicit multi-scale supervision described here.

Frequency-Domain Supervision and High-Frequency Detail Modeling. The standard pixel-space losses (L1, SSIM) used in 3DGS optimization constrain only the aggregate reconstruction error, permitting the redistribution of error across frequency scales. This often leads to oversmoothing and a loss of fine details—precisely the limitation that Gabor-based and Fourier-encoded primitives

aim to overcome. To address this, a significant body of work has shifted supervision into the frequency domain, providing an external constraint that complements the internal frequency expressivity of these enhanced kernels. A foundational strategy is progressive frequency regularization, as exemplified by FreGS [75]. FreGS performs coarse-to-fine Gaussian densification by exploiting low-to-high frequency components extracted via low-pass and high-pass filters in the Fourier space. By minimizing the discrepancy between the frequency spectrum of the rendered image and the ground truth, it effectively alleviates over-reconstruction and produces sharper images. Building on this, KC-3DGS [76] provides a theoretical analysis showing that pixel-space losses admit a family of indistinguishable perturbations under wavelet redistribution. To exclude these degenerate solutions, it introduces a multi-scale wavelet coefficient alignment loss to explicitly penalize missing high-frequency detail, a supervised kurtosis concentration loss to match the heavy-tailed frequency statistics of natural images, and a cross-band covariance penalty to promote frequency specialization. This wavelet-domain supervision leads to significant improvements in perceptual quality (e.g., DreamSim) with minimal impact on standard metrics.

The choice of frequency transform is crucial. While Fourier transforms provide global frequency information, they lack spatial locality. In contrast, the Discrete Wavelet Transform (DWT) offers both frequency and spatial information, making it more suitable for identifying and supervising localized high-frequency features. DWTGS [77] argues that Fourier-based regularization is difficult to tune and can be biased towards detrimental high-frequency learning in sparse-view settings. Instead, it proposes a wavelet-space loss that supervises only the low-frequency LL subbands at multiple DWT levels while enforcing sparsity on the high-frequency HH subband in a self-supervised manner. This low-frequency-centric strategy improves generalization and reduces high-frequency hallucinations. A further advancement is the use of learnable wavelet transforms. Learnable Multi-level DWT [78] introduces a multi-level DWT-based frequency modulation framework. By recursively decomposing the low-frequency subband, it constructs a deeper curriculum that provides progressively coarser supervision during early training, which consistently reduces the final Gaussian count while maintaining rendering quality. This method also simplifies the modulation to a single scaling parameter, improving efficiency and stability over learning the full filter bank. Beyond regularization, frequency-domain thinking can also inform geometry recovery. PSRGS [79] uses spectral residual maps to separate low-frequency and high-frequency regions, applying different densification thresholds and loss functions to each region. This progressive scheme ensures that geometry is initialized in low-frequency regions and refined in high-frequency areas, leading to better recovery of fine-grained details in large-scale remote sensing scenes. The synergy between these frequency-domain supervision losses and the frequency-adaptive primitives from the previous subsection is evident: the former provides the optimization signal, while the latter offers the representational capacity to absorb it effectively.

Multi-View Consistency and Advanced Transparency. Ensuring multi-view consistency is a prerequisite for coherent rendering, especially in dynamic or sparsely sampled scenes. While frequency-domain losses contribute to consistency by preventing the model from fitting view-specific high-frequency noise, explicit methods are also needed. For instance, the geometry-constrained framework in GSM-GS [80] employs a geometry-guided cross-view point cloud association method combined with a dynamic weight sampling strategy to construct 3D structural normal constraints across adjacent views, reinforcing multi-view consistency for reconstruction. This concern also directly connects to the primitive designs of the previous subsection, particularly dual-opacity primitives and factorization into geometric occupancy and optical opacity, which provide the structural mechanisms for consistent transparency handling.

Aliasing is not only a spatial but also a temporal problem, particularly when sorting transparent primitives. The standard alpha-blending pipeline in 3DGS relies on a fixed depth sorting, which

can lead to popping artifacts and incorrect transmittance. This is a specific instance of the order-independent transparency (OIT) problem. While STAR-NT [81] addresses OIT in a different context, the concept is highly relevant to 3DGS, where a high-quality rendering of overlapping surfaces requires a correct handling of transmittance. To eliminate artifacts arising from incorrect sorting, methods that focus on order-independent transparency are crucial. Finally, to further enhance high-frequency detail, Sparse-Up [82] addresses the “pixel-level pain point” of high-frequency detail loss in 3D asset generation by using surface anchoring to constrain voxels to the mesh surface, eliminating redundant voxels, and a view-domain partitioning scheme to supervise only visible local patches. This approach breaks through resolution constraints while maintaining multi-view consistency and enabling high-fidelity texture recovery. Similarly, SubSplat [83] introduces a Sub-pixel Gaussian Reparameterizer to subdivide primary Gaussians into finer-grained primitives, restoring structural density and capturing high-frequency details from low-resolution features in pixel-aligned pipelines. These methods also foreshadow the hybrid mesh-Gaussian strategies of the following subsection, where geometric anchoring and explicit surface constraints become central to achieving consistent, high-fidelity reconstructions. In summary, the field is moving towards a holistic approach where anti-aliasing, high-frequency detail recovery, and multi-view consistency are jointly addressed through a combination of scale-aware regularization, sophisticated frequency-domain supervision in the wavelet domain, and more robust rendering algorithms.

2.3 Hybrid Representations with Meshes, Implicit Fields, and Volumetric Grids

Hybrid representations have emerged as a cornerstone of modern 3D scene modeling, seeking to combine the complementary strengths of disparate primitives while mitigating their individual weaknesses. Within the realm of 3D Gaussian Splatting, this fusion strategy addresses fundamental tensions between rendering fidelity, geometric accuracy, storage efficiency, and compatibility with established graphics pipelines. By integrating 3D Gaussians with triangle meshes, implicit fields, and volumetric grids, researchers have developed unified frameworks that leverage the explicit, real-time rendering capabilities of Gaussians alongside the structural coherence and compactness of other representations. While the preceding discussions have focused on refining the Gaussian representation itself—through scale-aware regularization, frequency-domain supervision, and multi-view consistency—this subsection shifts the perspective outward, examining how Gaussians can be embedded within larger, multi-primitive ecosystems. Such hybridization not only addresses the limitations of pure Gaussian representations but also positions 3DGS as a versatile component compatible with established graphics pipelines, a theme that will be further explored in the subsequent discussion on non-pinhole camera adaptations and performance optimizations.

A significant body of work focuses on hybridizing Gaussians with triangle meshes, motivated by the need for explicit geometry that is compatible with traditional graphics engines and simulation pipelines. The core challenge lies in enabling joint optimization and seamless rendering between these fundamentally different primitives. [84] presents a unified framework that rasterizes mesh triangles and 3D Gaussian fragments in a single-pass, anti-aliased manner, using alpha-blending to achieve visually coherent results with precise occlusion handling. It further introduces a Gaussian-centric binding strategy that associates Gaussians with mesh faces for deformation, reducing artifacts and enabling the unified manipulation of objects represented by either modality. Similarly, [48] and [85] advocate for a “triangle comeback,” arguing that triangles are the fundamental primitive of computer graphics. They propose directly optimizing triangles within a differentiable splatting framework, treating them as splats to combine the efficiency of Gaussian-like rendering with the inherent compatibility of meshes. [86] develops a differentiable renderer that optimizes triangles via end-to-end gradients, rendering each triangle as a differentiable splat. This approach achieves higher

visual fidelity and faster convergence than Gaussian methods while producing outputs directly usable in standard graphics engines, as highlighted by [48]. [85] replaces 3D Gaussian primitives with 2D triangles that naturally form a discrete mesh-like structure, using controlled annealing of a compactness parameter to ensure differentiability during training while converging to fully opaque triangle meshes. [87] tightly integrates Gaussian and mesh representations by analytically deriving each Gaussian primitive from its corresponding mesh triangle, preserving structural fidelity and enabling smoother gradients more conducive to optimization with small batch sizes. [88] uses a collection of disconnected translucent triangle primitives, termed triangle soup, and introduces soft connectivity forces between them to encourage explicit surface continuity during optimization, improving geometric accuracy without sacrificing visual fidelity. [89] argues for a loose coupling of coarse triangles for geometry and Gaussians for appearance, yielding a compact and expressive model that adapts to varying scene complexity. [47] introduces a hybrid representation that uses textured meshes for texture-rich flat areas and retains Gaussians for intricate geometries, employing a joint optimization with transmittance-aware supervision to balance their contributions, resulting in higher FPS with fewer Gaussians. [90] binds 3D Gaussians to mesh faces for an end-to-end learnable representation, enabling direct information pathways for jointly learning explicit geometry and renderable appearance, which supports efficient local scene modifications. [91] presents a hybrid representation combining Bézier triangle-based vector graphics with Gaussian probabilistic models, maintaining accurate shape modeling, especially at sharp boundaries, while enabling resolution-independent differentiable rendering. [92] employs structured tetrahedral meshes, reparameterizing their geometric attributes into Gaussian primitives for differentiable rendering, and introduces an inversion-free homeomorphism to maintain a conformal, self-intersection-free mesh throughout optimization. In defense-related applications, [93] argues that a hybrid approach augmenting a mesh scaffold with neural representations like 3DGS offers the most promising trade-off between geometric integrity for functional tasks and photorealistic visuals for situational awareness. While meshes provide explicit, topologically sound structure, their integration with Gaussians often requires careful handling of transparency and occlusion, as well as robust optimization strategies to avoid conflicts between the two primitives, as addressed in [84] and [47].

Another major axis of hybridization integrates Gaussians with implicit fields, most notably signed distance fields (SDFs) and neural feature grids, to improve geometric fidelity and compactness. This direction is particularly complementary to the frequency-domain regularization discussed earlier, as implicit fields provide a natural mechanism for enforcing global structural coherence that complements the local, explicit nature of Gaussian primitives. [94] embeds SDF learning directly into the 3DGS pipeline by associating each Gaussian with a learnable SDF value, explicitly aligning primitives with underlying geometry and improving cross-view consistency, while employing a geometry-guided grid growth strategy for dense coverage. [95] tethers scaffold-anchored Gaussians to a jointly optimized sparse voxel scaffold, confining the Gaussians to a narrow band around surfaces defined by voxelized SDFs, thereby improving representation efficiency and condensing floating primitives without sacrificing geometric accuracy. [96] integrates explicit point clouds with implicit feature embeddings via a multi-level tri-plane architecture, using 2D feature grids at various resolutions to enhance spatial correlations among Gaussians, which enables high-quality rendering with only a few megabytes of storage. [97] and [98] propose a compact representation using a learnable mask to reduce the number of Gaussians and a grid-based neural field for view-dependent color, achieving substantial storage reduction while maintaining rendering quality. [99] introduces a decoupled neural field architecture that separately models geometry and view-dependent color, combining explicit Gaussians with grid-based neural fields for a compact yet expressive representation. [100] implicitly encodes a point cloud within a continuous octree-based probability field and a multi-resolution hash grid, combining the fast rendering of rasterization with the ability to preserve

fine geometric details without relying on structure-from-motion priors. These implicit integrations often face challenges in balancing the expressiveness of the neural field with the explicit control of the Gaussian primitives, and require carefully designed optimization schedules to ensure coherent learning across both components.

The integration of volumetric grids provides a third avenue for hybrid 3DGS systems, primarily targeting global coherence and improved initialization. [52] combines a grid-based volume with the 3DGS pipeline to eliminate the dependency on SfM point initialization and improve rendering in distant, sky, and low-texture areas, incorporating a point densification strategy for enhanced quality. [101] and [92] also explore grid-based structures for feature encoding and geometric regularization. Furthermore, hybrid approaches extend to other representations, such as superquadrics, as in [102], which integrates 2D Gaussians with superquadric meshes for interpretable, part-aware scene decomposition. The success of these hybrid systems highlights a key insight: the future of high-fidelity 3D reconstruction may not lie in a single universal primitive, but in the principled combination of multiple representations, each contributing its unique strengths to the unified goal of photorealistic and geometrically accurate scene modeling. This insight becomes particularly salient as we move to consider how these hybrid representations can be adapted to non-pinhole camera models and optimized for real-time performance on diverse hardware platforms, topics that form the foundation of the subsequent section.

2.4 Adaptations for Diverse Camera Models and Real-Time Rendering Efficiency

The default 3D Gaussian Splatting (3DGS) pipeline is fundamentally designed around the pinhole camera model, which assumes a linear perspective projection. While this model is appropriate for standard narrow field-of-view (FoV) images, it becomes a limiting factor when applying 3DGS to the increasingly prevalent non-pinhole camera systems found in consumer devices, robotics, and autonomous driving. This subsection surveys the key adaptations required to extend 3DGS to fisheye, panoramic, and omnidirectional cameras, and then examines performance-oriented enhancements that ensure real-time rendering efficiency on large-scale and mobile platforms. These efforts complement the hybrid representation strategies discussed previously, as they collectively broaden the applicability of 3DGS beyond controlled, pinhole-based scenarios, paving the way for deployment in diverse real-world settings.

A primary challenge in adapting 3DGS to non-pinhole cameras lies in the projection step. The naive approach of undistorting images before training introduces significant artifacts. For fisheye cameras, the wide FoV is a key advantage, yet undistortion can introduce black borders and resample the image, diluting high-frequency detail and leading to blurry or floating artifacts. To address this, methods like Fisheye-GS and DirectFisheye-GS [103, 104] have been developed to integrate the fisheye camera model directly into the 3DGS optimization pipeline, recalculating the projection transformation and its gradients natively. This approach preserves the full FoV and detail of the original images. However, DirectFisheye-GS [104] also identifies a novel issue: the standard per-iteration random-view selection in 3DGS neglects cross-view correlations, leading to extreme Gaussian shapes and floaters, particularly at the highly distorted image periphery. Their proposed feature-overlap-driven cross-view joint optimization establishes consistent constraints across views to mitigate this. This focus on cross-view consistency directly echoes the multi-view coherence strategies discussed in the context of hybrid representations, underscoring its importance regardless of the underlying primitive.

For panoramic and omnidirectional images, the challenges are even more pronounced. The equirectangular projection (ERP) used for these images lacks a single center of projection, making it incompatible with the perspective rasterizer in vanilla 3DGS. A common strategy to bridge this

gap is to project the 360° image onto a multi-face representation, such as a cubemap. However, this approach suffers from discontinuities and projection distortions at face boundaries. The work on TPGS [105] introduces a novel “Transition Plane” between adjacent cube faces to enable smoother splatting directions and reduce optimization ambiguity in boundary regions. Another line of work, exemplified by ODGS [106], takes a more geometrically principled approach. For each Gaussian, a tangent plane is defined on the unit sphere. The Gaussian is then projected onto this tangent plane using a standard perspective rasterizer, and the results are transformed and combined into the final omnidirectional image. This approach provides a formal mathematical interpretation and achieves significant speedups over NeRF-based methods. Similarly, UniTriSplat [107] proposes a unified framework that reformulates Gaussian splatting directly on the unit sphere using HEALPix discretization, leveraging its equal-area property to create a uniform optimization behavior across different camera models, from narrow-FoV to full 360-degree panoramas.

Beyond the core projection, achieving high geometric consistency is critical for panoramic 3DGS. Methods like Spherical-GOF [108] move away from projection-based rasterization altogether, performing ray sampling directly on the unit sphere within a Gaussian Opacity Field framework. This approach derives a conservative spherical bounding rule for efficient ray-Gaussian culling and a spherical filtering scheme that adapts Gaussian footprints to the distortion-varying pixel sampling of panoramic images, leading to substantially improved geometric consistency. The scalability of these methods to large outdoor scenes is addressed by frameworks like PanoLOG [109], which proposes a geometry and gradient-based partitioning strategy for large-scale panoramic reconstruction. This strategy uses parallax-driven uncertainty to build adaptive bounding volumes, avoiding the pitfalls of naive partitioning methods that fail with omnipresent visibility. Such geometric rigor is reminiscent of the structural coherence advocated by implicit-field hybridizations, suggesting a convergence of design principles across different adaptations.

In parallel with these representation-level adaptations, significant effort has been dedicated to improving the real-time rendering performance of 3DGS, which is a key advantage over implicit representations. This performance focus is essential not only for standalone Gaussian systems but also for hybrid representations, where the added complexity of multiple primitives must not compromise interactive frame rates. One major bottleneck is the rasterization stage, which is often compute-bound. A promising direction is the use of Tensor Cores, which are high-throughput units for low-precision matrix operations that sit idle in standard 3DGS implementations. Both TensorGS and TC-GS [39, 110] propose to “tensorize” the rasterization process by mapping the alpha-blending computation to matrix multiplications, enabling the use of FP16 arithmetic with negligible quality degradation. TensorGS [39] further introduces a cross-tile grouping strategy to improve Gaussian data reuse and increase Tensor Core utilization. These approaches demonstrate significant end-to-end rendering speedups. Related work on hardware acceleration includes an architecture-algorithm co-design for axis-shared rasterization and order-independent transmittance [111], which removes the need for explicit sorting, a costly operation for large scenes.

Another significant source of inefficiency in 3DGS rendering is poor GPU utilization due to load imbalance and redundant computation. Hierarchical tiling strategies decouple the spatial partitioning and rasterization tile sizes, allowing each to be optimized independently. HiGS [26] partitions space using coarse macro-tiles while rasterizing within fine render tiles, issuing work in proportion to the Gaussians in each macro-tile and preventing dense regions from serializing through a single tile. Similarly, Local-GS [112] focuses on improving SIMT (Single Instruction, Multiple Threads) execution by organizing Gaussian primitives with respect to warp boundaries, reducing warp divergence and redundant computation. At a higher level, CaT-GS [40] identifies inter-frame redundancy and severe tile-level load imbalance as major bottlenecks. It introduces a speculative multi-frame preprocessing method and an inter-frame caching mechanism to eliminate

redundant work across frames, refactoring the rasterization tasks to mitigate load imbalance. These optimizations are crucial for scaling 3DGS to large-scale scenes, and they are complemented by workload-balancing frameworks like LoBE-GS [113], which uses a load-balanced KD-tree partitioning scheme to ensure equal per-block camera counts across multiple GPUs.

For resource-constrained platforms like mobile devices, aggressive optimizations are required. Frameworks like SeeLe [114] combine a view-dependent scene representation with online filtering to reduce the number of relevant Gaussians before rasterization, and a contribution-aware rasterization that prioritizes high-contribution Gaussians. On-device training, a much more stringent requirement, is tackled by PocketGS [115], which uses geometry-faithful point-cloud priors and a backpropagation strategy with cached intermediates to enable training within strict memory and time budgets. Finally, exploring the boundaries of 3DGS rendering, CoherentRaster [116] adapts 3DGS for light field displays that require rendering an interlaced image encoding many views. By employing cross-view coherent attribute reuse and view-coherent remapping, it achieves real-time, high-quality light field synthesis on consumer-grade hardware. Together, these advancements in native camera model support and performance-oriented rendering techniques establish 3DGS as a versatile and practical technology across a wide array of real-world applications, while also reinforcing the trend toward principled, multi-representational systems that can adapt to both diverse capture hardware and stringent performance budgets.

3 Optimization, Densification, and Training Strategies

3.1 Adaptive Density Control and Primitive Management

Adaptive density control (ADC) constitutes the heart of the 3D Gaussian Splatting (3DGS) optimization pipeline, governing the birth, growth, and elimination of Gaussian primitives throughout training. The original 3DGS framework introduced a heuristic yet remarkably effective mechanism that periodically assesses the positional gradients of each Gaussian, leveraging them as a proxy for geometric under-reconstruction, and subsequently performs either cloning or splitting operations to densify the representation in regions of high error, while simultaneously pruning Gaussians whose opacity falls below a threshold [117]. Although this foundational approach has demonstrated impressive results, the research community has extensively analyzed its limitations and proposed a rich taxonomy of refinements that re-engineer various aspects of the densification and pruning pipeline. The primary deficiencies of the original ADC can be distilled into three broad categories: the criteria used to trigger densification, the operational mechanics of clone and split operations, and the strategies for pruning or eliminating redundant or overfitted Gaussians.

A significant body of work focuses on refining the trigger criterion that determines *when* and *which* Gaussians should be densified. The vanilla 3DGS uses the magnitude of the accumulated view-space positional gradient, which has been shown to be vulnerable to a failure mode known as “gradient collision.” In regions suffering from over-reconstruction with high-frequency details, conflicting gradient directions from different views can cancel out, preventing large, blurry Gaussians from meeting the splitting threshold. This issue is explicitly analyzed in [118], which proposes the homodirectional view-space positional gradient as a more reliable criterion. By aggregating only the component of the gradient that is coherent across views, this approach effectively identifies large Gaussians in over-reconstructed regions that require splitting, leading to the recovery of fine details. Similarly, the Gradient Coherence Ratio (GCR) is introduced to explicitly discriminate between Gaussians with concordant versus conflicting gradient directions, enabling gradient-direction-aware density control [119]. This allows for prioritized splitting of conflicting-gradient Gaussians to enhance geometry while suppressing the proliferation of redundant concordant-direction Gaussians. Another

line of work replaces the positional gradient with a more reliable statistical signal. For instance, an opacity-gradient-based trigger is proposed, where the gradient of opacity serves as a lightweight proxy for rendering error, which when paired with a conservative pruning schedule, yields significant improvements in compactness, especially in few-shot scenarios [120]. Furthermore, reinterpreting densification as a signal verification problem, the CAdam framework leverages the first moment of gradients to exploit the principle of destructive and constructive interference, effectively disentangling geometric signals from the stochastic noise floor introduced by generative guidance, thereby addressing the Densification Dilemma in optimization-based generative distillation [121]. Beyond gradient statistics, some methods incorporate multi-view photometric errors into a probabilistic framework. For example, a unified adaptive sampling approach within a Metropolis-Hastings framework dynamically inserts and relocates Gaussians based on aggregated multi-view errors and opacity scores, significantly reducing reliance on hand-crafted heuristics [122]. Alternatively, the identification of error-contributing zones through multiview geometry constraints and image rendering errors enables localized point management, where densification can occur in regions not adequately captured by global gradient heuristics, such as transparent objects [123].

Once the decision to densify is made, the mechanics of the clone and split operations themselves have been scrutinized. The original random splitting, which divides a large Gaussian into two smaller ones with fixed cardinality and random placement, is inefficient, often requiring many iterations to expose fine details and inducing geometric distortion. To address this, Long-Axis Split was proposed to precisely control the position, shape, and opacity of child Gaussians, minimizing the difference before and after splitting and thereby reducing distortions [124, 42]. A more principled, error-driven adaptive splitting operator, AdpSplit, determines the number of split children and initializes their parameters from L1-pixel-error region statistics, leading to faster geometry discovery and reduced training time [125]. Another technique, which it frames in terms of optimization theory, proposes that splitting is crucial for escaping saddle points in the loss landscape, establishing necessary conditions for densification, determining the minimal number of offspring Gaussians, and providing analytical solutions for normalizing offspring opacity—resulting in a steepest descent density control strategy [126]. A related reformulation treats Gaussian centers as samples from a learnable probability density function over an octree or hashed probability pyramid, integrating densification into a continuous, end-to-end gradient-based optimization process that replaces discrete heuristics entirely [127, 128]. Finally, the volumetric refinement process can be driven by exploring the volume of inertia associated with each Gaussian to guide the refinement process, offering an alternative to purely gradient-based geometry discovery [129].

On the other end of the spectrum, the pruning and elimination of redundant or detrimental Gaussians is critical for maintaining a compact, efficient, and high-fidelity model. A common observation is that naive opacity-based pruning can inadvertently remove beneficial Gaussians or fail to eliminate overfitted ones. Recovery-Aware Pruning leverages the differences in recovery speed after resetting opacity to identify and prune Gaussians that are overfitted to training views, thereby improving generalization [124, 42]. A significance-aware pruning strategy, which considers the contribution of Gaussians to the rendering of background regions, was proposed to avoid background artifacts [130]. More advanced methods formulate pruning as an optimization problem. For instance, Gradient-Driven Natural Selection models survival pressure as a regularization gradient field applied to opacity, allowing the optimization gradients themselves to autonomously determine which Gaussians to retain or prune, making the process fully learnable and human-free [131]. Similarly, a variational framework formulates Gaussian pruning as variational free energy minimization, using prediction-error gating for asynchronous pruning decisions [132]. Others have focused on the granularity of ranking, distinguishing between scene-level and pixel-level pruning; it has been shown that pixel-level pruning, which ranks primitives based on their contribution to specific

rendering equations, not only sustains higher quality under extreme decimation but also provides an inherent pruning boundary that prevents catastrophic scene destruction [133]. To further enhance spatial consistency, an evidence lower bound-based adaptive densification strategy combined with an opacity-aware progressive pruning strategy and a graph-based feature encoding module has been proposed to dynamically adjust the spatial distribution and remove low-opacity Gaussians [134]. Beyond individual heuristics, some frameworks treat the global primitive distribution as a resource allocation problem. For instance, the global budget-constrained optimization formulation in SPARE-GS derives an optimality condition requiring marginal utility to be balanced across spatial regions, dynamically aligning the distribution of primitives with regional representational demand and coordinating densification, pruning, and adaptive termination [135]. Another approach, by treating the existing Gaussian set as an empirical probability field and resampling centers from it, rebuilds the correlation among primitives, changing the gradient accessibility under alpha compositing and weakening local parameter coupling to enable further optimization of converged models [136].

In summary, the field of adaptive density control has evolved from a single, rigid heuristic towards a diverse ecosystem of sophisticated, often principled, mechanisms. These advances collectively aim to strike a delicate balance between representation fidelity and compactness, addressing long-standing issues like over-reconstruction, floaters, memory bloat, and slow convergence, and paving the way for more efficient and robust deployment of 3DGS across a wide range of applications.

3.2 Initialization Strategies and Coarse-to-Fine Optimization Schedules

The quality and convergence behavior of 3D Gaussian Splatting (3DGS) are profoundly influenced by how the initial set of Gaussian primitives is established and how the optimization process unfolds over time. While the adaptive density control mechanisms discussed previously govern the *birth*, *growth*, and *elimination* of Gaussians during training, the *starting conditions* and the *temporal schedule* of optimization determine the efficiency and stability of this entire process. Standard 3DGS relies on Structure-from-Motion (SfM) point clouds for initialization; however, the fidelity and density of these initial points often prove inadequate, particularly in large-scale, textureless, or sparsely observed regions. Recognizing the critical role of initialization, the research community has developed a spectrum of techniques ranging from enhanced SfM pipelines and learned initialization strategies to dense, correspondence-based seeding. Concurrently, the optimization schedule itself has been rethought, transitioning from monolithic, high-frequency supervision toward coarse-to-fine paradigms that modulate input signal frequency, rendering resolution, or wavelet subbands for more stable and efficient convergence, thereby establishing a favorable landscape for the parameter optimization strategies examined in the next section.

A primary line of inquiry addresses the deficiencies of standard SfM. In sparse-view settings, where feature matching becomes unreliable, SfM often yields poor point clouds, leading to overfitting and artifacts. To counter this, frequency-aware SfM has been proposed, which augments the input images with low-frequency versions and relaxes multi-view correspondences to improve coverage in low-texture areas [137]. However, even with enhanced SfM, the initial point cloud remains sparse. To supplement these regions, 3DGS self-initialization techniques have been developed, which lift photometric supervision from training images into additional learned Gaussian centers, thereby compensating for SfM-sparse regions [137]. Furthermore, point-cloud regularization using geometric and visibility priors can enforce multi-view consistency and uniform spatial coverage, yielding a cleaner and more reliable starting point [137].

A more radical departure from incremental densification is the concept of eliminating the densification phase altogether through dense initialization. EDGS proposes a one-step approximation

of scene geometry by triangulating pixels from dense image correspondences, providing each Gaussian with well-informed color, scale, and position [138]. This dense initialization ensures more uniform detail across the scene, particularly in high-frequency regions where sparse keypoint-based methods struggle, and dramatically shortens the optimization path [138]. In the realm of surface reconstruction, learned data-driven priors can generate dense initializations for 2D Gaussian splatting optimization, providing a strong starting point that accelerates convergence on large-scale indoor scenes and improves geometry on flat wall structures, as demonstrated by QuickSplat [139]. This work also proposes a learned densifier network that predicts new Gaussians based on rendering gradients, removing the need for heuristic-based densification [139].

Beyond initialization, the optimization schedule itself is pivotal. A major challenge in 3DGS training is the over-reconstruction of high-frequency image regions by a few large Gaussians, leading to blur and artifacts. To address this, coarse-to-fine optimization schedules have been widely adopted. A foundational approach is the use of progressive frequency regularization, which guides Gaussian densification from low-to-high frequency components by minimizing the discrepancy between the frequency spectrum of rendered and ground-truth images (FreGS) [75]. Signal structure-aware scheduling offers a more principled approach by deriving the average sampling frequency and bandwidth of 3D representations, then regulating both the training image resolution and the Gaussian densification process based on scene frequency convergence, a method that has shown state-of-the-art performance in large-scale scenes [140].

Wavelet-domain methods have emerged as a powerful tool for implementing these frequency-based schedules. The discrete wavelet transform (DWT) provides spatial and frequency localization simultaneously, making it ideal for constructing curriculum-based training. AutoOpti3DGS uses a learnable DWT with a fixed low-pass filter and a learnable high-pass filter, activating fine frequencies gradually in a coarse-to-fine manner that delays redundant Gaussian formation [141]. A subsequent extension demonstrates that modulating the low-frequency subband recursively with a multi-level DWT creates a deeper curriculum, consistently reducing Gaussian counts with only a single scaling parameter for the high-pass filter [78]. From a different angle, DWTGS rethinks sparse-view frequency regularization by supervising only the low-frequency LL subbands at multiple DWT levels while enforcing self-supervised sparsity on the high-frequency HH subband, avoiding the inherent bias of Fourier transforms toward detrimental high-frequency learning [77]. Building on such analyses, wavelet-domain supervision can be formulated as a plug-and-play regularization strategy that combines multi-scale wavelet coefficient alignment, kurtosis concentration, and cross-band covariance penalties to explicitly enforce frequency statistics of rendered images to match ground-truth, as proposed in KC-3DGS [76].

An alternative yet complementary scheduling paradigm involves dynamically modifying the rendering resolution during training. DashGaussian formulates 3DGS optimization as progressively fitting higher-frequency components of training views, using a dynamic rendering resolution scheme to reduce optimization complexity while synchronizing the growth of Gaussian primitives with the rendering resolution for a better trade-off between computing redundancy and fitting quality [142]. Energy-guided coarse-to-fine multi-resolution training frameworks further refine this idea by gradually increasing resolution based on the energy density in 2D images, cooperating with global-to-local densification strategies to promote efficient Gaussian growth across the scene space [143]. A particularly elegant synthesis of frequency and resolution scheduling is found in the Opti3DGS framework, which leverages image frequency modulation to enforce a coarse scene representation first, then progressively refines it by modulating frequency details; this approach yields an average 62% reduction in Gaussians and a 20% reduction in optimization time while naturally producing a level-of-detail representation [144]. Similarly, frequency-aware Gaussian decomposition organizes primitives into groups corresponding to Laplacian-pyramid subbands, enabling both interpretable

level-of-detail representations and progressive training schedules [145].

Taken together, these initialization and scheduling techniques represent a fundamental shift in how 3DGS models are trained. Rather than treating all Gaussians and all image frequencies equally, modern approaches recognize that a data-adaptive, frequency- and resolution-aware curriculum—coupled with geometric and semantic priors for initialization—substantially improves convergence speed, model compactness, and rendering fidelity, especially in challenging sparse-view, large-scale, and high-frequency regimes. Moreover, by establishing a well-conditioned optimization landscape and a structured progression of learning targets, these strategies create the necessary preconditions for the advanced first- and second-order optimizers and stability-enhancing techniques discussed in the subsequent section, which further refine the convergence trajectory and final model quality.

3.3 First- and Second-Order Optimization Methods

The optimization of 3D Gaussian Splatting (3DGS) parameters is a critical determinant of both the final reconstruction quality and the computational cost of training. Building upon the initialization and scheduling strategies discussed previously, the choice and design of the optimizer itself plays an equally pivotal role in determining how efficiently and effectively the Gaussian parameters converge to a high-fidelity solution. While the original 3DGS framework relies on the Adam optimizer, a substantial body of research has explored both first- and second-order alternatives to accelerate convergence, improve fidelity, and enhance robustness, often drawing on and adapting advances from the broader deep learning optimization literature. This subsection systematically examines these optimization strategies, connecting them to the training pathologies and mitigation techniques covered in the subsequent discussion on stable convergence.

First-Order Methods and Their Enhancements

The default choice for 3DGS training has historically been Adam and its variants, owing to their adaptive per-parameter learning rates and robust performance across diverse scenes [146]. However, the general-purpose nature of these optimizers, which produce independent parameter updates, fails to capture the structural and spatial relationships inherent in a 3D scene, leading to inefficient optimization and slow convergence [147]. In response, researchers have proposed a range of first-order refinements. For instance, a detailed re-examination of the 3DGS optimization pipeline has revealed two overlooked coupling issues: update-step coupling, which induces optimizer state rescaling and costly attribute updates outside viewpoints, and gradient coupling in the moment, which can lead to under- or over-effective regularization. The resulting recomposed framework, termed AdamW-GS, demonstrates that decoupling and then selectively re-coupling beneficial components yields better optimization efficiency and representation effectiveness [148]. Furthermore, the initialization of the second-order moment estimation is a known pain point; setting the initial value to zero in Adam has been identified as a significant factor contributing to suboptimal generalization and instability. Simple data-driven or random non-zero initialization strategies for the second moment can stabilize convergence and enhance final performance [149]. Building on this, the substitution of the raw gradient with its momentumized version in the second raw moment estimate—termed AdaMomentum—has been shown to improve generalization while maintaining training speed [150]. Memory constraints have also motivated innovations such as SlimAdam, which applies a layer-wise Signal-to-Noise Ratio analysis to identify dimensions where second-moment tensors can be safely replaced by their means, achieving up to 98% memory savings in the second moment without sacrificing performance [151]. Beyond these architectural tweaks, gradient regularization (GR) has been demonstrated to improve generalizability. In a broader context, the integration of explicit gradient regularization into natural gradient descent leads to a family of scalable second-order optimizers (GRNG) that outperform first-order baselines and some

second-order methods on vision and language benchmarks [152]. Moreover, in the specific context of 3DGS, a “warped gradient descent” concept from meta-learning can be integrated into Adam by introducing a learnable distortion matrix to linearly transform gradients, enhancing adaptation to distinct dataset characteristics [153].

Flat-Minima Optimization and Generalization

A significant challenge in sparse-view 3DGS is the tendency to overfit to the limited observed images and generalize poorly to unseen viewpoints, an issue that directly connects to the sparse-view regularization techniques elaborated in the following subsection. This issue has been addressed from the perspective of flat minima optimization, which seeks solutions that remain stable under small parameter perturbations [154]. By viewing Gaussian parameters as trainable weights, this approach regularizes optimization with controlled perturbations that account for each Gaussian’s anisotropy and the training progress. The incorporation of a periodic reinitialization step further stabilizes the process, leading to sharper and more stable reconstructions under sparse supervision [154]. This focus on flat minima aligns with broader findings in deep learning that adaptive methods often converge to sharper basins of the loss landscape, hurting generalization. Memory-augmented versions of Adam that promote exploration toward flatter minima by using a buffer of critical momentum terms have shown to improve performance on supervised tasks [155]. Similarly, research on quantization-aware training for domain generalization highlights a “scale-gradient conflict” problem, where task loss and smoothness loss induce conflicting gradients for quantizer scaling factors; selectively freezing these gradients stabilizes training and enhances out-of-domain generalization, a principle applicable to the mixed-precision training of 3DGS models [156].

Second-Order Optimization for 3DGS

To address the slow convergence of first-order methods, a growing body of work has introduced second-order optimizers tailored to the 3DGS problem, offering a complementary strategy to the coarse-to-fine scheduling discussed earlier for improving computational efficiency. The Levenberg-Marquardt (LM) algorithm has emerged as a particularly promising approach. The 3DGS-LM framework replaces the Adam optimizer with a tailored LM method, enabling efficient GPU-parallelized computation of update directions from multiple image subsets. This approach achieves a 20% speedup over the original 3DGS while maintaining identical reconstruction quality, and is agnostic to other acceleration methods, thus enabling further compound speedups [157]. A further advancement, 3DGS²-TR, proposes a scalable second-order trust-region method that approximates curvature using only the diagonal of the Hessian matrix, computed efficiently via Hutchinson’s method. This fully matrix-free approach has the same $O(n)$ computation and memory complexity as Adam, but achieves better reconstruction quality in 50% fewer iterations, with a peak GPU memory overhead of less than 1GB [158]. Another line of work exploits the significant sparsity in the 3DGS Jacobian—since each Gaussian affects only a limited number of pixels—to develop a matrix-free, GPU-parallelized LM optimizer with Conjugate Gradient (CG). This method, which also introduces sampling strategies for camera views and loss functions, achieves roughly a 4x speedup over standard LM and approximately 5x over Adam when the Gaussian count is low, while using 3.5x less memory than 3DGS-LM [159]. These developments reflect a broader trend in deep learning toward quasi-Newton and Hessian-free methods. For instance, symmetric rank-one quasi-Newton approaches with adaptive regularized cubics allow for indefinite Hessian approximations, enabling the exploitation of directions of negative curvature [160]. Hessian-free methods using block-diagonal approximations of the generalized Gauss-Newton matrix have also shown better convergence and generalization compared to standard approaches [161].

Trust-Region and Learned Optimizers

Trust-region methods have demonstrated strong performance in non-convex optimization, particularly for ill-conditioned problems [162]. In the context of deep learning, stochastic trust-

region and cubic regularization methods have been shown to escape saddle points and exhibit robust convergence to second-order stationary points [163]. Sub-sampled variants of these methods reduce computational overhead while retaining strong convergence guarantees [164]. For 3DGS, the parameter-wise trust-region technique based on the squared Hellinger distance, as used in 3DGS²-TR, regularizes updates to Gaussian parameters, ensuring stable optimization despite the strong non-linearity of the rasterization process [158]. In parallel, learned optimizers offer a compelling alternative, especially in the context of the learned initialization strategies discussed in the previous subsection. Unlike conventional optimizers, learned optimizers can predict correlated updates informed by inter-parameter and inter-Gaussian dependencies. The Learn2Splat framework introduces a learned optimizer that avoids degradation over extended optimization horizons by using a checkpoint buffer and an optimizer rollout strategy, and it demonstrates improved early novel view synthesis quality with zero-shot generalization to unseen reconstruction settings [147]. Furthermore, optimization-aware training frameworks like ForeSplat have been developed to ensure that the predicted initializations from feed-forward 3DGS models are explicitly designed for rapid, effective refinement by downstream optimizers. At its core is a lightweight multi-anchor meta-gradient training rule that bypasses costly higher-order differentiation, enabling high-quality reconstruction within seconds after a few refinement steps [165]. This concept is extended by Diff3R, which explicitly bridges feed-forward prediction and test-time optimization by incorporating a differentiable 3DGS optimization layer into the training loop, computing gradients via the Implicit Function Theorem and a scalable, matrix-free PCG solver, while simultaneously predicting uncertainty to mitigate overfitting [166].

Hybrid and Adaptive Optimizer Designs

Given the complementary strengths and weaknesses of first- and second-order methods, hybrid designs have become increasingly common, often motivated by the same instability concerns that drive the densification and pruning refinements in the next subsection. For example, dynamic momentum and learning rate adjustments, as seen in DWMGrad, allow optimizers to adapt to changing environments and task complexities [167]. The use of projected gradients combined with adaptive moment estimation, as in PadamP, accelerates training and improves generalization for unconstrained non-convex problems [146]. The observation that loss functions often transition from non-convexity to local convexity near optima has led to two-phase algorithms that detect the transition point and switch from Adam to conjugate gradient methods, yielding improved convergence and accuracy [168]. In the realm of second-order optimization, many practical implementations remain computationally prohibitive due to large matrix inversions. Techniques that approximate the Fisher information matrix using a diagonal representation, integrated with an adaptive trust-region mechanism, reduce the complexity from $O(n^2)$ to $O(n)$ and demonstrate faster, more stable convergence compared to Adam [169]. Similarly, the efficient computation of matrix square roots and the use of low-rank approximations in full-matrix preconditioning methods have made them more practical for large-scale problems [170]. The Frequent Directions sketch can be used to maintain low-rank matrix preconditioners, reducing memory and compute requirements while matching the regret guarantees of full-matrix adaptive methods up to an additive error in the bottom eigenvalues [171]. In summary, the landscape of 3DGS optimization is evolving rapidly, moving from general-purpose first-order methods toward specialized, second-order, and learned optimizers that exploit the unique structure of the Gaussian representation to achieve faster convergence, lower memory footprints, and higher reconstruction fidelity. These advances in the optimizer itself, when combined with the initialization and scheduling techniques of the previous subsection, set the stage for the targeted strategies to mitigate over-reconstruction and degenerate convergence that are examined next.

3.4 Mitigating Over-Reconstruction and Degenerate Convergence

The optimization of 3D Gaussian Splatting (3DGS) is fundamentally a non-convex, high-dimensional problem. As discussed in the previous subsection, the choice of optimizer—whether first-order, second-order, or learned—largely dictates the efficiency and trajectory of parameter convergence. However, even with a well-designed optimizer, standard training pipelines remain susceptible to a range of training pathologies that can trap the optimization in poor local minima, leading to visible artifacts and degraded rendering quality. These pathologies include over-reconstruction, persistent floater artifacts, unstable densification cycles, and overfitting in under-constrained settings. This subsection surveys the key strategies developed to diagnose and mitigate these issues, ensuring more stable and robust convergence for 3DGS models. Where the previous subsection focused on the optimizer as the primary driver of convergence, the techniques discussed here act as complementary safeguards, targeting the structural and statistical failure modes that optimizers alone cannot address.

One of the most fundamental challenges is the implicit gradient bias that causes 3DGS optimization to converge to suboptimal local solutions, a phenomenon characterized as the “Blur Trap” [172]. This trap is rooted in the optimization formulation, where the gradient signal for large Gaussians can vanish when a pseudo-equilibrium is reached, preventing further refinement. To escape these sharp minima, simple explicit exploration operators, such as random seeding to mitigate the “Far-Side Blur Trap” and random splitting to address the “Near-Side Blur Trap,” have been proposed [172]. This finding highlights that deterministic densification strategies are insufficient and that stochastic exploration is crucial for navigating the complex loss landscape.

A closely related problem is over-reconstruction, where regions with high-frequency details become blurry. This issue is often traced back to a flaw in the adaptive density control strategy, where “gradient collision” prevents large Gaussians in over-reconstructed regions from being split [118]. To overcome this, novel densification criteria have been introduced, such as the homodirectional view-space positional gradient, which more effectively identifies and splits under-split Gaussians [118]. Other approaches refine the densification process itself, for instance, by using a pixel-aware gradient that considers the number of pixels a Gaussian covers, prompting the growth of large Gaussians that are observable from many viewpoints [173]. Furthermore, techniques like Edge-Aware Score for selecting split candidates, a Long-Axis Split strategy to reduce geometric distortions, and Recovery-Aware Pruning to remove overfitted Gaussians have been shown to collectively enhance rendering fidelity [42]. The use of an opacity-gradient-based densification trigger, paired with a more conservative pruning schedule, has also proven effective in preventing destructive optimization cycles and promoting more compact reconstructions in few-shot scenarios [120].

The instability of the densification process itself is another major source of degenerate convergence. Standard 3DGS often suffers from “stubborn floater” artifacts whose root cause is the vanishing opacity gradients when a floater’s color blends into a canceling error with the background. To resolve this, a Dual Opacity architecture has been proposed, which decouples geometric regularization from final appearance rendering, allowing a “Geometric Regularization Path” to bear strong depth-based constraints and an “Appearance Refinement Path” to focus on high-fidelity details [174]. Similarly, when a converged model is poorly parameterized due to high-opacity floaters that attenuate gradients to true surfaces, an equivalent distribution reorganization technique has been suggested. This method resamples Gaussian centers from the existing set, estimates local anisotropic covariances, and re-initializes low opacity, effectively rebuilding the visibility structure to improve gradient accessibility and reduce parameter coupling [136]. Another strategy proposes localizing point management (LPM) based on multiview geometry constraints and rendering errors to identify error-contributing zones for densification and opacity reset, which is a versatile plugin for both static and dynamic

models [123].

Sparse-view settings are particularly prone to overfitting, where the model memorizes training views and produces artifacts like blurring and floaters in novel views. A significant body of work addresses this through regularization, often drawing inspiration from dropout. Early approaches like DropoutGS proposed a Random Dropout Regularization (RDR) to benefit from the lower overfitting of smaller models, combined with an Edge-guided Splitting Strategy (ESS) to recover high-frequency details [175]. However, a more recent framework, PairDropGS, identifies that random dropout can lead to inconsistencies between different dropped subsets, causing unstable reconstruction. It introduces a Paired Dropout-induced Consistency Regularization (PairDropGS) that constructs pairs of dropped subsets and applies a low-frequency consistency regularization between them, along with a progressive scheduling strategy for stability [176]. Another perspective on this issue frames the core challenge as the unobservability of Gaussian primitive reliability. The DOC-GS framework models this via a Continuous Depth-Guided Dropout (CDGD) strategy and uses a Dark Channel Prior (DCP) from the observation domain to identify and prune low-confidence, haze-like Gaussians [177]. To further stabilize the optimization, some methods use Fisher Information to provide principled guidance for both prior-based augmentation and dropout regularization, actively selecting the most informative supporting views and adapting the removal probability of each Gaussian [178].

Beyond regularization, efficient training strategies have also been developed to mitigate overfitting and promote convergence. For instance, an alternating densification framework interleaves high-densification phases to capture details with low-densification phases that focus on aggressive opacity pruning and geometry regularization through pseudo-view consistency [179]. The initialization of Gaussians is also critical; a dense initialization from triangulated pixels eliminates the need for incremental densification, bypassing a lengthy optimization path and reducing the chance of converging to a poor local optima [138]. During the refinement phase after densification, significant redundancy in backward computation has been observed. A view-adaptive backward gating mechanism called SkipGS selectively skips backward passes for views whose losses have plateaued, reducing post-densification training time by 42% without sacrificing quality [180].

Finally, co-regularization approaches offer a complementary means to identify and suppress inaccurate reconstruction. By training two 3D Gaussian radiance fields simultaneously and observing their point and rendering disagreements, one can predict and identify inaccurate regions, which can then be pruned or regularized [181]. For in-the-wild scenes, where transient distractors and appearance inconsistencies generate conflicting cross-view gradients, a conflict-aware approach harmonizes gradients from different views by rotating them into an orthogonal configuration, reducing directional interference and leading to more stable optimization [182].

In summary, mitigating over-reconstruction and degenerate convergence in 3DGS requires a multi-faceted approach that complements the optimizer-level strategies discussed previously. It involves refining the densification and pruning criteria to be more sensitive to detail and artifacts, introducing explicit exploration and stochastic perturbation to escape local traps, and applying sophisticated regularization techniques, often guided by uncertainty, consistency, or geometric priors, to counter overfitting in data-sparse scenarios. The strategies discussed here are not mutually exclusive and are often combined to create robust, efficient, and high-fidelity 3D reconstruction systems.

4 Geometry Reconstruction and Surface Extraction

4.1 Surface Regularization and Geometric Priors for Gaussian Alignment

The raw optimization of 3D Gaussian Splatting (3DGS) is driven predominantly by photometric reconstruction loss. While this approach yields impressive novel-view synthesis, it frequently leads to geometrically inaccurate reconstructions. The explicit, point-based nature of Gaussians provides no inherent surface constraint; the optimizer is free to place primitives anywhere that minimizes the rendering error, which can result in non-physical configurations. A significant body of research has therefore emerged to introduce geometric priors and regularization terms that steer the optimization process toward a more accurate alignment of Gaussian primitives with the underlying scene surfaces. These techniques address a range of artifacts, from the formation of “needle-like” splats to the presence of floating, semi-transparent debris (“floaters”), which plague naive 3DGS reconstructions. This line of work forms the foundation for the subsequent development of more sophisticated hybrid representations, which couple explicit Gaussians with implicit surface models to achieve globally coherent geometry.

A primary source of geometric guidance comes from monocular depth and normal estimators. Off-the-shelf models provide dense priors that can be used to supervise the geometry of the reconstructed scene. However, these priors are not without their challenges; they suffer from scale ambiguity, cross-view inconsistency, and local inaccuracies, as highlighted by research into reliable monocular supervision [183, 184]. A naive application of these priors as hard ground-truth can actually degrade the reconstruction. To address this, a common strategy is to distill confidence or uncertainty from the monocular predictions and use this to weight the geometric loss, preventing the propagation of inaccurate depth values into well-constrained regions [183, 184, 185]. Similar confidence-based filtering approaches are proposed in systems designed for indoor reconstruction, where supervision is adaptively applied by comparing the consistency of prior estimates during optimization to identify and filter out low-quality depth and normal information [186].

Beyond simple depth supervision, higher-order geometric constraints have proven powerful. The effective rank of a Gaussian’s covariance matrix is a revealing statistic; many optimized Gaussians degrade into anisotropic “needle-like” shapes, which are suboptimal for surface representation [9]. This phenomenon is directly linked to the geometry of the splats themselves, as an extreme anisotropy with one dominant variance contributes little to surface fidelity and can produce inaccurate normals. To combat this, the effective rank has been introduced as a regularization term, discouraging extreme anisotropy and promoting more balanced, disk-like forms that better align with flat surfaces [9, 187]. This regularization has been shown to not only reduce needle-like artifacts but also to improve the quality of reconstructed normals and geometry across various 3DGS variants [9]. Such per-primitive shape control is a critical step toward making individual Gaussians more surface-compliant, yet it remains a local constraint; the next logical step is to couple these regularized primitives with a global surface representation.

The integration of explicit surface representations with the Gaussian cloud presents another avenue for enforcing global geometric coherence. By coupling Gaussian primitives with a neural Signed Distance Function (SDF), optimization can be guided by a continuous surface prior. The SDF is regularized with an Eikonal loss to maintain proper distance properties, which in turn provides a global context that guides the Gaussians toward a consistent and coherent surface, addressing issues of local misalignment [187, 188]. This hybrid methodology has demonstrated significant improvements in generating watertight and topologically consistent meshes compared to using Gaussians alone [188]. This same philosophy—leveraging an external global structure to correct local Gaussian behavior—extends beyond SDFs to other forms of scene-level priors, such as

multiview geometry and planar constraints.

Multiview consistency serves as a crucial, self-supervised signal for correcting geometric errors. The principle is that points on the true surface should be consistently visible and accurately reconstructed across different viewpoints. However, relying on depth-based reprojection to infer visibility can create a circular dependency where visibility estimation requires accurate depth, and depth supervision is conditioned on visibility [189]. To break this cycle, more recent approaches model visibility explicitly at the level of Gaussian primitives rather than pixels. By aggregating cross-view visibility of the same Gaussian, one can construct reliable supervision over co-visible regions, mitigating the ambiguity and improving surface reconstruction accuracy [189]. Other methods leverage multi-view stereo (MVS) priors that are robust in regions with significant color variation, complementing the strengths of 3DGS near object boundaries and low-texture regions to produce a more reliable overall geometry [190]. This complementary nature of priors is also exploited in dynamic scenes, where the distinct advantages of monocular and multi-view geometric depths are combined for more robust 4D reconstruction [191].

Stereoscopic priors offer a powerful alternative to monocular cues, providing absolute scale and binocular consistency that monocular methods inherently lack. By constructing virtual stereo pairs during optimization and using a foundation stereo model, methods like StereoGS enforce stronger geometric constraints, leading to more accurate geometries in sparse-view settings [192]. This is in contrast to scale-agnostic monocular supervision, which can lead to defective geometry due to its inherent ambiguity [192]. This approach can be further augmented with a dense initialization strategy that uses zero-shot multi-view depth estimation to anchor primitives to accurate scene surfaces [192].

Planar priors are particularly effective in man-made environments like indoor and urban scenes, where surfaces are often dominated by flat regions. By leveraging off-the-shelf segmentation and normal prediction models, systems like GSPlane can extract robust planar priors to structure the representation of Gaussian coordinates on these planes, enforcing geometric consistency during training [193]. Such methods advocate for representing scenes using planes as fundamental primitives, which is beneficial not only for geometric accuracy but also for downstream applications like scene editing and physical simulation [193].

Finally, a key challenge in applying external priors is dealing with their inherent unreliability, especially in real-world scenarios where geometric information, such as LiDAR data, is sparse and unevenly distributed. Instead of enforcing geometry as a hard constraint, recent methods propose modeling the reliability of these priors. This is achieved by representing partially observable geometry as an energy field, where the confidence of the prior guides the optimization process, providing soft guidance that steers the Gaussians without directly restricting the solution space. This prevents potentially detrimental constraints from harming the reconstruction in areas where the prior is untrustworthy [194]. This philosophy of treating geometric priors as informative but uncertain signals, rather than ground truth, is a central theme in creating robust and geometrically accurate 3DGS frameworks. Collectively, these regularization strategies—ranging from local shape control to global surface coupling—establish the critical groundwork for the hybrid Gaussian-SDF approaches that follow, which seek to unify the strengths of explicit and implicit representations for high-fidelity surface reconstruction.

4.2 Hybrid Gaussian-SDF Representations for Implicit Surface Reconstruction

The discrete, explicit nature of 3D Gaussian Splatting (3DGS) provides exceptional rendering efficiency and quality but introduces significant challenges for recovering accurate, continuous 3D geometry. The absence of an implicit global field means that surfaces extracted from raw Gaussians

are often fragmented, noisy, and riddled with floaters, limiting direct mesh generation. To bridge this gap, a prominent line of research integrates implicit Signed Distance Fields (SDFs) with the explicit Gaussian representation, creating hybrid frameworks that aim to combine the rendering strengths of 3DGS with the geometric robustness of implicit surfaces. These hybrids enforce global surface continuity, improve watertight mesh extraction, and enable accurate reconstruction across diverse indoor, outdoor, and reflective scenarios. This paradigm directly extends the regularization strategies discussed in the previous subsection, moving from external geometric priors as supervision toward a fully coupled, differentiable joint representation where the SDF serves as an internal, learnable prior that continuously guides Gaussian placement and shape.

A foundational approach in this domain involves linking Gaussian primitives to an SDF through a differentiable SDF-to-opacity transformation. The 3DGSR method introduces a transformation function that converts SDF values into Gaussian opacities, enabling joint optimization of the SDF and the Gaussians within a unified framework [195]. While optimization of the Gaussians provides supervisory signals for the SDF, these signals are sparse and limited to locations occupied by discrete Gaussians. To address this, 3DGSR incorporates volumetric rendering and aligns the rendered geometric attributes—depth and normal—with those derived from the Gaussian splatting, providing dense supervision for learning a continuous SDF and eliminating redundant surfaces outside the Gaussian sampling range [195]. Similarly, a line of work attaches a discretely sampled SDF value to each Gaussian, linking it to opacity via an SDF-to-opacity transformation to avoid the computational cost of ray marching [196]. This strategy introduces a projection-based consistency loss that projects Gaussians onto the zero-level set of the SDF, ensuring alignment between the discrete samples and the underlying continuous surface [196]. Another hybrid technique, GSurf, integrates SDFs into the splatting pipeline, leveraging the continuous nature of SDFs to regularize Gaussian primitives, which effectively suppresses noise and fills geometric holes, resulting in compact and high-quality surfaces [197]. Furthermore, DiGS directly embeds SDF learning into the 3DGS pipeline by associating each Gaussian with a learnable SDF value, which aligns primitives with underlying geometry and improves cross-view consistency, complemented by a geometry-guided grid growth strategy for dense coverage [94].

A common strategy to enhance the mutual benefits of both representations is to employ bidirectional guidance between the Gaussian and SDF branches. For instance, GSDF proposes a dual-branch architecture where the efficient 3DGS representation and the neural SDF leverage and enhance each other’s strengths through mutual guidance and joint supervision [198]. In the context of reflective objects, GS-ROR² guides 3DGS and SDF bidirectionally: an SDF-aided Gaussian splatting improves the efficiency of relighting optimization through mutual depth and normal supervision, while a GS-guided SDF refinement enhances the quality of the final mesh [199]. This approach also introduces an SDF-aware pruning strategy to remove Gaussian outliers that are distant from the SDF-defined surface, mitigating floater artifacts [199]. A similar bidirectional philosophy is seen in GaussianRoom, which integrates neural SDFs with 3DGS for indoor scenes. The SDF guides the densification and pruning of Gaussians, allowing for accurate modeling even with poor point cloud initialization, while the Gaussians improve SDF efficiency by piloting its point sampling. Additionally, it introduces normal and edge priors to resolve geometric ambiguities in textureless areas [200]. In dynamic urban environments, leveraging 2D priors for depth and point tracking, coupled with an SDF representation for dynamic objects, has been shown to enhance geometric accuracy and enable editing tasks such as scene decomposition and composition [201]. A more tightly coupled approach aligns Gaussians directly onto the SDF’s zero-level set. This technique, termed neural pulling, dynamically positions 3D Gaussians on the surface defined by the SDF; the aligned Gaussians are then rendered through differentiable rasterization, and the neural SDF is updated by pulling neighboring space towards these aligned Gaussians. This joint optimization

of Gaussians and SDF, guided by both RGB and geometry constraints, progressively refines the signed distance field near the surface, achieving accurate and complete reconstructions [202]. To improve the surface alignment of individual Gaussians, ARGS builds upon SuGaR by introducing an effective rank regularization that discourages extreme anisotropy in Gaussian shapes, promoting more stable “disk-like” forms, and integrating a neural SDF regularized with an Eikonal loss to provide a continuous global surface prior that guides Gaussians toward better geometric alignment [187].

In complex scenes with strong lighting or specular highlights, hybrid Gaussian-SDF representations are particularly valuable. The GS-Octree method combines an octree-based implicit surface representation with Gaussian splatting. In this approach, a coarse SDF is first reconstructed via volume rendering and encoded in a low-resolution octree. 3D Gaussians are then introduced as additional degrees of freedom, guided by the SDF, to recover finer geometric details, and the refined SDF in turn optimizes the Gaussians further [203]. To improve robustness, MonoGSDF couples Gaussian primitives with a neural SDF, where the SDF guides the Gaussians’ spatial distribution during training, and the Gaussians serve as priors for surface reconstruction at inference, eliminating the need for memory-intensive Marching Cubes. A multi-resolution training scheme and monocular geometric cues further enhance the reconstruction quality [188]. A related framework, Gaussian-Voxel Duet, uses a sparse voxel scaffold that is jointly optimized with scaffold-anchored Gaussians, confining them to a narrow band around surfaces defined by voxelized SDFs, thus improving representation efficiency and condensing floaters while maintaining fast convergence [95]. For large-scale urban scenes, the GSSR method has also shown the potential for accurate geometric precision by distributing Gaussians evenly along latent surfaces, guided by per-Gaussian single-view normal consistency and multi-view photometric consistency, along with a surface-aligned optimization objective [204].

The integration of depth and normal priors is a critical component in stabilizing the joint optimization of Gaussians and SDFs. SplatSDF accelerates the convergence of an SDF-NeRF by fusing a pre-trained 3DGS model at the architectural level, injecting neural embeddings of the Gaussians into the SDF-NeRF around the object surface [205]. This method demonstrates faster convergence to a given geometric accuracy compared to baselines, showcasing a practical use of 3DGS as a geometric prior. Another approach, GSDF, which uses depth and normal alignment between the rendered Gaussians and the SDF-derived geometry, has shown state-of-the-art performance in both rendering and reconstruction, validating the effectiveness of geometric consistency regularizers [198]. Furthermore, methods like GLS jointly optimize surface reconstruction and open-vocabulary segmentation, where a surface normal prior and depth error optimization are used to improve the geometric quality of the Gaussian field [206]. The EnerGS framework takes a different approach by treating partial geometric priors from LiDAR as a continuous energy field that provides soft guidance for Gaussian optimization, rather than a hard constraint, thus steering the optimization towards geometrically plausible solutions [194].

Hybrid Gaussian-SDF representations also offer strong advantages in robotics and SLAM applications, where both dense reconstruction and rendering are required. GS-SDF exploits the geometry from a trained SDF to initialize Gaussians physically and provide comprehensive geometric regularization, achieving superior reconstruction accuracy and rendering quality across different trajectories [207]. Similarly, in the context of robotic navigation, frameworks that parameterize Gaussian elements to directly recover a distance field, such as SplatlessDF, demonstrate that Gaussian-style representations can be used not only for photometric rendering but also as a continuous mapping representation suitable for downstream tasks [208]. The CarGS framework learns an adaptive contribution for Gaussian primitives, squeezing the knowledge from geometry regularization into a compact MLP, which improves the mutual learning between rendering and

reconstruction tasks in a unified model [209]. Finally, methods that leverage normal information extracted from an SDF to regularize Gaussian splatting have been shown to improve rendering quality while also facilitating direct mesh extraction [210]. Collectively, these works demonstrate that the integration of SDFs within the Gaussian splatting framework is a powerful paradigm for overcoming the geometric limitations of explicit representations, establishing a trajectory towards high-fidelity 3D reconstruction that is both accurate and computationally efficient. This hybrid paradigm directly sets the stage for the geometry-aware rendering and meshing techniques discussed in the following subsection, which further refine how continuous fields are extracted and optimized for direct, high-quality surface generation from these coupled representations.

4.3 Geometry-Aware Rendering and Differentiable Rasterization for Surface Extraction

The efficacy of surface extraction from 3D Gaussian Splatting (3DGS) is fundamentally contingent upon the design of the rendering pipeline and the geometric interpretation of its primitives. While the standard alpha-blending pipeline of 3DGS is engineered for photorealistic appearance, it does not inherently provide a mechanism for querying a continuous geometric field, such as a signed distance or occupancy function. This subsection reviews the key advances in geometry-aware rendering that bridge this gap, enabling the extraction of accurate, watertight, and high-fidelity meshes, and thereby complementing the implicit SDF-based hybrids discussed earlier. These innovations span from modifying the shading and attenuation models to developing novel, differentiable rasterization strategies and associated mesh extraction algorithms.

A central challenge in deriving geometry from 3DGS is that the standard Gaussian kernel and its blend of opacity, color, and depth do not naturally align with a well-defined surface. To address this, several works have re-engineered the rendering process to be geometry-aware. One prominent approach involves deriving an analytical occupancy field directly from the Gaussian primitives. For instance, [22] introduces a principled occupancy field for Gaussian Splatting by incorporating a learnable oriented normal for each Gaussian and modifying the attenuation formulation. This yields closed-form expressions for the normal and occupancy at arbitrary spatial locations, providing a continuous geometric field over which an isosurface can be extracted. Similarly, [211] defines a GOF derived from ray-tracing-based volume rendering of 3D Gaussians. This formulation enables direct geometry extraction by identifying the level-set of the opacity field, bypassing the need for error-prone post-processing steps like Poisson reconstruction or TSDF fusion. The key is that these methods move beyond the simple aggregation of depth maps to establish a coherent, global, and differentiable volumetric field based on the Gaussian distribution.

The precision of the extracted geometry is also heavily influenced by the shading and normal models used during rendering, especially for scenes with complex lighting and view-dependent effects. Methods that rely on opaque Gaussians often struggle to separate a surface’s albedo from its lighting-induced appearance, leading to inaccurate normal estimation and flawed geometry. To mitigate this, [212] proposes a self-supervised confidence framework that dynamically balances photometric and geometric supervision, along with losses that penalize per-primitive color and normal variance. This approach improves surface extraction by reducing ambiguity in areas with abundant view-dependent effects. Another notable direction is the integration of physically-based shading into the rasterization pipeline. [213] explicitly integrates normals into the rendering equation by re-parameterizing surface colors as a product of normals and an Integrated Directional Illumination Vector (IDIV), thereby ensuring that the optimized normals are directly linked to the rendering process and not merely an auxiliary output. This direct coupling, exemplified by [214], is crucial for ensuring that geometric attributes and visual appearance are jointly optimized without conflict,

which is particularly important for robust reconstruction of highly reflective surfaces.

Beyond the shading model, the differentiable rasterization step itself is a focal point for innovation. The standard per-pixel alpha-blending of sorted Gaussians is an approximation that can produce artifacts at object boundaries and is not a direct measure of a surface intersection. To overcome these limitations, [215] introduces hierarchical resorting and a robust formulation of Gaussian depth that aligns better with the level-set, in conjunction with a parallelized Marching Tetrahedra algorithm tailored to their opacity formulation. This improves both the accuracy of the extracted surface and the speed of the meshing step. Furthermore, the emergence of order-independent transparency (OIT) techniques offers a significant advantage for surface extraction in complex scenes. Unlike depth-sorted methods, which can be prone to errors with overlapping primitives, OIT rendering as seen in [32] provides a more robust foundation for computing and reasoning about depth ordering, potentially leading to more stable and consistent geometric inference. The challenge of reconstructing semi-transparent surfaces is a testament to the need for more sophisticated rendering strategies. For instance, [216] moves away from the single-depth-per-pixel assumption, uniformly sampling transmittance to model a multi-modal distribution of opacity and depth. This probabilistic approach is critical for accurately reconstructing surfaces where multiple layers are visible, such as laboratory glassware, an issue that standard depth-accumulation methods fundamentally cannot resolve. Relatedly, [71] disentangles each Gaussian’s geometric occupancy from its optical opacity to create a unified surface-volume representation. This factorization captures high-frequency reflections as surfaces while preserving clear transmission through the volume, and their hybrid renderer, augmented with specular-aware gradient gating, enables robust reconstruction of scenes with complex reflections.

Finally, the ultimate goal of geometry-aware rendering is to facilitate direct and high-quality mesh generation, thereby bridging the gap between the discrete Gaussian representation and the continuous surface outputs required by downstream applications. The strategies for extracting the final surface are diverse and have evolved significantly. Early and still common approaches treat rendered depth maps as geometry proxies and fuse them using Truncated Signed Distance Functions (TSDF). Methods like [186] refine this process by developing scale-aware meshing strategies inspired by TSDF and octree-based isosurface extraction to recover finer details. Others have proposed using the Gaussian primitives’ points as a dense point cloud to be processed by Marching Cubes. For example, [217] demonstrates that with its optimized normal calculation and depth-guided densification, direct mesh extraction from the dense Gaussian point cloud using Marching Cubes becomes a viable and high-fidelity option. However, the most advanced methods establish a continuous field from the Gaussians prior to meshing. [22] introduces Primal Adaptive Meshing for Region-of-Interest meshing at arbitrary resolutions from their continuous occupancy field, allowing for the recovery of thin structures like bicycle spokes. Similarly, [193] leverages the optimized planar priors to refine mesh layouts, significantly improving topological structure and reducing vertex counts, showcasing that integration of structural cues can directly enhance the meshing output. For scenarios requiring topology-adaptive optimization, methodologies such as [218] estimate a continuous opacity field for differentiable element pruning to construct a unified, topologically coherent tetrahedral mesh, streamlining downstream physical simulations. When viewed together, these approaches illustrate a clear progression from geometric proxy fusion toward direct, field-based isosurface extraction, enabled by increasingly geometry-aware rendering and rasterization pipelines that are deeply integrated with the final meshing stage. This progression not only addresses the limitations of raw Gaussian geometry but also establishes a foundation for the domain-specific adaptations and system-level integrations explored in the following subsection, wherein these geometry-aware rendering and meshing techniques are tailored to large-scale, dynamic, and real-time applications.

4.4 Domain-Specific Geometry Reconstruction and System Integration

Domain-specific adaptations of Gaussian-based surface reconstruction have become essential for translating general-purpose methods into practical systems that operate under real-world constraints. While the geometry-aware rendering and meshing techniques discussed in the preceding subsection establish a robust foundation for extracting high-fidelity surfaces in controlled settings, their direct application to large-scale outdoor environments, dynamic scenes, and sensor-constrained platforms often requires substantial architectural and algorithmic modifications. This subsection examines these domain-specific extensions, focusing on large-scale urban and road surface modeling, dynamic object reconstruction via deformation fields, sparse-view and feed-forward pipelines leveraging multi-modal priors, and system-level integration with SLAM and real-time frameworks.

Large-Scale Urban and Road Surface Reconstruction. Urban and road environments present unique challenges for surface reconstruction, including elongated camera trajectories, sparse and occluded observations, textureless regions, and the need for scalable memory management—conditions that push the limits of the field-based isosurface extraction methods previously described. Several methods have been developed to address these issues. For instance, [219] introduces a hybrid implicit-explicit approach that decomposes unbounded street views into close-range, distant, and sky regions with aligned cuboid boundaries, using cuboid hash-grids and road-surface initialization to achieve fine-grained reconstruction without LiDAR. Building on the explicit advantages of Gaussian primitives, [220] employs a planar-based octree representation with segmented training to reduce memory costs and handle long, narrow camera trajectories, augmented by guided smoothing and dual-step matching strategies for depth accuracy and multi-scale robustness. [221] and [222] both propose placing 2D Gaussian surfels on a uniform meshgrid, which aligns with the thin-surface property of roads, reduces primitive overlap, and significantly accelerates reconstruction. To further respect structural semantics, [221] incorporates a road-structure-aware adaptive meshgrid that allocates denser surfels to complex regions such as lane markings and boundaries, alongside a trajectory-consistency-guided pose-refinement strategy. Complementing these geometric approaches, [223] integrates segmentation-guided video inpainting and semantic-aware color correction to handle occlusions from dynamic agents and static obstacles, producing visually coherent and geometrically faithful road surfaces. For large-scale scene completeness, [224] proposes a hybrid Gaussian representation that models different scene parts (e.g., near, far, sky) with distinct Gaussian properties, achieving complete reconstruction beyond LiDAR coverage while using 66% of the Gaussians and a 20% faster reconstruction speed. Similarly, [225] introduces depth-consistent normal regularization and spatially adaptive pruning to balance geometric accuracy and memory efficiency, while [226] leverages a distributed 2D Gaussian representation, structured dense enhancement from SfM priors, and progressive hybrid geometric optimization for stable reconstruction across complex urban scenes. For city-scale digital twins, [227] combines textured proxy geometry with residual Gaussians, enabling real-time rendering on mobile GPUs with significantly reduced memory and energy consumption. Meanwhile, feed-forward approaches like [228] demonstrate that geometric foundation models can predict dense Gaussian attributes directly from multi-view images and pose/depth observations, eliminating per-scene optimization and enabling scalable road surface mapping. Similarly, [229] predicts geometry and texture parameters from a single frame in a bird’s-eye-view paradigm, achieving real-time road reconstruction with elevation errors reduced to 1.73 cm.

Dynamic and Articulated Object Reconstruction. Extending Gaussian surface reconstruction to dynamic scenes requires modeling temporal deformations while preserving the surface fidelity that geometry-aware rendering pipelines strive to maintain. [230] tackles this by incorporating Gaussian features from 4D neural voxels into planar-based Gaussian splatting, combined with normal

regularization and an as-rigid-as-possible constraint to maintain local rigidity and smooth surfaces across timesteps. For articulated objects, [231] introduces a category-agnostic framework that infers optimal sparse viewpoints via a Gaussian information field, constrains Gaussians to planar primitives for accurate normals and depth, and leverages a Vision-Language Model for open-vocabulary part segmentation and joint parameter estimation. In the context of urban dynamics, [232] distinguishes static and potentially dynamic regions using 2D semantic maps, applying global consistency for definite static areas and learnable time embeddings for dynamic objects, thereby preserving static content while capturing deformations. Another approach, [233], injects task-aware semantic priorities into the pipeline, employing semantically guided pruning and priority-driven rendering to allocate resources to safety-critical objects while maintaining over 350 FPS. [234] further addresses dynamic scenes in highway settings by enabling both image and LiDAR rendering, with improvements in reconstruction through LiDAR supervision in sparse-view and monotone-background environments.

Sparse-View and Feed-Forward Reconstruction with Multi-Modal Priors. In many practical settings, dense multi-view observations are unavailable, necessitating sparse-view or feed-forward approaches that bypass the per-scene optimization typical of the methods discussed earlier. [139] accelerates large-scale indoor reconstruction by learning data-driven priors for Gaussian initialization and densification, achieving an 8x runtime speedup and a 48% reduction in depth errors compared to state-of-the-art methods. In the autonomous driving domain, [235] formulates Gaussian prediction as a joint learning framework of visual and sparse LiDAR depth features, with multi-modal feature matching and multi-scale decoding for zero-shot novel-view synthesis. [236] reformulates generalizable dynamic reconstruction as an iterative process of optimization and densification, using self-supervised policies and geometric regularization to improve fidelity with fewer primitives. For surround-view reconstruction, [237] distills geometric priors from a pre-trained VGGT model into a lightweight geometry branch, uses a Gaussian head to fuse multi-scale tokens, and jointly supervises semantic refinement, significantly improving novel-view quality on nuScenes. [238] moves beyond per-pixel paradigms by unifying volume-based and pixel-based Gaussian prediction across three branches (close-range static, dynamic actors, far-field), ensuring stable 4D reconstruction despite noisy motion priors. [239] employs a controllable LiDAR generation model conditioned on coarsely extrapolated rendering, producing geometry-consistent scans for expansive reconstruction and improving extrapolative novel view synthesis. [240] initializes dense point clouds via multi-view metric depth predictions, refines them with a progressive pruning strategy, and jointly optimizes Gaussian geometry with diffusion-prior-enhanced dense depths, outperforming methods that use ground-truth LiDAR. [241] integrates a multi-modal video diffusion model with 3DGS, synthesizing RGB, depth, and semantic outputs for novel viewpoints to provide comprehensive supervision and robustness under extreme viewpoint changes.

System Integration with SLAM and Real-Time Pipelines. The integration of Gaussian-based surface reconstruction into SLAM systems has enabled simultaneous tracking, mapping, and photorealistic rendering in real time, thereby operationalizing the continuous-field and meshing strategies from the previous subsection within an online, resource-constrained framework. [242] proposes a real-time system with robust sparse-to-dense tracking and geometry-aware Gaussian surfel mapping, achieving a surface reconstruction error of 0.6 cm on Replica and ScanNet++ while running at 24 FPS. [243] leverages feed-forward visual geometry models for camera and scene priors, jointly optimizing poses and map with photometric and geometric losses, and achieves superior rendering and tracking accuracy online. Conversely, [244] demonstrates that geometry-only Gaussians can reconstruct scene geometry without appearance modeling, reducing per-primitive parameters by over 80% and enhancing robustness to illumination variations. For large-scale outdoor environments, [245] introduces a decoupled system with sparse anchor grids, geometry-based flow-densification, and scene partitioning, achieving a 15% improvement in reconstruction quality while maintaining real-

time performance. [246] integrates dense SLAM with adaptive densification and geometry-guided optimization, achieving state-of-the-art PSNR, SSIM, and LPIPS on Replica and TUM-RGBD. [247] extends Gaussian SLAM to multi-camera setups with multi-camera bundle adjustment and scale consistency, enabling reconstruction of side-view regions that monocular setups miss. [248] proposes a hybrid Gaussian-SDF representation that combines a colorized SDF for smooth geometry with 3D Gaussians for details, achieving over 150 FPS on real-world sequences—an order-of-magnitude improvement over existing methods. [249] combines NeRF priors with 3DGS, using sparse keyframes to densely populate maps and fill gaps, while integrating loop closure and bundle adjustment for improved tracking. In the LiDAR-centric domain, [250] tightly couples LiDAR-inertial-visual data with surface Gaussian scenes, while [251] introduces plane-constrained multimodal Gaussian Mixture Models for continuous geometric supervision, improving accuracy by 52.6% over LiDAR-based and 68.7% over Gaussian-based methods. [252] directly aligns sparse LiDAR data with continuous Gaussian maps, achieving 7.98 FPS novel view synthesis and demonstrating cross-modal localization potential. Finally, [253] and [254] illustrate system-level integration where Gaussian-based reconstructions are incorporated into broader multi-modal pipelines for dynamic urban scenes and configurable sensor fusion. Collectively, these domain-specific adaptations underscore the importance of tailoring Gaussian representations to application constraints, ensuring that geometric accuracy, training efficiency, and rendering quality are balanced for practical deployment across robotics, autonomous driving, and large-scale scene understanding—thereby extending the geometry-aware principles established earlier into operational, real-world systems.

5 Robustness and Handling Imperfect and Sparse Inputs

5.1 Sparse-View and Few-Shot Reconstruction

Sparse-view and few-shot reconstruction represents one of the most challenging and practically significant frontiers in 3D Gaussian Splatting (3DGS) research. The fundamental difficulty stems from the ill-posed nature of the problem: with only a handful of input images, the geometric and appearance information required to constrain the optimization is severely insufficient, leading to overfitting, structural artifacts, and poor novel-view generalization [255]. The foundational 3DGS pipeline relies on dense multi-view inputs and Structure-from-Motion (SfM) point clouds for initialization, both of which degrade catastrophically under sparse-view conditions—SfM often fails to produce sufficient reliable feature matches, resulting in sparse or incomplete initial point clouds that propagate errors throughout the optimization [256, 257]. Building upon the pose-estimation challenges discussed in the previous section, sparse-view reconstruction introduces an additional layer of difficulty: even when camera poses are known, the sheer lack of visual overlap and geometric redundancy renders the optimization severely under-constrained. Consequently, the research community has developed a diverse arsenal of strategies that can be broadly categorized into robust initialization methods, depth and stereo prior injection, and advanced regularization techniques aimed at mitigating overfitting while suppressing characteristic artifacts such as floaters and background collapse [258]. These categories are deeply interconnected with the pose-free and feed-forward paradigms, as many of the learning-based initialization tools introduced for SfM-free reconstruction also serve as powerful priors for sparse-view scenarios.

Robust Initialization Strategies. The quality of the initial point cloud is widely recognized as the decisive factor for sparse-view 3DGS performance. Controlled ablations have demonstrated that initialization determines the attainable performance band, while training-time constraints provide only modest improvements within that band [137]. Since SfM inevitably struggles with sparse inputs, a prominent line of work replaces or supplements it with learning-based alternatives, echoing

the feed-forward approaches highlighted in the pose-free reconstruction paradigm. One approach leverages DUS3R, a vision foundation model, to generate dense and non-redundant Gaussian point clouds during initialization, effectively overcoming the limitations of traditional SfM under sparse-view constraints [256]. Similarly, the π^3 network, a reference-free point cloud estimation network, has been integrated into pipelines to provide dense initialization without requiring reliable reference views, combined with uncertainty-guided depth supervision, normal consistency losses, and depth warping regularization to handle geometric inaccuracies [259]. Another strategy proposes eliminating densification entirely by using triangulated pixels from dense image correspondences to estimate scene geometry in a single step, providing each Gaussian with well-informed color, scale, and position while achieving uniform detail coverage across the scene [138]. Segmentation-driven approaches offer a complementary direction: by leveraging region-based segmentation to identify structurally significant regions, these methods selectively downsample dense point clouds, reducing Gaussian counts by up to 50% while preserving scene fidelity and substantially lowering memory footprints [260]. For extreme cases where SfM fails entirely, SfM-free frameworks jointly estimate camera poses and reconstruct scenes by using dense stereo modules to progressively recover pose information and build global dense point clouds, supplemented by coherent view interpolation modules that generate viewpoint-consistent content as additional supervision [261]. Additionally, frequency-aware SfM techniques that augment low-texture regions via low-frequency view augmentation and relaxed multi-view correspondences, combined with 3DGS self-initialization that lifts photometric supervision into additional points, have proven effective in compensating SfM-sparse regions [137]. This tight coupling between robust initialization and pose estimation underscores the importance of addressing both problems jointly, as the accuracy of one directly impacts the feasibility of the other.

Depth and Stereo Priors for Geometric Supervision. Geometric priors, particularly monocular depth estimates and stereo-derived cues, have become indispensable for constraining the ill-posed sparse-view optimization. These priors serve as a form of external geometric knowledge that compensates for the missing multi-view redundancy, and they resonate closely with the dense correspondence and trajectory-based methods discussed in the pose-free section. Early methods such as SparseGS incorporate depth priors with novel depth rendering techniques and pruning heuristics to mitigate floater artifacts, alongside unseen viewpoint regularization modules to alleviate background collapse [258]. The integration of stereo geometry-texture alignment has been shown to bridge rendering quality and geometry estimation, jointly enhancing both surface reconstruction and novel view synthesis, while pseudo-feature-enhanced geometry consistency enforces multi-view consistency by incorporating both training and unseen views [262]. For large-scale outdoor scenes, geometric supervision from LiDAR or monocular depth is often spatially incomplete and uneven; energy-based frameworks model partially observable geometry as a continuous energy field induced by geometric evidence, providing soft geometric guidance that steers optimization without directly restricting the solution space [194]. Depth-based regularization has also been enhanced through confidence-aware mechanisms that leverage multi-cue confidence maps from monocular depth estimators and sparse SfM depth to adaptively adjust supervision during optimization, improving geometric detail preservation and convergence stability [263]. The depth-and-density guided dropout strategy addresses two distinct failure modes in sparse-view settings: overfitting in regions with excessive Gaussian density near the camera, and underfitting in distant areas with insufficient coverage, using adaptive masking and distance-aware fidelity enhancement modules [264]. A particularly promising direction combines stereo priors for robust initialization with diffusion-based refinement, where stereo priors aid in recovering camera poses and reliable point clouds, and diffusion priors preserve intricate scene details during optimization [257]. Additionally, hierarchical depth-guided frameworks employ cascade Pearson correlation losses to progressively refine geometry from coarse to fine scales, enforcing multi-scale depth consistency that improves structural fidelity [265]. Probabilistic joint

flow-depth optimization frameworks further leverage the complementarity between optical flow and depth through pixel-level mechanisms that scale information fusion based on matching probability, combined with multi-view depth-consistency losses that suppress misleading gradients in uncertain areas [266]. These approaches collectively illustrate how dense geometric cues not only enable pose-free reconstruction but also provide the robust supervision needed to stabilize sparse-view optimization.

Regularization Techniques for Overfitting Mitigation. A rich body of work addresses the overfitting problem through sophisticated regularization schemes that impose structural, statistical, or frequency-domain constraints on the Gaussian representation. The unobservability of Gaussian primitive reliability has been identified as a core challenge, motivating dual-domain observation and calibration frameworks that model reliability through the synergy of optimization-domain inductive bias and observation-domain evidence; these employ continuous depth-guided dropout strategies as explicit proxies for primitive reliability, combined with dark channel priors as structural consistency cues to identify and accumulate anomalous regions for reliability-driven geometric pruning [177]. Fisher information provides a principled foundation for guiding both the utilization of geometric priors and regularization in sparse-view settings. By leveraging Fisher information to actively select the most informative supporting views and quantitatively measuring the uncertainty of each Gaussian to adaptively adjust removal probabilities, these methods reduce randomness in augmentation and improve stability and rendering fidelity [178]. Structured Gaussian representations that enforce spatial coherence have proven effective, including supergaussian groupings of Gaussians based on local geometric cues and appearance features, with inter-group global self-attention and sparse local attention enabling the integration of global and local spatial information [267]. Similarly, coherent Gaussian representations constrained through implicit convolutional decoders and total variation losses, supported by monocular depth-based initialization, prevent Gaussians from moving independently during optimization [268]. Loop-based frameworks introduce progressive Gaussian initialization that iteratively densifies the initial point cloud using rendered pseudo-images, combined with depth-alignment regularization that leverages both reliable SfM depth and window-based dense monocular depth for precise geometric supervision [269]. Multimodal-prior-guided importance sampling fuses photometric rendering residuals, semantic priors, and geometric priors to produce robust local recoverability estimates that directly drive where fine Gaussians should be injected, enabling hierarchical coarse-to-fine representations that selectively refine geometrically critical regions [270]. Signal-structure-aware scheduling frameworks address the mismatch between supervising low-frequency sparse points with high-frequency images by deriving the average sampling frequency and bandwidth of 3D representations and regulating training image resolution and densification processes based on scene frequency convergence, achieving frequency-consistent and floater-free training in large-scale scenes [140]. Finally, comprehensive surveys of sparse-view reconstruction emphasize that while geometry-based, neural implicit, and generative (diffusion-based) methods each offer distinct trade-offs between reconstruction accuracy, efficiency, and generalization, the persistent challenges of domain generalization and pose-free reconstruction remain open, pointing toward the development of 3D-native generative priors and real-time unconstrained sparse-view reconstruction as key future directions [255]. These directions align directly with the feed-forward and learning-based pose-estimation trends highlighted in the following subsection, suggesting that the convergence of robust pose estimation, sparse-view reconstruction, and generative modeling will define the next generation of 3DGS systems.

5.2 Pose-Free and SfM-Free Pipelines

The reliance of 3D Gaussian Splatting (3DGS) on accurate camera poses, typically derived from Structure-from-Motion (SfM) pre-processing, constitutes a critical bottleneck that limits its application to casual, unconstrained, and large-scale real-world data. Classical SfM pipelines are notoriously fragile in the presence of textureless regions, repetitive patterns, large viewpoint changes, or significant object motion, often leading to catastrophic failure or pose drift. This fragility is further exacerbated under the sparse-view conditions discussed in the previous subsection, where the limited visual overlap provides insufficient feature matches for reliable pose estimation, creating a compounding failure mode: without trustworthy poses, the already ill-posed sparse-view optimization becomes intractable. To overcome this dependency, a significant body of research has emerged to eliminate the need for a separate, offline SfM stage, giving rise to pose-free and SfM-free pipelines that jointly estimate camera poses and scene geometry. Notably, many of the learning-based initialization tools developed in these pose-free paradigms serve a dual purpose: they not only enable SfM-free reconstruction but also provide robust priors for sparse-view scenarios, forging a tight coupling between pose estimation and sparse-view reconstruction. These methods can be broadly categorized into joint optimization frameworks, dense correspondence-based approaches, and feed-forward or learning-based paradigms, each offering distinct trade-offs between robustness, accuracy, and computational efficiency that become particularly consequential when deployed under the sparse-view and degraded-imaging conditions examined in this survey.

A foundational class of pose-free methods formulates the problem as a joint, end-to-end optimization where camera poses are treated as learnable parameters alongside the scene representation. This approach has been widely explored in the context of neural rendering and extends naturally to 3DGS, enabling the simultaneous refinement of camera poses and Gaussian parameters from a purely photometric loss. For instance, in the context of dynamic scenes, a unified optimization framework that incorporates camera poses as learnable parameters complementary to 3DGS attributes has been proposed, where both camera and object motion are recast as per-primitive $SE(3)$ affine transformations on 3D Gaussians [271]. This design allows for the joint refinement of poses and scene structure, which is particularly beneficial when motion blur compromises the accuracy of pre-estimated poses—a scenario directly connected to the degraded-imaging challenges detailed in the following subsection. Such joint optimization strategies are conceptually rooted in the classical analysis-by-synthesis paradigm, which casts the inference of 3D structure and camera poses as a single optimization problem aimed at explaining the observed pixels. This classical framework has been modernized with expressive neural field representations and gradient-based pose refinement, and recent instantiations that explicitly incorporate generative priors for novel view synthesis have proven effective in sparse-view settings [272]. This highlights that even in the absence of reliable initial poses, a well-initialized and regularized joint optimization can converge to accurate solutions, a property that becomes especially valuable when SfM fails entirely, such as in low-texture or heavily blurred inputs.

While joint optimization approaches are promising, their performance is often contingent on good initialization. To mitigate this, a second category of methods leverages dense geometric cues, such as optical flow and point trajectories, as a more robust alternative to sparse keypoint matching. The use of dense correspondence is a powerful strategy for pose-free reconstruction, especially in dynamic scenes where classical SfM methods fail. One prominent line of work exploits long-range point trajectories, which are optimized and then classified into static and dynamic parts using a specifically designed neural network architecture; the static trajectories are then used for robust bundle adjustment and camera pose estimation [ParticleSfM Exploiting Dense Point Trajectories for Localizing Moving Cameras in the Wild]. This trajectory-based approach is more resilient to

dynamic objects than sparse keypoint matching and offers a strong foundation for pose-free 3DGS. The concept of leveraging dense visual correspondence as a substitute for explicit pose estimation has been further advanced by the rise of “tracking-any-point” and even dense tracking models. Building on these foundations, a reconstruction framework has been introduced that makes cross-frame 3D correspondence an explicit prediction by estimating camera-coordinate 3D trajectories, coupling sparse trajectories, per-frame local point maps, and relative camera poses with geometric consistency objectives to ensure coherent global alignment [273]. By explicitly avoiding the need for accurate initial poses, these methods provide a robust alternative for deriving camera motion directly from dense pixel-level relationships. This dense-correspondence paradigm resonates strongly with the depth and stereo priors discussed in the previous subsection, as both strategies leverage dense geometric cues to compensate for the absence of reliable sparse feature matches, underscoring a broader theme of replacing fragile sparse correspondences with dense, learnable geometric signals.

The most disruptive shift in pose-free reconstruction, however, has come from the emergence of large-scale, feed-forward 3D foundation models. Unlike traditional iterative pipelines, these models are trained to directly regress camera poses and dense geometry from a set of images in a single forward pass. This paradigm, popularized by models like DUST3R, has led to a new family of SfM systems that are orders of magnitude faster and more robust than their classical counterparts, particularly in textureless and low-overlap scenarios [274]. A prominent example is the MAST3R-SfM pipeline, which leverages a foundation model to robustly produce local 3D reconstructions and accurate matches, using a low-memory approach to align these local reconstructions into a global coordinate system, and even serving as an efficient image retriever to reduce complexity [275]. However, these feed-forward models are not perfect; they often generate incorrect correspondences for non-overlapping image pairs, which can degrade the final pose estimation. To address this, the G-MAST3R-SfM pipeline introduces a graph-based view pruning module to geometrically prune outlier views and a multi-stage optimization module that refines camera parameters progressively from local to global consistency [276]. Furthermore, a fully differentiable deep pipeline, VGGSfM, has been proposed where every component—from point tracking to bundle adjustment—is differentiable, enabling end-to-end training of the entire SfM process [277]. The robustness of these feed-forward models to challenging imaging conditions—such as extreme motion blur, as seen in methods like GeMS [278]—demonstrates a direct synergy with the degraded-imaging challenges explored in the next subsection, where learning-based pose estimation often serves as the only viable initialization strategy.

These feed-forward models have also inspired hybrid strategies that combine the robustness of learning-based initialization with the precision of classical optimization. A key insight is that while feedforward approaches excel in challenging conditions, they often fall short of classical methods in standard settings; therefore, a systematic combination of both strengths has been proposed to achieve state-of-the-art results across a wide range of scenarios [279]. Similarly, the “Benchmarking Efficient & Effective Camera Pose Estimation Strategies” study demonstrates that using feed-forward networks to obtain initial estimates and refining them with classical SfM techniques leads to the best efficiency-effectiveness trade-off [280]. This suggests a growing consensus that the integration of learning-based initialization with geometric refinement is the most practical pathway for robust, pose-free reconstruction. This is particularly relevant for 3DGS, where the quality of the final reconstruction is highly sensitive to the accuracy of the camera poses used for optimization—a sensitivity that is amplified in sparse-view settings, where the optimization has little redundancy to absorb pose errors, and in degraded-imaging scenarios, where the photometric loss itself is unreliable.

Finally, a key challenge in pose-free pipelines is handling long video sequences, where small per-frame errors can accumulate into significant pose drift. To ensure global consistency over hundreds of frames, a novel system has been developed that combines the strong sequential constraints of

SLAM with the flexibility and global optimization of offline SfM, treating temporal ordering as a first-class citizen for reliable loop closure and augmenting global optimization with metric monocular depth priors [281]. In the context of dynamic scenes, a large-scale reconstruction framework has also been proposed that processes long videos in overlapping chunks, connects chunk-local coordinate systems through confidence-weighted registration, and associates dynamic identities across chunk boundaries to recover coherent long-range motion [282]. These system-level approaches are crucial for scaling pose-free pipelines to real-world, casual video capture, where the lack of a pre-computed SfM model means that pose estimation and scene reconstruction must be performed jointly and robustly over time. The ability to maintain global consistency without an initial pose estimate is a testament to the maturity of these methods and their suitability for powering next-generation 3DGS applications, particularly as they must contend with the temporal variations in illumination and transient distractors discussed in the following subsection. Collectively, pose-free reconstruction represents a fundamental paradigm shift in 3DGS, transforming the pipeline from a two-stage process dependent on robust offline preprocessing into a unified, joint optimization that is inherently more resilient to the imperfections of real-world capture. The convergence of joint optimization, dense correspondence, and feed-forward foundation models is not only enabling SfM-free reconstruction but also providing the robust initialization and geometric priors that are indispensable for sparse-view reconstruction—and, as the next subsection will demonstrate, for handling degraded imaging and adverse conditions.

5.3 Robustness to Degraded Imaging and Adverse Conditions

While the previous subsection addressed the challenge of unknown or inaccurate camera poses, a complementary and equally critical challenge for deploying 3D Gaussian Splatting (3DGS) in the wild is the degradation of the input imagery itself. In real-world deployment, 3DGS is frequently confronted with input imagery that deviates significantly from the idealized sharp, well-exposed, and static conditions assumed by its core optimization pipeline. Robustness to degraded imaging and adverse conditions is therefore a critical requirement for practical applications ranging from handheld phone captures to autonomous driving. Research in this area addresses a spectrum of challenges, including motion blur, low-light and high-dynamic-range (HDR) scenes, adverse weather, noise, and the presence of transient distractors that violate the multi-view consistency assumption central to 3DGS.

Motion Blur and Event-Camera Assistance. Motion blur, often caused by camera shake or object movement during the exposure time, is one of the most pervasive forms of degradation. Early approaches attempted to model the blur process directly within the 3DGS optimization. For instance, a common paradigm involves modeling the camera trajectory during exposure using continuous neural ordinary differential equations (ODEs) and applying rigid body transformations to the Gaussians to synthesize a blurred image, as demonstrated in CoMoGaussian [283]. Similarly, CRiM-GS extends this concept with an adaptive distortion-aware transformation to account for nonlinear effects like rolling shutter [284]. Another approach, BSGS, employs a bi-stage strategy that first refines camera poses and then applies a global rigid transformation to correct residual motion-induced distortions [285]. However, these methods often struggle with severe blur, where conventional pose estimation like COLMAP fails, leading to unreliable initialization. GeMS addresses this by leveraging a deep learning-based SfM pipeline (VGGSfM) and 3DGS-MCMC for more robust scene initialization directly from extremely blurred inputs [278]. This connection to learning-based pose estimation echoes the feed-forward paradigms discussed in the previous subsection, highlighting how advances in pose-free reconstruction directly bolster robustness to degraded inputs.

Event cameras, with their microsecond temporal resolution and high dynamic range, have emerged

as a powerful complement to standard frame-based cameras for handling motion blur. These sensors capture per-pixel brightness changes asynchronously, providing the temporal information lost during exposure integration. A significant body of work integrates event streams into the 3DGS framework. EBAD-Gaussian constructs a blur formation model by synthesizing multiple latent sharp images during the exposure time and uses the event-based double integral (EDI) prior to supervise intensity changes, leading to sharper reconstructions [286]. DiET-GS builds on this by using a two-stage training strategy with event double integral constraints and a diffusion prior to further enhance edge details [287]. EvFlow-GS proposes an end-to-end learnable double integral (LDI) and a novel event-residual prior to strengthen supervision, while jointly optimizing camera poses and 3DGS [288]. The joint optimization of poses and scene in these methods also aligns with the pose-free optimization strategies surveyed earlier, demonstrating a convergence of solutions across different robustness challenges.

Addressing the practical limitations of existing event-based methods, AsyncEvGS introduces a flexible, high-resolution asynchronous RGB-Event dual-camera system, reconstructing sharp images from event data and using a cross-domain pose estimation module for robust 3DGS initialization [289]. JADE-GS takes a different perspective by jointly allocating deblurring evidence from two complementary priors—the analytical EDI inversion and a learned restoration network—using a lightweight Spatial Prior Router to fuse them into a supervision target, achieving leading perceptual quality without generative decoding at inference [290]. For dynamic scenes, BARD-GS explicitly decomposes motion blur into separate camera and object motion components, enabling robust reconstruction from handheld monocular captures [291], while Deblur4DGS extends this concept to 4D reconstruction from blurry monocular video by estimating exposure time and using variable canonical Gaussians [292]. UPGS further proposes a unified pipeline where camera poses are learnable parameters optimized jointly with 3DGS attributes using a three-stage training schedule [271].

Low-Light, Exposure, and Illumination Robustness. Low-light conditions present a distinct set of challenges, including high noise, significant information loss, and color distortion. Direct application of 3DGS to such inputs often fails due to the limited number of SfM points and the ambiguity of the optimization. LITA-GS tackles this with an illumination-agnostic novel view synthesis method that extracts an illumination-invariant physical prior, develops a lighting-agnostic structure rendering strategy, and includes a progressive denoising module to achieve normally-exposed representations without reference images [293]. The reliance on physical priors here parallels the integration of geometric priors discussed in the following subsection, illustrating a broader trend of using domain knowledge to regularize ill-posed inverse problems.

To handle the combination of low-light and motion blur, FLED-GS proposes a fast framework that reformulates 3D scene restoration as an alternating cycle of enhancement and reconstruction, using intermediate brightness anchors for progressive recovery and a noise-aware 3DGS reconstruction to suppress noise [294]. The use of event cameras is also explored for this challenging scenario. Dark-EvGS is a framework specifically designed for event-assisted 3DGS in low-light environments, introducing triplet-level supervision and a color tone matching block to ensure color consistency in the reconstructed bright frames [295]. E-3DGS leverages a transmittance adjustment device at the hardware level to capture both motion and exposure events, enabling high-quality reconstruction in conditions where frame-based cameras fail [296]. Thermal imaging provides another avenue for robustness; DTGS couples Retinex-inspired illumination decomposition with thermal-guided 3DGS, where the thermal supervisory branch stabilizes both color restoration and geometry learning under extreme low-light conditions [297]. This multi-modal fusion strategy foreshadows the LiDAR- and radar-augmented approaches detailed in the next subsection, where auxiliary sensors provide complementary cues that are robust to the failure modes of RGB-only pipelines.

Beyond low-light, the varying exposure and tone mapping across different cameras in a multi-camera setup, common in urban reconstruction, can also degrade 3DGS quality. P2GS addresses this by jointly decomposing a view-invariant linear HDR radiance field along with per-view exposure scales and tone-mapping functions, grounded in the physical image-formation process, to ensure photometric consistency [298].

Adverse Weather and Transient Distractors. Outdoor scenes are often subject to adverse weather conditions like rain, snow, and haze, as well as transient objects such as pedestrians or vehicles that violate the static scene assumption. WeatherGS explicitly categorizes multi-weather artifacts into dense particles (snowflakes, raindrops) and lens occlusions, proposing a dense-to-sparse preprocessing strategy to remove them and a mask-based training procedure to reconstruct a clear scene [299]. For transient distractors, a wide range of methods has been developed to identify and suppress their influence. A common strategy is to use semantic cues from vision-language models to identify distractors. For example, a semantic filtering framework accumulates CLIP similarity scores per Gaussian and applies opacity regularization and periodic pruning to remove transient objects [300].

Another line of work focuses on modeling the transient elements explicitly with additional learnable components, such as ForestSplats, which introduces a deformable transient field and a superpixel-aware mask to separate static scenes from distractors without heavy vision foundation models [301]. To address the circular dependency between accurate transient detection and clean reconstruction, DualSplat converts first-pass reconstruction failures into explicit priors, using photometric residuals, feature mismatches, and SAM2 instance boundaries to construct object-level pseudo-masks and guide a clean second-pass optimization [302]. More advanced optimization-based techniques, such as 3DGS-HPC, combine a patch-wise classification strategy with a hybrid photometric and perceptual metric to more reliably separate static and transient regions [303]. HarmoGS addresses conflicting cross-view gradients caused by transient distractors and illumination inconsistencies through a dual-view Conflict-Aware Gradient Harmonization strategy, which reconciles view-specific gradients by rotating them into an orthogonal configuration [182]. With a focus on large-scale unconstrained environments, GS-Diff leverages a multi-view diffusion model to guide the 3DGS optimization, transforming under-constrained reconstruction problems into well-posed ones and integrating appearance embedding, depth priors, and dynamic object modeling to handle both transients and appearance variability [304]. Finally, for photo collections with significant appearance variability, Wild-GS decouples the appearance of each Gaussian into material attributes, global illumination, and per-image camera properties, explicitly aligning pixel appearance features to 3D space using triplanes extracted from reference images [305].

Collectively, these methods demonstrate a clear trend toward integrating physical priors, complementary sensor modalities (e.g., events, thermal), and powerful generative priors to build 3DGS systems that are robust to the imperfections and variability of real-world captures. Moreover, the strategies surveyed here—particularly the use of auxiliary sensors and pre-trained priors—establish a natural bridge to the next subsection, which explores how multi-modal sensor fusion and geometric priors from LiDAR, radar, and foundation models further enhance the geometric fidelity and metric accuracy of 3DGS reconstructions. Together, these two dimensions of robustness—degraded imaging and geometric grounding—form a comprehensive toolkit for deploying 3DGS in challenging real-world scenarios.

5.4 Multi-Modal Sensor Fusion and Geometric Priors

The reconstruction of accurate and robust 3D scenes from imperfect or sparse inputs is fundamentally an ill-posed problem, and the integration of auxiliary sensor data and pre-trained geometric models

has emerged as one of the most effective strategies to address this challenge. In the context of 3D Gaussian Splatting (3DGS), multi-modal sensor fusion and geometric priors serve a dual purpose: they provide metric-scale anchors that mitigate the inherent scale ambiguity of monocular pipelines, and they supply dense or structured supervision that regularizes the optimization of Gaussian primitives in under-constrained regions. While the previous subsection examined robustness to degraded image quality—where complementary sensors like event cameras and thermal imaging were introduced to compensate for RGB failures—this subsection extends that multi-modal philosophy by systematically leveraging range sensors and geometric priors to anchor the metric scale and structural fidelity of Gaussian reconstructions. This subsection surveys the landscape of techniques that leverage LiDAR, depth sensors, radar, and foundation model priors to enhance the robustness and geometric fidelity of 3DGS-based reconstruction.

A primary avenue of research involves the use of LiDAR and depth-sensor-derived supervision to anchor the metric scale and geometry of Gaussian scenes. A fundamental insight is that monocular depth foundation models, while providing dense and coherent relative geometry, often lack reliable metric scale [306, 307]. Conversely, LiDAR provides precise but sparse metric measurements. Several approaches have been developed to bridge this gap, and notably, these strategies directly complement the deblurring and low-light methods discussed earlier—where RGB-only reconstruction may fail, LiDAR-anchored geometry provides a stable scaffold that is invariant to photometric degradation. For instance, the GTLR-GS framework explicitly incorporates metric geometric priors into the entire Gaussian optimization process, treating LiDAR not as a passive initialization source but as a geometry-conditioned allocation and refinement problem. It introduces a geometry-texture-aware allocation strategy and a confidence-aware metric depth regularization that anchors the reconstructed geometry to absolute scale, demonstrating state-of-the-art performance on metric-scale reconstruction [308]. Similarly, the Geometry-Preserving in 3D Gaussian Splatting for LiDAR-Camera Extrinsic Calibration work addresses a critical issue in using 3DGS as a geometric proxy for calibration: since 3DGS is designed for novel view synthesis, it tends to prioritize rendering quality, causing the proxy geometry to drift from the true LiDAR structure. Their solution aggregates multi-view LiDAR observations for dense depth supervision while blocking photometric gradients from updating Gaussian spatial parameters, thus preserving metric geometry and improving calibration accuracy [309]. Another approach, EnerGS, tackles the challenge of partial geometric priors in outdoor scenes. Recognizing that LiDAR measurements are often spatially incomplete and uneven, EnerGS models partially observable geometry as a continuous energy field, providing soft geometric guidance to the optimization of Gaussian primitives rather than enforcing geometry as a hard constraint [194]. The Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting method also exemplifies this fusion, jointly optimizing sensor poses with a neural Gaussian-based scene representation, freezing reliable LiDAR points as anchor Gaussians to preserve global structure, and using auxiliary Gaussians to prevent local overfitting [310].

Beyond LiDAR, the fusion of other range sensors such as radar has been explored to improve depth completion and scene reconstruction, particularly in conditions where cameras and LiDAR are unreliable—echoing the adverse weather robustness challenges surveyed previously. The RaCalNet framework proposes a novel paradigm that eliminates the need for dense supervision by using sparse LiDAR to supervise the learning of refined radar measurements, achieving a supervision density of merely around 1% compared to dense-supervised methods. It uses a Radar Recalibration module for point screening and displacement refinement, followed by a Metric Depth Optimization module that fuses radar priors with monocular depth predictions for metrically accurate outputs [311]. Similarly, the work on Sparse Beats Dense explores why sparse supervision can outperform dense supervision in radar-camera depth completion. The authors identify the Projection Transformation Collapse phenomenon, where sparse supervision leads the model to learn collapsed projection transformations

between Image/Radar/LiDAR spaces, and propose a Disruption-Compensation framework to handle this collapse while leveraging 3D spatial and 2D semantic information [312]. Additionally, the “Sparse-LiDAR Prompting of Monocular Geometry Foundations” study introduces SLIM, the first adaptation of a point-map foundation model (MoGe-2) to accept truly sparse LiDAR input. By integrating a partial-convolution sparse encoder with a multi-scale fusion neck, SLIM demonstrates substantial improvements in absolute relative error at long-range distances (50-150m), highlighting the value of geometric priors in extreme sparsity scenarios [313].

Multi-modal registration and calibration form another critical pillar of this research area, as accurate extrinsic parameters are a prerequisite for effective sensor fusion. This concern directly parallels the pose-estimation challenges addressed in the robustness-to-pose-errors subsection, yet here the emphasis shifts from recovering camera trajectories from degraded imagery to aligning complementary sensor modalities that jointly constrain the scene. Many targetless calibration methods have been developed, often leveraging differentiable scene representations like 3DGS. The UniCalib work reformulates calibration as an intra-modality depth flow estimation problem, estimating a dense depth map from the camera image and completing the sparse LiDAR projected depth map, while introducing a reliability map to prioritize valid pixels [314]. The iterative surrogate diffusion framework demonstrates a versatile approach to enhance any calibration method without architectural modifications, using the original method as a denoiser to iteratively refine extrinsic estimates, showing improvements in accuracy and robustness across multiple state-of-the-art baselines [315]. In a related vein, the Native-Domain Cross-Attention for Camera-LiDAR Extrinsic Calibration work addresses the limitations of projecting LiDAR points into 2D depth maps, which distorts 3D geometry, by directly aligning image patches and LiDAR point groups in their native domains, injecting extrinsic hypotheses into the correspondence modeling process for geometry-consistent interaction [316]. Furthermore, the DST-Calib network introduces a dual-path, self-supervised calibration framework with double-sided data augmentation to enhance generalization, demonstrating that cross-modal feature association can be improved through difference map construction instead of traditional dual-branch feature extraction [317]. The simultaneous estimation of spatial and temporal parameters is addressed by SST-Calib, which uses segmentation masks for pixel-wise bidirectional loss and incorporates velocity information from optical flow for temporal parameter estimation [318].

The fusion of diffusion or vision-language model priors represents a frontier for geometric and appearance regularization in under-constrained scenarios, and it naturally bridges the generative-prior strategies used for deblurring and transient-object removal with the geometric grounding goals of this subsection. The Veila framework, for instance, introduces a conditional diffusion framework for panoramic LiDAR generation from monocular RGB images, integrating a Confidence-Aware Conditioning Mechanism to strengthen RGB conditioning and a Geometric Cross-Modal Alignment for robust alignment under noisy diffusion [319]. The AURORA-KITTI benchmark introduces a distillation-based baseline that leverages depth foundation models to inject clean structural priors into in-the-wild depth completion training, demonstrating that weather-aware, physically consistent data contributes more to robustness than architectural modifications alone [320]. The COPS framework for thermal-LiDAR depth completion uses contrastive learning and pseudo-supervision, leveraging a depth foundation model to enforce a depth-aware contrastive loss that sharpens depth boundaries and provides dense depth priors to mitigate incomplete supervision in adverse weather [321]. The “Depth Anything with Any Prior” framework presents Prior Depth Anything, which combines incomplete but precise metric information from depth measurements with relative but complete geometric structures from depth prediction. It introduces pixel-level metric alignment and distance-aware weighting to pre-fill diverse metric priors, followed by a conditioned monocular depth estimation model to refine prior noise and implicitly merge complementary depth

sources [322]. This is complemented by the AnchorD method, which proposes a training-free depth grounding framework that anchors monocular depth estimation priors from a foundation model in raw sensor depth through factor graph optimization, performing patch-wise affine alignment to locally ground monocular predictions in metric depth while preserving geometric detail [323]. Finally, the Prompting Depth Anything framework introduces the new paradigm of prompting depth foundation models with low-cost LiDAR, guiding the Depth Anything model for accurate metric depth output up to 4K resolution, achieving state-of-the-art results on ARKitScenes and ScanNet++ [307]. Together, these works demonstrate that the strategic integration of multi-modal sensor data and pre-trained geometric models is essential for overcoming the inherent limitations of 3DGS in sparse, degraded, and under-constrained real-world scenarios, and they collectively establish a geometric grounding layer that complements the photometric robustness achieved by the methods in the previous subsection.

6 Dynamic Scenes, 4D Reconstruction, and Temporal Modeling

6.1 Temporal Representation Paradigms: Canonical Deformation Fields, Per-Frame Models, and Explicit 4D Primitives

The extension of 3D Gaussian Splatting (3DGS) to dynamic scenes represents one of the most active and consequential research frontiers in modern computer graphics and vision. Building upon the static representation foundations established in earlier sections, the central challenge here lies in augmenting a static, explicit representation with a temporal dimension while preserving the real-time rendering capabilities and high visual fidelity that have made 3DGS the de facto standard for static novel view synthesis. A survey of the literature reveals that approaches to this problem can be broadly categorized into three foundational paradigms: structure-guided methods that rely on canonical deformation fields, per-frame or per-primitive temporal models, and native 4D primitive designs that treat time as an intrinsic spatial dimension. Each paradigm embodies distinct trade-offs among reconstruction fidelity, rendering speed, storage requirements, and the ability to represent complex or abrupt motions, and recent work has increasingly sought to hybridize these strategies to exploit their complementary strengths. The choice of paradigm also fundamentally shapes the downstream motion modeling strategies discussed in the next section, determining whether motion is learned implicitly through deformation networks, explicitly through per-primitive dynamics, or inherently through the slicing of higher-dimensional primitives.

Structure-Guided Paradigms: Canonical Deformation Fields and Embeddings. The most widely adopted paradigm for dynamic 3DGS is the structure-guided approach, which maintains a canonical set of 3D Gaussians in a reference space and learns a deformation field that maps these canonical primitives to each observation frame. The seminal formulation of this idea, exemplified by deformable Gaussian frameworks, established a template in which a coordinate-based neural network takes a canonical position and a time index as input and outputs per-Gaussian deformations [324]. This design inherits the compactness of a single canonical representation while leveraging the expressiveness of neural fields for motion modeling. However, subsequent studies have identified a critical architectural weakness: modeling deformation purely as a function of coordinates is conceptually mismatched with the mixture-of-fields nature of 3DGS. To address this, [325] introduce per-Gaussian embeddings as an additional input to the deformation network, effectively allowing each primitive to maintain a learnable identity that conditions its motion, along with a coarse-to-fine decomposition of temporal embeddings to separately capture slow and fast dynamics. This embedding-based strategy is further refined by [326], which accelerates rendering through early fusion of temporal embeddings and introduces depth- and error-guided sampling to improve robustness to

sparse initializations, as well as opacity modulation to escape local optima in poorly lit regions.

Despite their elegance, deformation-based methods encounter persistent challenges. A primary issue is optimization difficulty in scenes with high-frequency or abrupt motions, where a single unified deformation network struggles to represent diverse motion patterns. [327] directly confronts this by proposing a motion-aware partitioning strategy that recursively duplicates deformation networks for temporally segmented groups of high-dynamic Gaussians while treating low-dynamic primitives as static, augmented by a cross-frame consistency loss to mitigate discontinuities. An alternative line of work addresses redundancy and computational efficiency. [328] introduces a deformable anchor grid, a hierarchical scene representation in which each anchor derives multiple Gaussians within a local spatiotemporal region, paired with a deformation-aware densification strategy that adaptively grows anchors only in under-reconstructed dynamic areas; this achieves an average 60% model size reduction and a 100% FPS improvement over state-of-the-art methods. Similarly, [329] proposes a global-to-local motion decomposition, employing Global Canonical Scaffolds with anchor deformation to model coarse motion and Local Canonical Scaffolds with explicit Gaussian deformation for fine corrections, together with temporal interval adjustment for optimal segment allocation, yielding approximately a 70% reduction in model size on real-world dynamic videos. In a related vein, [330] combines a hierarchical anchor scaffold with per-Gaussian deformation, using a static warm-up stage and a stop-gradient operation to stabilize canonical positions while enabling fine-grained local motion predictions; this framework achieves 36.98 dB PSNR on synthetic monocular benchmarks at 435.6 FPS with only 4.67 MB of storage. For long-range sequences, [331] offers a novel perspective by treating motion as continuous rigid-body transformations, integrating SE(3) transforms with hierarchical Bézier residuals and learnable local anchors, which preserves temporal identity and avoids the unbounded Gaussian proliferation characteristic of piecewise-linear or short-window approximations.

Gaussian-Centric and Per-Frame Temporal Models. A second paradigm shifts the locus of temporal modeling from a global deformation field to the individual primitives themselves. In these Gaussian-centric approaches, each Gaussian is endowed with an explicit time-dependent behavior, either through learnable temporal activation profiles or through per-primitive motion functions. The primary motivation is flexibility: because each primitive carries its own dynamics, the representation can adapt more readily to spatially heterogeneous motion and sudden changes. For instance, [332] proposes a representation in which Gaussian primitives are allowed to appear at arbitrary positions and times, rather than being anchored in a canonical space, and are each assigned a motion function that enables movement to neighboring regions over time, substantially improving the modeling of complex dynamic scenes. The concept of temporal lifespan has also emerged as a powerful tool. [333] introduces a learnable lifespan parameter that reforms temporal visibility into a flat-top profile, allowing primitives to remain consistently active over their duration; the lifespan simultaneously modulates primitive motion, decoupling motion magnitude from temporal duration and improving long-term stability without sacrificing dynamic fidelity. A grow-and-prune philosophy underlies several other works; [334] proposes temporal sensitivity pruning to remove low-impact Gaussians based on temporally aggregated sensitivity metrics, along with temporal sensitivity sampling to suppress floaters and improve coherence, delivering up to a $13.71\times$ rendering speedup with $10\times$ fewer primitives. Similarly, [335] identifies two causes of temporal redundancy—short-lifespan Gaussians and inactive Gaussians at render time—and introduces a Spatial-Temporal Variation Score to prune the former and an active mask to skip the latter, resulting in a $41\times$ storage reduction and a $9\times$ faster rasterization speed. Hybrid 3D-4D approaches have also been proposed to exploit static-dynamic separation; [336] begins with a fully 4D representation and iteratively converts temporally invariant Gaussians into static 3D primitives, significantly reducing parameters and accelerating training while maintaining high visual quality. Likewise, [337] employs radial basis function modeling with

time-dependent RBFs for dynamic regions and shared temporally invariant parameters for static regions, achieving up to a 98% model size reduction and real-time rendering at 125 FPS at 4K resolution.

Native 4D Primitives: Time as a Spatial Dimension. A third, more radical paradigm treats a dynamic scene as an intrinsically 4D spatiotemporal volume, where Gaussian primitives are defined in XYZT space rather than in a canonical 3D space with a separate motion model. This approach is epitomized by [338], which introduces anisotropic 4D XYZT Gaussians. In this formulation, dynamics at each timestamp are obtained by temporally slicing the 4D Gaussians, naturally composing dynamic 3D Gaussians that can be projected into images via a tailored rendering routine. A key advantage is its capacity to handle abrupt motions and scene discontinuities that often confound deformation-based methods, and the implementation achieves real-time inference speeds of up to 277 FPS on an RTX 3090 GPU. A subsequent work by the same group, [339], extends the representation with 4D spherindrical harmonics for view- and time-dependent appearance and introduces compact variants that substantially reduce the memory footprint, tackling the primary bottleneck of this paradigm. Building on the native 4D formulation, [340] makes strides in storage efficiency by decomposing color attributes into a three-parameter direct color component and a shared lightweight alternating-current color predictor, eliminating the need for high-dimensional spherical harmonic coefficients; combined with an entropy-constrained deformation technique and opacity-based entropy loss, this framework achieves approximately a $190\times$ storage reduction on the Technicolor dataset and a $125\times$ reduction on Neural 3D Video datasets while preserving comparable rendering speed and quality.

Comparative Analysis and Emerging Synthesis. Across these paradigms, a clear pattern of trade-offs emerges. Structure-guided methods that rely on canonical deformation fields generally yield superior reconstruction fidelity and compact model sizes, because the shared canonical space imposes a strong regularization that improves generalization across frames [341]. However, this fidelity comes at the cost of rendering speed and can be brittle when confronted with complex or abrupt motions. Gaussian-centric and per-frame models, by contrast, offer considerably higher rendering speeds and greater flexibility for locally heterogeneous motion, though they may suffer from quality variability and, in some cases, voluminous storage requirements [341]. Native 4D primitives provide a conceptually elegant and highly parallelizable framework, but historically have struggled with large model footprints due to the explosion of primitives in the temporal dimension. The most recent trends, however, point toward hybrid and adaptive designs that blur the boundaries between these categories. Multi-level competitive allocation frameworks, such as [342], deliberately distribute modeling capacity across three structured levels—static structure, persistent dynamic geometry, and transient appearance primitives—and use shared rasterization with residual-driven optimization to adaptively specialize each level, preserving long-term motion consistency while capturing fine dynamic detail. Dynamic-static decomposition strategies have also proven fruitful; [343] integrates a probabilistic dynamic perception coefficient directly into the Gaussian pipeline to adaptively separate static and dynamic components, enabling explicit motion learning for dynamic elements while maintaining geometric stability for static structures. Finally, [344] offers a plug-and-play module that learns spatio-temporal probability distributions for each Gaussian, disentangling spatial and temporal patterns to improve both reconstruction quality and spatio-temporal consistency when inserted into existing frameworks. Together, these works indicate that the field is moving toward unified, adaptive representations that can dynamically allocate resources based on scene content, promising robust and efficient performance across the full spectrum of dynamic scene complexity. The hierarchical, frequency-domain, and ODE-based motion decomposition strategies explored in the subsequent section build directly upon these representation paradigms, offering complementary tools for capturing the intricate temporal dynamics that these foundational architectures enable.

6.2 Motion Modeling and Decomposition Strategies for Dynamic Gaussians

Motion modeling represents the central challenge in extending 3D Gaussian Splatting (3DGS) from static scenes to dynamic, time-varying environments. Building upon the representation paradigms established in the previous subsection—canonical deformation fields, per-primitive temporal models, and native 4D primitives—the choice of how to represent, decompose, and predict motion determines not only visual fidelity but also the capability for temporal extrapolation, physical plausibility, and long-range consistency. While the representation paradigms define *where* motion is encoded, the strategies reviewed here address *how* motion is modeled, decomposed, and predicted. A broad spectrum of strategies has emerged, ranging from frequency-domain decompositions and ordinary differential equation (ODE)-based formulations to hierarchical, rigid-body, and flow-guided approaches. These approaches directly complement the training strategies described in the following subsection, as the choice of motion model fundamentally shapes the optimization schedules, regularization terms, and robustness requirements discussed there.

Fourier-Based and Frequency-Domain Motion Models

A natural approach for capturing periodic or quasi-periodic dynamics is to represent motion in the frequency domain. In the context of generative image dynamics, [345] proposes a Fourier-domain prior learned from real video sequences, where a spectral volume encodes dense, long-term motion trajectories for natural oscillatory dynamics such as swaying trees or flickering candles. Extending this to Gaussian-based dynamic scenes, FAST-GS [346] introduces a Fourier Motion Modeling module that decomposes complex motion into frequency-based sinusoidal components. This decomposition separates low-frequency global trajectories from high-frequency local motion details, enabling more accurate fitting of complex dynamic scenes while suppressing temporal jitter through frequency-dependent regularization. The frequency-view also benefits motion representation learning more broadly; FLD [347] learns structured motion representations in a parameterized latent space, enhancing interpolation and generalization for periodic motion patterns.

ODE-Based and Dynamical-System Formulations

Beyond frequency analysis, continuous-time dynamical systems modeled by neural ordinary differential equations (ODEs) provide a principled framework for motion extrapolation and physically plausible dynamics. ParticleGS [348] reformulates dynamic 3D scenes as physically grounded particle systems, employing a Neural ODE-based evolver to learn continuous-time dynamics for predicting future scenes beyond observed timeframes. Similarly, ODE-GS [349] integrates latent neural ODEs with 3DGS, modeling Gaussian parameter trajectories as continuous-time latent dynamics. By aggregating past trajectories into a latent state and evolving it via an ODE solver, ODE-GS enables arbitrary future timestamp rendering without explicit timestamp conditioning. The latent ODE paradigm has been further refined by path-minimizing formulations [350], where an ℓ_2 penalty on the latent path length improves extrapolation accuracy and inference efficiency. At the world-model level, ODEWorld [351] learns a continuous latent velocity field parameterized by an ODE in physical time, enabling arbitrary temporal resolution, backward prediction, and improved planning-oriented representation for robotic control.

Global-to-Local and Hierarchical Motion Decomposition

Real-world scenes often exhibit motions at multiple scales, motivating hierarchical decomposition strategies that align naturally with the multi-level motion representations introduced in the previous subsection. HEIR [352] learns structured, interpretable motion hierarchies from data, representing observed motions using graph-based structures where vertices denote elemental motions and directed edges capture parent-child dependencies via graph neural networks. This approach explicitly decomposes global absolute motions into parent-inherited patterns and local residual motions, enabling more realistic dynamic Gaussian deformation. RiGS [353] introduces three types of

Gaussian primitives—static, rigid, and transient—to represent long-term low-frequency motions and short-term high-frequency dynamics separately, with rigid Gaussians transitioning into transient Gaussians based on temporal duration. DynaSplat [354] also classifies elements as static or dynamic through deformation offset statistics and 2D motion flow consistency, then applies a hierarchical motion modeling strategy capturing both coarse global transformations and fine-grained local movements. A related frequency-domain hierarchical approach, the Long-Range Trajectories system [355], integrates motion from local and global representations to produce long-range trajectories that outperform trajectory-based methods for human activity analysis.

Rigid-Body and SE(3)-Transformation-Based Dynamics

A powerful inductive bias for motion decomposition is the assumption that scene elements undergo rigid-body transformations governed by the Special Euclidean group SE(3). This geometric perspective connects directly to the canonical deformation fields discussed earlier, offering a more constrained and physically interpretable alternative to free-form neural deformations. Shape of Motion [356] represents scene motion with a compact set of SE(3) motion bases, expressing each point’s motion as a linear combination of these bases and thereby enabling soft decomposition into multiple rigidly-moving groups. LieFlow [357] explicitly models motion within the SE(3) Lie group, unifying translation and rotation learning in a coherent geometric space, which enforces physically inspired constraints for motion continuity and geometric consistency. SIRE [358] learns intrinsic rigidity embeddings through a self-supervised framework, where an image encoder estimates scene rigidity and geometry, and a least-squares solver lifts 2D point tracks into SE(3) tracks for supervision—providing strong data efficiency for object segmentation and rigid motion estimation. The rigid-deformation decomposition framework [359] decouples global rigid-body motion from local deformation using two scale-specific networks, employing a frozen-anchor training strategy with quaternion-incremental schemes to improve kinematic separation and prediction accuracy. More generally, the structured motion geometry perspective [360] argues that real-world objects obey rigid-body kinematics, representing scene dynamics as a compact set of motion bases, each a temporal sequence of 6D twists in SE(3), with dense motion recovered via sparse, time-shared per-pixel assignment weights.

Flow-Guided and Multi-Scale Motion Decomposition

Optical flow and scene flow provide dense motion cues that can guide Gaussian deformation, offering a complementary signal to the photometric losses emphasized in the training strategies of the next subsection. DynMF [361] decomposes a dynamic scene into a few neural trajectories, binding each point to motion coefficients that enforce sharing of basis trajectories; a sparsity loss on coefficients enables motion disentanglement and novel motion combinations. H3D-DGS [362] proposes heterogeneous 3D control points obtained via a hybrid strategy combining optical flow back-projection for directly observable components and gradient-based optimization for occluded ones, achieving rapid convergence. SplitGaussian [363] explicitly decouples static and dynamic components, preventing motion artifacts in static regions while allowing view- and time-dependent appearance refinement in the dynamic branch. At the video level, decomposition into static scenes and dynamic objects with quasi-3D canonical volumes, as in DecoMotion [364], facilitates affine transformations for camera motion and rectified non-rigid transformations for dynamic content. The training-free motion factorization framework [365] decomposes complex motion into motionlessness, rigid motion, and non-rigid motion, modulating synthesis of distinct motion categories in a disentangled manner. The Multi-Level Semantic Motion Encoder in [366] progressively masks features to capture multi-granularity spatiotemporal relationships, while the decoupled spatio-temporal framework DeST [367] performs once unified spatial modeling with joint-decoupled temporal modeling to capture distinctive motion patterns of each joint.

Motion Composition and Interpretable Representations

Beyond decomposition, motion can be composed from reusable primitives or represented via interpretable embeddings, offering a bridge to the semantic and control-oriented applications discussed later in this survey. The Motion Poke Transformer [368] directly predicts distributions of local motion conditioned on sparse interactions termed “pokes,” enabling interpretable, multimodal scene dynamics. Lagrangian Motion Fields [369] treat each joint as a Lagrangian particle with uniform velocity over short intervals, condensing motion into “supermotions” that integrate static spatial information with interpretable temporal dynamics, enabling efficient long-term generation. Motion Inversion [370] introduces motion embeddings—explicit, temporally coherent one-dimensional embeddings—that integrate with temporal transformers of video diffusion models, allowing complex motion manipulation through vector arithmetic. The decomposition of video content into foreground, background, and motion flow components [371] further enables fine-grained control over each component. Finally, the Flow Poke Transformer [368] and the compositional flow matching framework [372] highlight the benefit of explicit, reusable motion primitives for embodied trajectory generation, improving sample efficiency and accuracy. These diverse motion modeling and decomposition strategies collectively provide a rich toolkit for representing dynamic Gaussian scenes, balancing expressiveness, physical plausibility, interpretability, and computational efficiency. Critically, the choice among these motion models directly impacts the training dynamics and optimization strategies detailed in the next subsection—frequency-domain approaches lend themselves to regularized optimization, ODE-based formulations require careful temporal sampling schedules, and rigid-body models benefit from geometric consistency losses—thereby establishing a tight coupling between motion representation and the practical trainability of dynamic 3DGS systems.

6.3 Training Dynamics and Optimization for Long-Range and Sparse-Input Dynamic Scenes

Training 4D Gaussian representations for dynamic scenes presents optimization challenges that extend well beyond the static 3DGS case, yet the strategies developed to address these challenges are what ultimately make the downstream applications discussed in the next subsection—animatable avatars, dynamic SLAM, and real-time streaming—practically achievable. The core difficulty lies in the joint estimation of appearance, geometry, and—critically—motion over time, all while managing the computational and memory costs associated with extended sequences. This subsection reviews the practical training strategies that have emerged to address these issues, focusing on adaptive densification, progressive optimization schedules, motion-aware losses, and robustness to sparse or imperfect inputs.

Adaptive Densification and Primitive Management for Motion-Dense Regions. A fundamental issue in dynamic 3DGS training is the uneven distribution of motion across a scene. Naively applying densification strategies designed for static scenes often leads to a proliferation of Gaussians in static regions and insufficient capacity in fast-moving areas. A growing body of work addresses this imbalance. [373] introduces a lifespan-aware densification strategy that explicitly allocates more Gaussian capacity to regions with pronounced motion while keeping static areas compact, mitigating optimization imbalance. [374] tackles this by decomposing the scene into static, persistent dynamic, and transient appearance primitives, allowing the optimization to dynamically allocate modeling capacity based on photometric errors. This competitive allocation prevents the over-parameterization of static regions while preserving dynamic fidelity. [375] proposes separating static and dynamic Gaussians during training and explicitly sampling positions and rotations of dynamic Gaussians at sparse timestamps, which are then interpolated, using a point-backtracking technique to detect and remove erroneous Gaussians. Similarly, [376] shows that exploiting hierarchical structural patterns—where groups of Gaussians share similar transformations—

allows for more efficient convergence with fewer iterations. [377] directly tackles the allocation mismatch by using semantic and motion priors from vision foundation models to concentrate control points in dynamic regions while suppressing redundancy in static backgrounds. [143], while focused on static scenes, provides a foundational analysis of split and clone operations that informs these dynamic approaches.

Progressive and Coarse-to-Fine Optimization Schedules. Long-range dynamic scenes are notoriously difficult to optimize from scratch; the loss landscape is highly non-convex, and the optimizer can get trapped in poor local minima. Progressive and coarse-to-fine strategies have proven crucial for stable convergence. [375] proposes a progressive training scheme where the model is initially trained using short timestamps and then progressively extended, allowing it to work effectively with sparse point clouds. [378] employs a hierarchical divide-and-conquer learning strategy to guide optimization toward globally coherent motion solutions. [379] introduces a progressive optimization strategy with two decoupled propagation stages—a background extension stage and a foreground propagation stage—to ensure robust reconstruction over long durations. For sparse-frame inputs, [380] introduces texture-aware deformation regularization and canonical optimization to handle the ill-posed nature of the problem, focusing on texture-rich areas that are particularly prone to failure. The need for such schedules is also highlighted in [381], which notes that initialization problems are exacerbated in fast-moving regions where SfM fails.

Motion-Aware Consistency Losses and Regularization. Recent methods emphasize that photometric loss alone is insufficient to learn physically plausible motion. Imposing motion-aware consistency losses is critical for temporal coherence and preventing drift. [382] proposes a mutual information shaping technique that enforces movement resonance between correlated Gaussians, an idea that can be integrated into the training loss to improve compositional control. [383] introduces a view-space ray grouping strategy to explicitly preserve local geometric structure of Gaussians over time, eliminating reliance on external flow priors. [384] argues that uncertainty matters—Gaussians with recurring observations should act as reliable anchors—and propagates reliable motion cues to enhance 4D reconstruction. [385] explicitly models per-Gaussian motion and uses optical flow on sparse key views as a lightweight motion cue to regularize against the tendency of photometric loss to chase pixel residuals. [386] introduces a physics-grounded supervised fine-tuning and a simulation-grounded reinforcement learning stage to correct residual physical violations, representing an extreme form of motion-aware regularization. [387] uses Cubic Hermite Splines with Temporal Curvature Regularization to ensure smooth motion evolution and prevent artifacts during temporal interpolation.

Handling Sparse, Monocular, and Noisy Inputs. A major driver of instability in dynamic 3DGS is the under-constrained nature of monocular or sparse-view input. Several methods explicitly address this challenge. [378] demonstrates that existing 4DGS methods dramatically fail under monocular settings, introducing isotropic Gaussian “marbles” to reduce degrees of freedom and adding image-level and geometry-level priors, such as tracking losses, to constrain optimization. [381] introduces uncertainty-aware regularization to identify uncertain regions with few observations and selectively imposes diffusion-based and depth-smoothness priors on them. [380] is specifically designed for scenarios where only sparse frames are accessible, proposing new regularization terms for both the deformation network and the canonical Gaussian field. [388] uses an occlusion-aware optimization with a view-conditioned diffusion prior to *sculpt* an initial reconstruction, dealing with severe occlusions, a common failure mode in monocular input. The challenge of noisy pose estimates is addressed by [389], which uses a symmetry-aware temporal encoding to absorb camera estimation drift. [390] introduces an adaptive densification strategy specifically to mitigate unstable Gaussian gradients during optimization of 4D generation, a challenge also present in reconstruction tasks.

Optimization Stability and Long-Range Coherence. Beyond the specific techniques above, ensuring stable convergence and temporal continuity over extended sequences is a critical concern, particularly as these reconstructed sequences are increasingly targeted for streaming and compression pipelines. [391] proposes progressive construction of local canonical anchor spaces to enhance temporal coherence and mitigate memory explosion and flickering. [392] explores keyframe-constrained streaming optimization under a max-points budget to improve long-horizon stability. [393] points out that piecewise linear velocity approximations lead to temporal fragmentation, and proposes a continuous SE(3) formulation that preserves temporal identity and mitigates unbounded memory growth. [394] presents a framework to construct a compact set of salient Gaussians via progressive pruning and merging, directly addressing the storage and optimization overhead of large dynamic models. [395] highlights the importance of jointly optimizing representation with rate-distortion trade-offs to reduce redundancy. Finally, the work on fast spatial memory of [396] is motivated by the observation that plastic inference-time updates are vulnerable to catastrophic forgetting, and proposes an elastic test-time training with a Fisher-weighted prior to stabilize long-context reconstruction, a concept directly applicable to online dynamic scene training. Collectively, these training strategies—adaptive capacity allocation, progressive schedules, motion-aware regularization, and robustness to imperfect inputs—form the optimization backbone that enables the efficient, stable, and deployable dynamic 3DGS systems explored in the following subsection.

6.4 Applications, Temporal Compression, and System-Level Integration

The practical utility of dynamic 3D Gaussian Splatting (4DGS) is ultimately determined by its ability to power real-world applications while overcoming the substantial storage, bandwidth, and computational overheads inherent to spatio-temporal representations. Building on the training strategies discussed previously—adaptive capacity allocation, progressive optimization, and motion-aware regularization—this subsection examines the primary application domains—animatable avatars, dynamic SLAM, and real-time streaming—alongside the critical enabling technologies of temporal compression and system-level integration that make these applications feasible on resource-constrained platforms.

Animatable Avatar Reconstruction. One of the most compelling applications of dynamic 3DGS is the creation of high-fidelity, animatable human avatars, directly leveraging the deformation and motion modeling capabilities developed in the training phase. These systems replace static representations with deformable Gaussian primitives that can be driven by pose parameters or expression codes. For clothed avatars, methods such as [397] leverage 2D Gaussian surfels to better align with surface geometry, improving both geometric accuracy and appearance fidelity over 3DGS-based counterparts. For head avatars, the challenge lies in modeling the distinct dynamics of deformable regions (e.g., skin) versus rigid regions (e.g., teeth). The Conditionally-Adaptive Gaussian Avatars (CAG-Avatar) framework [398] addresses this by using a cross-attention fusion module where each Gaussian acts as a query, selectively extracting relevant driving signals from a global expression code based on its canonical position. This localized conditioning, reminiscent of the motion-aware consistency losses discussed earlier, significantly reduces blurring and distortion in challenging regions. In the generative domain, AGORA [399] extends GAN-based generation to produce animatable avatars, combining spatial shape conditioning with a dual-discriminator strategy and achieving real-time performance on both GPU and mobile platforms. For streamable avatars, HGC-Avatar [400] introduces a hierarchical compression framework that disentangles the representation into a structural layer (StyleUNet-based generator) and a motion layer (SMPL-X), enabling layer-wise compression and progressive decoding for efficient transmission.

Dynamic SLAM and Robotics. Dynamic 3DGS is increasingly integrated into Simultaneous Localization and Mapping (SLAM) systems to enable dense, photorealistic mapping in dynamic environments, where the robustness to sparse and noisy inputs emphasized in the previous subsection becomes critical. Early methods like Dy3DGS-SLAM [401] were the first to use 3DGS SLAM with monocular RGB input, fusing optical flow and depth masks to filter dynamic objects. Building on this, D4DGS-SLAM [402] represents the first SLAM method based on a 4DGS map representation, incorporating a dynamics-aware InfoModule to filter unstable points and applying isotropic regularization based on dynamic characteristics. The challenge of computational efficiency is tackled by RTGS [403], an algorithm-hardware co-design framework that reduces redundancies via adaptive Gaussian pruning and dynamic downsampling, achieving real-time performance (≥ 30 FPS) on edge GPUs. For dynamic object handling, DAGS-SLAM [404] maintains per-Gaussian motion probability states and triggers semantic segmentation on demand via an uncertainty-aware scheduler, balancing accuracy and compute. UP-SLAM [405] further improves robustness with a probabilistic octree for primitive management and a training-free uncertainty estimator for open-set dynamic object detection. Memory-efficient variants like MemGS [53] address GPU memory constraints by merging redundant Gaussians in voxel space, while DiskChunGS [406] enables large-scale reconstruction by storing inactive spatial chunks on disk. The survey by [407] provides a comprehensive technical reference for this rapidly evolving area.

Real-Time Streaming and On-the-Fly 4D Reconstruction. The ability to stream dynamic scenes in real-time is critical for volumetric video and immersive communication, and it directly benefits from the long-range coherence and optimization stability techniques described earlier. Online reconstruction methods have made significant strides in reducing per-frame training time. 3DGStream [408] achieves on-the-fly per-frame reconstruction within 12 seconds using a Neural Transformation Cache. HiCoM [409] improves efficiency by introducing a Hierarchical Coherent Motion mechanism and perturbation smoothing, reducing training wall time to less than 2 seconds per frame. For arbitrary-length sequences, SwinGS [410] integrates spacetime Gaussians with Markov Chain Monte Carlo (MCMC) and a sliding window strategy, reducing transmission costs by 83.6% and enabling WebGL-based playback on mobile devices. ClipGStream [411] offers a hybrid clip-level stream optimization for long sequences, while StreamSplat [412] provides a fully feed-forward framework that instantly transforms uncalibrated videos into dynamic 3DGS representations. The DASS pipeline [413] introduces a three-stage framework (selective inheritance, dynamics-aware shift, error-guided densification) for iterative streamable reconstruction. For bandwidth-adaptive delivery, PD-4DGS [414] proposes progressive compression with hierarchical deformation decomposition, reducing first-frame latency from 73-930 seconds to ~ 1.7 seconds on a 2 Mbps link, while TIGAS [415] offloads rasterization to a backend and streams view-dependent 2D projections over QUIC for thin-client rendering.

Temporal Compression and Rate-Distortion Optimization. A central challenge for deployment is the enormous storage footprint of 4DGS models, which the memory-management strategies of the previous subsection help mitigate but do not fully resolve. The survey by [416] provides a unified overview of compression techniques, categorizing them into parameter and restructuring compression. Significant progress has been made in rate-distortion (RD) optimized compression. P-4DGS [417] designs a 3D anchor point-based spatial-temporal prediction module combined with adaptive quantization and context-based entropy coding, achieving up to $90\times$ compression. RD4DGS [418] employs a wavelet transform to exploit temporal smoothness priors, achieving up to $91\times$ compression. MEGA [419] decomposes color attributes into a direct component and a shared predictor, eliminating spherical harmonics, and reduces storage by approximately $190\times$. TC3DGS [420] selectively prunes Gaussians based on temporal relevance and uses gradient-aware mixed-precision quantization along with an adapted Ramer-Douglas-Peucker algorithm for

trajectory interpolation, achieving up to $67\times$ compression. 4DGC [395] introduces a motion-aware representation with a compact motion grid and an end-to-end compression scheme for variable bitrate support. For motion decoupling, 4D-MoDe [421] separates static backgrounds from dynamic foregrounds, achieving as low as 11.4 KB/frame, while ComGS [422] uses keypoint-driven motion representation to achieve over $159\times$ storage reduction. Additionally, CDGS [423] formulates dynamic scene reconstruction as a budget-constrained optimization problem, enforcing strict Gaussian budgets for edge deployment. Finally, 4DGS-CC [424] provides a contextual coding framework using neural voxel and vector quantization coding, while EvolvingGS [425] exploits temporal coherence between frames to achieve over $50\times$ compression. The system-level latency analysis by [426] provides critical insights into optimizing the application, transport, and network layers for end-to-end responsiveness. Together, these applications and compression frameworks demonstrate a clear trajectory toward scalable, real-time, and deployable dynamic 3DGS systems, closing the loop from the optimization foundations established in the previous subsection to practical, resource-aware deployment.

7 Compression, Acceleration, and Scalability

7.1 Model Compression Techniques for Static 3DGS

The substantial storage footprint of 3D Gaussian Splatting (3DGS) remains a primary barrier to its widespread deployment. A single high-fidelity scene, represented by millions of Gaussian primitives each carrying up to 59 floating-point parameters, can consume roughly one gigabyte of memory [427, 98]. This challenge has driven a rich body of research on reducing model memory and storage requirements while preserving rendering quality—a pursuit that is closely intertwined with the Level-of-Detail (LOD) and progressive rendering techniques discussed in the following subsection. Indeed, effective compression not only enables efficient storage and transmission but also underpins scalable rendering by providing compact, multi-resolution representations that can be selectively activated based on viewing conditions. This subsection offers a comprehensive overview of model compression techniques for static 3DGS, broadly categorized into primitive-level pruning, attribute-level quantization and coding, and hybrid frameworks that integrate multiple strategies.

Pruning Strategies

A foundational approach to reducing model size is eliminating redundant or insignificant Gaussian primitives. A common strategy is to prune Gaussians that contribute little to the final rendering. This can be achieved through post-hoc, gradient-informed iterative pruning, which identifies and removes redundant information to enable memory and computational scalability [428]. Similarly, ELMGS integrates differential quantization and entropy coding estimators into the optimization strategy alongside an iterative pruning approach to enhance both compressibility and scalability [429].

More sophisticated pruning methods move beyond simple heuristics. For instance, some frameworks learn an importance score for each Gaussian to automatically identify the least significant primitives for a given scene, enabling more effective pruning than manual criteria [430]. A similar philosophy is adopted by the Gradient-Driven Natural Selection framework, which models survival pressure as a regularization gradient field on opacity, allowing the optimization process to autonomously determine which Gaussians to retain [131]. VEDAL formalizes this concept further by casting pruning as a variational free energy minimization problem, using a prediction-error gating mechanism to activate pruning asynchronously [132]. Other work focuses on the *timing* and *method* of densification, introducing an Edge-Aware Score for selecting candidates for splitting and a Long-Axis Split strategy to improve the quality of the cloned Gaussians, coupled with overfitting mitigation techniques like Recovery-Aware Pruning to achieve a more compact and higher-fidelity

model [42].

An interesting direction is geometry-driven pruning, where primitives are ranked based solely on their 3D attributes without relying on 2D image information, as seen in the PointSplat framework, which uses a dual-branch encoder to re-weight geometry and appearance [431]. Furthermore, pruning can be combined with more expressive primitives, such as the Difference-of-Gaussians, to reduce the total number of primitives required while maintaining visual quality [432]. Some methods also exploit redundancy in the function space, where strongly overlapping and non-orthogonal Gaussian basis functions contribute to representational redundancy, using quantum-inspired eigendecomposition to select a representative subset of primitives [433].

Quantization and Attribute Coding

After reducing the number of primitives, the remaining storage overhead is dominated by the 45–59 floating-point parameters per Gaussian. Quantization is a powerful tool to reduce the bit-width of these attributes. A simple yet effective technique is vector quantization based on K-means, which clusters similar Gaussian parameters and stores a small codebook along with the index for each Gaussian, achieving significant compression while maintaining quality [434]. To ensure consistency between the continuous optimization and discrete parameter assignment, the noise-substituted vector quantization technique is used to jointly train codebooks and model features [427].

More advanced methods explore different quantization spaces and strategies. For instance, a data-independent approach based on random rotations has been proposed, which transforms parameter vectors into a space with a known distribution, enabling near-optimal Lloyd-Max quantization without any per-scene training or calibration data [435]. Another line of work replaces uniform scalar quantization with lattice vector quantization (LVQ), which can be adapted to the scene-specific characteristics of the data, improving rate-distortion efficiency while maintaining low complexity [436]. More comprehensive approaches, such as GETA-3DGS, propose an end-to-end automatic framework for joint structured pruning and mixed-precision quantization, treating each Gaussian as a group with heterogeneous per-attribute bit-widths optimized under a target storage budget [437].

Entropy Coding and Context Modeling

To squeeze out the final bits, many compression pipelines incorporate entropy coding. This is often enhanced by sophisticated context models that estimate the probability distribution of the quantized attributes for more efficient arithmetic coding. A pioneering work in this direction introduced a hash-grid assisted context framework, which leverages the structured nature of a binary hash grid to model the spatial relations between unorganized anchors or Gaussians [438]. This was later extended to HAC++, which further captures intra-anchor contextual relationships and uses an adaptive quantization module for higher-precision quantization, achieving over 100x compression [439]. Building on these concepts, ContextGS introduced the first autoregressive model at the anchor level, dividing anchors into different levels and predicting uncoded anchors based on already-coded ones to improve coding efficiency [440].

Another line of work focuses on integrating prediction techniques borrowed from video compression. For example, a spatial condition-based prediction module can be used to predict attributes of Gaussian primitives from the surrounding scene context, with residual compensation learning fine-grained missing information [441]. A more recent framework, MesonGS++, proposes a size-aware post-training codec that combines joint importance-based pruning, octree geometry coding, attribute transformation, selective vector quantization for higher-degree spherical harmonics, and group-wise mixed-precision quantization with entropy coding [442]. Prior to this, MesonGS laid the groundwork with a post-training codec that used a measurement criterion and attribute transformations like replacing quaternions with Euler angles to improve compression rates [443].

The choice of entropy model is critical. The Mixture of Priors (MoP) strategy addresses limitations in existing entropy models by using a Mixture-of-Experts paradigm to process hyperprior

information, and introduces a prior-guided Coarse-to-Fine Quantization (C2FQ) strategy for element-wise quantization [444].

Codebook-Based and Hybrid Frameworks

Many state-of-the-art methods combine the aforementioned techniques into a unified, hybrid framework. The compact 3D Gaussian representation C3DGS proposes a learnable mask to prune redundant Gaussians and a grid-based neural field to replace spherical harmonics, before applying residual vector quantization to geometric attributes [98, 97]. ContraGS takes a different approach by enabling training directly on codebook-compressed representations, using MCMC sampling to solve the problem of learning non-differentiable parameters, which drastically reduces the peak memory during training [445]. Another training-in-the-loop method, OMG (Optimized Minimal Gaussians), focuses on reducing the number of Gaussians to a minimal set while using a compact and precise attribute representation with sub-vector quantization [446]. MEGS² also contributes to this area by replacing memory-intensive spherical harmonics with spherical Gaussian lobes and proposing a unified soft pruning framework that jointly optimizes the number of primitives and parameters [447].

Beyond these, there are hybrid approaches that integrate diffusion priors for image-level restoration after heavy compression [448], or leverage the powerful representational capacity of learned image priors to recover compression-induced quality degradation [449]. Finally, some methods use dictionary learning to compress the parameters of a pre-trained 3DGS model directly, without re-training or architectural modifications [450]. Collectively, these techniques demonstrate a clear trend toward more holistic and integrated compression pipelines, effectively balancing compression ratios, rendering fidelity, and computational efficiency [451]. Importantly, the compact and hierarchical representations produced by these compression methods naturally complement the LOD and progressive rendering strategies explored next, enabling adaptive quality scaling and efficient bandwidth utilization across diverse deployment scenarios.

7.2 Level-of-Detail (LOD) and Progressive Rendering

While the compression techniques discussed in the previous subsection primarily target static storage reduction, real-time interactive applications impose an additional, orthogonal challenge: rendering must gracefully adapt to changing viewing conditions, hardware capabilities, and bandwidth constraints without compromising the user experience. Level-of-Detail (LOD) techniques address this by managing the computational and memory costs of rendering complex 3D scenes at runtime. Within the context of 3D Gaussian Splatting (3DGS), LOD frameworks have been developed to enable scalable rendering across diverse hardware capabilities and bandwidth constraints, leveraging the explicit primitive-based nature of Gaussians to achieve what was difficult with implicit neural representations. These approaches generally fall into two broad paradigms: discrete LOD, which offers a finite set of quality levels, and continuous LOD, which provides a seamless spectrum of detail from a single unified model. Notably, the compact hierarchical representations produced by the compression methods in Section 7.1—such as learnable masks, codebooks, and multi-resolution structures—provide a natural substrate upon which many LOD systems build, while the progressive streaming mechanisms described here also complement the storage-oriented goals of compression by enabling adaptive bandwidth utilization.

Discrete LOD methods traditionally build distinct, pre-optimized model versions that are swapped based on viewing conditions. However, this paradigm suffers from significant storage overhead and visual “popping” artifacts during transitions. In contrast, the explicit and primitive-based nature of 3DGS enables a more elegant solution. For instance, CLoD-GS introduces a continuous LOD mechanism by attaching a learnable, distance-dependent decay parameter to each Gaussian

primitive. This parameter dynamically modulates the primitive’s opacity based on viewpoint proximity, allowing for the progressive and smooth filtering of less significant primitives and creating a continuous spectrum of detail within a single model. The framework employs a virtual distance scaling mechanism and a coarse-to-fine training strategy with rendered point count regularization to ensure robustness across all distances [452]. Similarly, Matryoshka Gaussian Splatting (MGS) achieves continuous LOD through stochastic budget training, where each optimization iteration samples a random splat budget and optimizes both the corresponding prefix and the full set of ordered Gaussians. This strategy allows the model to render any prefix of the Gaussian set as a coherent reconstruction with smoothly improving fidelity as the budget increases, effectively establishing a continuous speed-quality trade-off without requiring architectural modifications [453].

Hierarchical multi-level representations form another key strategy for LOD. These approaches typically construct a pyramid of detail, where each level represents the scene at a different granularity, and they bear a strong conceptual resemblance to the multi-resolution quantization and codebook hierarchies introduced in Section 7.1. The Iterative Gaussian Synopsis framework, for example, employs a top-down “unfolding” scheme. It begins with a full-resolution 3DGS model and iteratively derives coarser LODs using an adaptive, learnable mask-based pruning mechanism. This process constructs a multi-level hierarchy that integrates hierarchical spatial grids with a shared Anchor Codebook to capture both global structure and localized details, promoting inter-layer reusability and minimizing data overhead for progressive refinement [454]. FLoD proposes a flexible level-of-detail approach that constructs a multi-level 3DGS representation through level-specific 3D scale constraints. Each level independently reconstructs the entire scene, offering varying detail and GPU memory usage, which allows for selective rendering of different image regions at different detail levels and enables real-time rendering under diverse memory constraints [455]. LODGE presents a hierarchical LOD representation for large-scale scenes by iteratively selecting optimal subsets of Gaussians based on camera distance. It constructs each LOD level by applying depth-aware smoothing filters followed by importance-based pruning and fine-tuning, and partitions the scene into spatial chunks for dynamic loading to reduce rendering time and GPU memory [456]. A notable system-level implementation, Virtualized 3D Gaussians, draws inspiration from game engines to construct hierarchical 3D Gaussian clusters and dynamically select only the perceptible clusters during rendering, balancing efficiency and visual quality for composed scenes requiring billions of Gaussians [457]. CityGaussian also incorporates an LoD strategy with block-wise detail level selection for efficient large-scale scene rendering, following a divide-and-conquer training approach [458].

Continuous LOD is also being explored through alternative primitive designs. Fourier Splatting introduces an inherently scalable primitive by parameterizing planar surfels with Fourier encoded descriptors. This allows a single trained model to be rendered at varying levels of detail simply by truncating Fourier coefficients at runtime, providing a versatile solution for bandwidth-constrained rendering [65]. Frequency-aware Gaussian Splatting Decomposition offers another perspective by organizing 3D Gaussians into groups corresponding to Laplacian-pyramid subbands of input images. This explicit frequency decomposition enables dynamic LOD rendering, progressive streaming, and foveated rendering, as well as artistic filtering [145].

Progressive and scalable compression schemes are essential for adaptive streaming and deployment, serving as a bridge between the static compression objectives of Section 7.1 and the real-time rendering demands addressed here. GoDe (Gaussians on Demand) introduces a framework for scalable compression and progressive LOD control. It reorganizes Gaussian primitives into a fixed progressive hierarchy based on aggregated gradient sensitivity, enabling multiple rate-distortion operating points from a single trained model without retraining, and achieves rate-distortion performance comparable to state-of-the-art single-rate methods [459]. For dynamic scenes, PD-

4DGS (Progressive Decomposition of 4DGS) presents the first framework for progressive compression and on-demand transmission, decomposing the motion hierarchy into independently transmittable layers to enable any prefix of the bitstream to be immediately renderable and compatible with modern adaptive-bitrate streaming [414]. LapisGS constructs a layered 3DGS structure for cumulative representation, incorporating dynamic opacity optimization and occupancy maps to support adaptive streaming and progressive rendering, effectively balancing visual fidelity with model compactness for bandwidth-aware applications [460]. The concept of progressive rendering can also be extended to neural representations beyond Gaussians, as demonstrated by methods for encoding continuous levels of detail in light field networks [461].

Beyond the core LOD paradigms, several works address the challenge from complementary angles. 3DGS³ proposes a unified post-rendering framework that performs gradient-aware super sampling and temporal frame interpolation on low-resolution outputs to achieve high-resolution and high-frame-rate rendering, a strategy that complements LOD by decoupling rendering resolution from the underlying scene complexity [462]. Techniques from the broader compression field, such as region-adaptive learned hierarchical encoding, can also contribute to scalable representations by designing codecs that overfit latents to each Gaussian attribute under a global rate constraint, exploiting octree structures for hierarchical multi-resolution representation [463]. Finally, the analysis of converged 3DGS solutions has revealed that dense regions of a scene exhibit geometry-correlated parameters amenable to render-free prediction, while sparse regions require multi-view constraints, providing density-aware strategies that can inform how LOD systems allocate primitives for improved robustness and efficiency [464].

In summary, the field of LOD and progressive rendering for 3DGS is rapidly evolving from simple discrete model variants towards sophisticated, single-model continuous representations that support seamless quality scaling, progressive loading, and efficient streaming across a wide range of devices and network conditions. These capabilities not only complement the storage-centric compression methods of Section 7.1 by enabling adaptive runtime selection of detail, but also set the stage for the training-time scalability challenges addressed next in Section 7.3, where efficient production of high-quality large-scale models becomes the enabling factor for the LOD systems described here to operate at their full potential.

7.3 Training Acceleration, Out-of-Core, and Distributed Optimization

The training of 3D Gaussian Splatting (3DGS) models, while significantly faster than NeRF-based counterparts, remains a computationally intensive process, particularly as scene complexity and the number of Gaussian primitives scale. The core bottlenecks are threefold: the cost of the rasterization pass itself, the memory footprint required to store parameters, gradients, and optimizer states for millions of primitives, and the system-level inefficiencies in how data is accessed and computed. This subsection surveys the key innovations in algorithm-system co-design, out-of-core training, and distributed optimization that collectively aim to break these barriers, complementing the level-of-detail and progressive rendering techniques discussed previously by addressing the upstream challenge of efficiently producing the high-quality, large-scale models those methods rely on.

At the foundational level, the efficiency of the rendering and gradient computation pipeline is paramount. The conventional 3DGS training pipeline sequentially executes Gaussian densification, projection, and color splatting, a process that suffers from skewed density allocation, imbalanced computation workloads during projection, and fragmented memory access during color splatting. To address these, algorithm-system co-design has proven highly effective. For instance, **BalanceGS** proposes a holistic optimization: at the algorithm level, it introduces heuristic workload-sensitive Gaussian density control to automatically balance point distributions by removing redundant

Gaussians in dense regions while filling gaps in sparse areas, which rectifies the skewed density allocation [465]. At the system level, it replaces the static one-to-one thread-pixel mapping with a similarity-based Gaussian sampling and merging strategy, allowing threads to dynamically process variable numbers of Gaussians based on local cluster density, thereby balancing the computation workload [465]. Furthermore, it employs a reordering-based memory access mapping strategy to restructure RGB storage, enabling efficient batch loading in shared memory and mitigating fragmented memory access [465]. This co-design approach achieves a significant $1.44\times$ training speedup on an NVIDIA A100 GPU with negligible quality degradation [465].

Building upon these principles, more aggressive system-level frameworks have been developed to push training performance further. **LiteGS** presents a high-performance framework that systematically optimizes the training pipeline through a multi-level strategy. At the low-level computation layer, it designs a warp-based raster associated with two hardware-aware optimizations to significantly reduce gradient reduction overhead. At the mid-level data management layer, it introduces dynamic spatial sorting based on Morton coding to enable a highly performant “Cluster-Cull-Compact” pipeline, which improves data locality and reduces cache misses. At the top-level algorithm layer, it establishes a new robust densification criterion based on the variance of the opacity gradient, paired with a more stable opacity control mechanism, to achieve more precise parameter growth. Experimental results demonstrate that LiteGS accelerates the original 3DGS training by up to $13.4\times$ with comparable or superior quality [466]. These works highlight that performance gains are not solely derived from algorithmic improvements but from a deep co-design of the algorithm with the underlying hardware architecture.

While the aforementioned methods target in-memory training, the memory wall presents an insurmountable barrier for scaling to extremely large-scale scenes. The primary limitation is that each Gaussian primitive carries a large attribute vector, and the aggregate parameter table quickly exceeds GPU capacity, limiting prior systems to tens of millions of Gaussians. A key observation is that 3DGS training is inherently sparse and trajectory-conditioned: each iteration activates only the Gaussians visible from the current camera batch. This insight has enabled the GPU memory to be treated as a working-set cache rather than a persistent parameter store. This principle is the foundation of out-of-core training frameworks. **TideGS** introduces an out-of-core training framework that manages parameters across an SSD-CPU-GPU hierarchy. It achieves this via three synergistic techniques: block-virtualized geometry for SSD-aligned spatial locality, a hierarchical asynchronous pipeline to overlap I/O with computation, and trajectory-adaptive differential streaming that transfers only incremental working-set deltas between iterations. This system enables training with over one billion Gaussians on a single 24 GB GPU, achieving the best reconstruction quality among evaluated single-GPU baselines on large-scale scenes [467].

Similarly, **GS-Scale** addresses GPU memory limitations by storing all Gaussians in host memory and transferring only a subset on-demand for each forward and backward pass. To mitigate the slowdowns introduced by CPU-based frustum culling and optimizer updates, GS-Scale employs three system-level optimizations: selective offloading of geometric parameters for fast frustum culling, parameter forwarding to pipeline CPU optimizer updates with GPU computation, and a deferred optimizer update to minimize unnecessary memory accesses for Gaussians with zero gradients. This approach lowers GPU memory demands by $3.3\text{--}5.6\times$ while achieving training speeds comparable to GPU-only training, thereby enabling large-scale 3DGS training on consumer-grade GPUs [468]. These out-of-core methods are reminiscent of analogous challenges in large language model (LLM) training, where host memory offloading and heterogeneous memory hierarchies are also crucial. For instance, systems have been developed to offload optimizer states to host memory to reduce GPU memory utilization [469], and more recent works redefine the roles of CPU and GPU for large-model optimization by using host memory as the authoritative parameter store [470]. The success of these

approaches in LLMs provides a strong precedent for the ongoing shift towards memory-centric and out-of-core execution paradigms in 3DGS.

At an even larger scale, training must be distributed across multiple GPUs. This introduces challenges related to load balancing, communication overhead, and data partitioning. While specific multi-GPU training systems for 3DGS are still nascent, the principles of load-balanced partitioning and hybrid parallelism from the broader machine learning community are directly applicable. For example, systems like **Zorse** demonstrate that efficient training on heterogeneous clusters requires the integration of pipeline and data parallelism in a manner that is both communication- and memory-efficient, with the capability to flexibly partition GPUs into asymmetric pipeline stages [471]. Similarly, **HETHUB** tackles heterogeneous clusters by introducing a distributed unified communicator and an automatic parallel planner to efficiently train models across AMD, Nvidia, and other GPU types [472]. Furthermore, the **HexiScale** system formalizes the allocation of asymmetric partitioned training computations over heterogeneous GPUs as a constrained optimization problem, proposing an efficient hierarchical graph partitioning algorithm to fully leverage available computational power [473]. These approaches to distributed and heterogeneous training, though developed for LLMs, provide a critical blueprint for adapting 3DGS to multi-GPU and multi-node environments, ensuring that the increased scale of Gaussian primitives can be matched by proportional compute resources without being bottlenecked by communication or load imbalance.

In summary, the path to scalable 3DGS training is being paved through a three-pronged approach: fine-grained hardware-aware kernel and scheduling optimizations (e.g., warp-based rasterization, Morton coding), out-of-core hierarchical memory management to break the GPU memory wall, and large-scale distributed system designs adapted from the broader deep learning community. The convergence of these strategies promises to make the training of billion-primitive, photorealistic scenes both memory-feasible and time-efficient. These training-time advancements are not only essential for producing the high-quality models used in the level-of-detail and progressive rendering systems described earlier, but they also lay the groundwork for the deployment and streaming optimizations discussed in the next subsection, ultimately unlocking new possibilities for large-scale spatial intelligence applications.

7.4 Hardware Acceleration, Mobile Deployment, and System Integration

The deployment of 3D Gaussian Splatting in real-world, resource-constrained environments—ranging from embedded devices and mobile platforms to edge servers—represents the final frontier of the system-level challenges introduced in the previous subsection. While the core 3DGS pipeline is designed for GPU-friendly rasterization, achieving interactive frame rates under tight power, memory, and bandwidth budgets necessitates a holistic approach that spans hardware architecture, software optimization, and network-aware delivery—building upon the memory-centric and distributed paradigms previously discussed by addressing the equally demanding constraints of the deployment side.

Hardware-Accelerated Rasterization and Custom Accelerators

A primary avenue for enabling real-time 3DGS on edge devices is the development of specialized hardware accelerators, paralleling the algorithm-system co-design philosophy that proved essential for training acceleration. Field-Programmable Gate Arrays (FPGAs) and Application-Specific Integrated Circuits (ASICs) offer the potential for significant gains in both performance and energy efficiency over general-purpose GPUs, and their viability for 3DGS is underpinned by the same hardware-software co-design principles that have driven the training-time optimizations discussed earlier. The design of such accelerators requires careful consideration of the underlying dataflow

and memory access patterns, a challenge well-documented in the broader context of neural network acceleration [474, 475]. For 3DGS, this involves architecting pipelines that can efficiently handle the tile-based rasterization, Gaussian sorting, and alpha-blending stages. Custom accelerators can integrate compression-aware pipelines directly into the hardware, reducing memory traffic and power consumption, much like the Morton coding and memory access restructuring used to optimize training. The viability of such approaches is supported by general-purpose edge-AI accelerator designs that demonstrate substantial improvements in throughput and energy efficiency by exploiting algorithmic properties like sparsity or reduced precision [476, 477]. Furthermore, a hardware-software co-design philosophy is crucial, as the most effective solutions jointly optimize the algorithm and the architecture [478, 479]. Techniques such as RISC-V ISA extensions for specific acceleration functions offer a programmable yet efficient middle ground for edge inference [480].

Mobile and Edge Platform Optimization

Beyond custom silicon, optimizing 3DGS for existing mobile and embedded GPUs is a critical practical concern on platforms like those from NVIDIA’s Jetson family, and these efforts mirror the workload-balancing and cache-locality strategies that proved essential for training efficiency. Profiling studies on these platforms reveal significant underutilization of key computational resources, such as streaming multiprocessors (SMs) and tensor cores, despite high overall GPU utilization, indicating that software optimizations like warp-coherent rasterization and memory access scheduling are essential to unlock true hardware potential [481]. The inherent constraints of such devices, including thermal throttling and limited memory bandwidth, can lead to significant performance degradation during sustained operation, a phenomenon often overlooked in benchmark-centric evaluations [482]. Emulation-based studies that systematically under-clock and power-cap a high-end GPU provide valuable insights into the performance-energy trade-offs that will govern the viability of edge-deployed 3DGS systems, particularly for standalone XR headsets [483]. The “Memory Wall” is frequently a more significant bottleneck than raw compute, especially for high-resolution outputs. Hardware-algorithm co-design strategies that decouple compute-bound and memory-bound components can effectively mask latency and enable higher-resolution real-time generation, a principle directly applicable to optimizing 3DGS pipelines [484].

On-Device Training and Adaptation

While the primary goal is often high-performance inference, the ability to adapt or fine-tune a 3DGS model on-device extends the out-of-core and memory-efficient training principles from the previous subsection into the deployment domain, opening up new possibilities for personalization and continual learning. This is an extremely demanding task given the memory and compute constraints of edge hardware, and it inherits the same memory-wall challenges that motivated host-memory offloading and working-set caching in large-scale training. Lightweight adaptation techniques, such as those explored for fine-tuning large language models on a single edge-class GPU [485], can inform strategies for efficient 3DGS fine-tuning. Similarly, frameworks designed for on-device continual learning, which use hyperdimensional computing to provide fast, non-iterative online updates with a small memory footprint, could inspire novel approaches for updating Gaussian parameters in response to new data or changing environments without full retraining [486]. The requirement for timely decision-making in autonomous systems also demands self-adaptive runtimes that can balance real-time performance, task reward, memory, and energy consumption for on-device training and inference loops [487]. The training process itself can be made more efficient and deployable through algorithmic compression, such as tensor-train decomposition, with a co-designed hardware engine to accelerate the computationally intensive intermediate steps [479].

Streaming and Adaptive Transmission

For interactive and immersive applications, the ability to stream 3DGS content over bandwidth-constrained networks is as critical as local rendering performance, and it presents a complementary

challenge to the out-of-core memory hierarchy discussed in training: just as training moves data between SSD, CPU, and GPU based on access patterns, deployment must move model data between servers and devices based on network conditions. A monolithic 3DGS model must be fully downloaded before any rendering can occur, leading to unacceptable black-screen waits on mobile connections. Progressive compression and transmission frameworks address this by decomposing the model into hierarchy of layers—such as a static scaffold, global deformation, and local refinement—so that any prefix of the bitstream is immediately renderable [414]. This approach, analogous to modern adaptive-bitrate (ABR) video streaming, enables true on-demand delivery. Strategies for rendering content on a server and streaming the resulting video to clients can also be enhanced by adaptively adjusting frame rate and resolution based on scene content, motion, and network bandwidth to optimize perceived quality [488]. At the system level, the high computational demands of 3DGS may necessitate edge offloading, where a portion of the rendering or computation is performed on a nearby server, echoing the distributed and heterogeneous training strategies from the previous subsection but now applied to inference. Reinforcement learning-based frameworks can dynamically allocate network resources and make offloading decisions based on the headset’s local computing capabilities, network conditions, and the trade-offs between frame loss rate and energy efficiency, thereby extending coverage area and reducing energy consumption on the mobile device [489]. To further reduce latency, models can be partitioned and partially executed on the edge device with a lightweight on-device encoder, transmitting only compact feature tensors to a remote server for the heavier processing [490]. Finally, for orchestrated systems serving multiple clients, adaptive scheduling frameworks that jointly optimize operator parallelism, placement, and configuration transitions across heterogeneous compute and bandwidth constraints can significantly improve end-to-end throughput for related data-preparation and rendering pipelines [491].

In summary, the deployment-side optimizations for 3DGS form a coherent ecosystem that mirrors and extends the training-side system challenges: custom accelerators and hardware-algorithm co-design address compute and energy efficiency, mobile optimizations tackle memory and thermal walls, on-device adaptation borrows from memory-efficient training paradigms, and streaming frameworks address network bandwidth constraints. Together with the training acceleration, out-of-core, and distributed optimization techniques discussed previously, these deployment strategies complete the full lifecycle of 3DGS—from billion-primitive model creation on multi-GPU clusters to real-time interactive experiences on resource-constrained edge devices—thereby enabling the large-scale spatial intelligence applications that motivate this entire body of work.

8 Semantic Understanding, Editing, and Scene Manipulation

8.1 Semantic Feature Distillation and Representation

The integration of semantic understanding into 3D Gaussian Splatting (3DGS) represents a pivotal advancement, transforming scenes from mere geometric and appearance models into queryable, interactive environments. The cornerstone of this transformation is the ability to embed high-dimensional semantic features, typically derived from powerful 2D foundation models, into the explicit 3D Gaussian primitives. This subsection surveys the methodologies developed for this crucial process of semantic feature distillation and representation, a field that navigates the inherent challenges of aligning 2D view-dependent features with a 3D world model while maintaining efficiency and fidelity. The approaches explored here—ranging from direct feature injection to compact, decoupled, and hierarchical representations—collectively establish the representational substrate upon which higher-level tasks such as open-vocabulary and instance-level segmentation, detailed in the following subsection, are built.

The prevailing paradigm in the initial wave of research involves distilling features from pre-trained 2D models such as CLIP, SAM, and LSeg into the 3D Gaussian field. The foundational work in this area, [492], demonstrated the feasibility of this approach by establishing the first general framework for distilling arbitrary-dimensional feature fields into 3DGS. This work tackled the initial hurdles of spatial resolution and channel consistency mismatches between RGB images and feature maps, laying the groundwork for subsequent methods. Building on this, approaches like [493] sought to integrate semantics more efficiently, proposing a Semantic Attribute Compactness (SAC) approach to learn compact yet effective semantic representations, achieving real-time rendering speeds (over 100 FPS).

However, a central challenge in this distillation process is the inherent cross-view inconsistency of 2D semantic features. The task of supervising a 3D feature field with inconsistent 2D predictions across different viewpoints is a significant source of error. To mitigate this, methods like [494] integrate vision-language embeddings into 3DGS and introduce a pixel alignment loss to ensure that rendered feature distances for the same semantic entities are close, thereby fostering multi-view consistency and enabling significant speedups in inference. Similarly, [493] introduces a 3D Coherent Self-training (3DCS) strategy, which leverages the multi-view consistency inherent to the 3D model itself to refine pseudo-labels and impose cross-view consistency constraints. Another significant challenge is the granularity inconsistency in 2D segmentations from models like SAM, where an object might be segmented as a whole in one view but into its constituent parts in another. To address this, [495] proposes constructing local graphs to propagate contextual features across Gaussians, reducing noise from inconsistent granularity, and employs mask-centric contrastive learning to smooth features across views. More uniquely, some methods explicitly tackle the view-dependent nature of semantics by decomposing the 3D scene into objects and constructing view-aggregated semantic representations, as proposed in [496].

The storage and computational overhead of embedding high-dimensional features is a significant practical bottleneck. Attaching a high-dimensional CLIP feature to each of millions of Gaussians inflates scene sizes to gigabytes, hindering storage and deployment. This has spurred a wave of research into feature compression and compact representation. One common approach is dimensionality reduction through specialized networks. For instance, [497] not only addresses multiview inconsistency by associating SAM prompt density with camera distance but also learns a granularity factor to guide which 2D features are most consistent for distillation. More advanced methods treat the problem as a rate-distortion challenge. Notably, [498] introduces a paradigm shift by decoupling field construction from storage. It uses a closed-form transmittance-weighted lift to create spatially grounded semantic anchors without per-scene training, and then compresses the per-Gaussian binding to these anchors using a spatially predictive entropy coder. This approach achieves sub-megabyte storage while maintaining high accuracy, demonstrating the power of treating semantic field storage as a data compression problem.

Further avenues for compactness include parametric feature grids and codebook-based learning. [499] proposes a semantic feature grid to save multi-view CLIP features and then maps the grids to low-dimensional features for training, enabling high-quality and real-time open-vocabulary queries. Similarly, [500] introduces extremely low-dimensional semantic bottleneck features processed by a multi-resolution hash encoder, achieving significant improvements in runtime and memory efficiency for large-scale scenes. Codebook-based approaches offer another compelling alternative by learning a shared set of semantic primitives, or “words,” to compose semantics. For example, [501] introduces a cross-attention module with learnable codebooks that encode view-independent semantic embeddings, ensuring intra-instance feature similarity while reducing memory. In the domain of 3D segmentation, [502] employs a Contrastive Codebook Learning module to distill discriminative semantic features by enforcing intra-class compactness and inter-class distinctiveness. The integrate-and-fire model of

codebooks is pushed further in [503], which encodes each Gaussian’s category with a single integer via binary-to-decimal mapping, drastically reducing memory and enabling a progressive training strategy for fine-grained segmentation.

Given the semantic sparsity in a scene, a major insight is that per-Gaussian high-dimensional features are often unnecessary. This has led to the decoupling of geometry and semantics, a theme explored in several cutting-edge works. [504] capitalizes on this by using a sparse set of semantic Gaussians to represent language information, using only 5% of the language embeddings compared to dense methods. The “extrinsic paradigm” represents a bold step in this direction, as exemplified by [505]. Instead of embedding features, ExtrinSplat clusters Gaussians into 3D object groups and uses a Vision-Language Model to generate lightweight textual hypotheses, creating an extrinsic index layer that supports polysemy and reduces adaptation time from hours to minutes. This is complemented by object-level aggregation methods like [506], which bypass rendering for semantics by using predecomposed object-level Gaussians and aggregating multiview CLIP features into a “bag of embeddings” per object. A training-free framework for scalable retrieval is presented in [507], which decouples semantic understanding from geometric representation by using an instance-level query mechanism and lifting 2D masks into 3D, enabling efficient retrieval on city-scale scenes.

To further improve fidelity, some methods align language features with surface geometry. [508] introduces a Language-Embedded Surface Field (LangSurf) that flattens language Gaussians onto object surfaces using geometry supervision, creating a more accurate 3D language field aligned with object boundaries. This is complemented by a Hierarchical-Context Awareness Module for extracting fine-grained features from SAM masks at different hierarchies. Finally, the concept of hierarchical understanding is crucial for a complete semantic representation. [509] achieves view-consistent 3D hierarchical semantics by constructing a 3D semantic tree with layered instance clustering, capturing both instance-wise and part-wise information. [510] similarly fosters a cohesive, context-aware hierarchical scene representation by disentangling segmentation and language field distillation, using “super-Gaussians” to lift and distill 2D language features efficiently. These hierarchical and geometry-aware representations not only refine the quality of the embedded semantics but also seamlessly bridge into the subsequent stage of open-vocabulary and instance-level segmentation, where such multi-scale, structured features are directly leveraged for object discovery and language-driven querying. In summary, the field of semantic feature distillation has evolved from naive feature injection to sophisticated methods involving compression, decoupling, codebook learning, hierarchical structuring, and a deeper understanding of the geometry-semantics relationship, addressing the core challenges of consistency, efficiency, and fidelity.

8.2 Open-Vocabulary Segmentation and Object-Level Perception

Open-vocabulary and instance-level segmentation within 3D Gaussian Splatting (3DGS) represents a critical frontier in scene understanding, moving beyond closed-set paradigms toward flexible, language-driven perception. Building upon the semantic feature distillation techniques discussed previously, which establish the representational substrate of language-aligned features within Gaussian primitives, this subsection addresses the higher-level challenge of organizing those features into discrete, queryable entities. The explicit, primitive-based nature of 3DGS provides a unique substrate for these tasks, enabling methods that range from class-agnostic grouping to hierarchical semantic decomposition. A fundamental challenge lies in reconciling the continuous, appearance-driven nature of Gaussian primitives with the discrete, semantic entities that define objects and their parts.

A dominant paradigm in this area is the aggregation of class-agnostic masks followed by semantic labeling, which directly leverages the distilled feature fields described in the previous subsection.

Early approaches in this vein often rely on distilling features from 2D foundation models into the 3D representation. For instance, the method proposed in [511] introduces a zero-shot approach that first generates class-agnostic 3D instance masks, then aggregates per-mask features via multi-view fusion of CLIP-based image embeddings. This decoupling of geometric grouping from semantic classification allows the system to segment arbitrary objects, including those in the long-tail distribution, by querying the aggregated features with free-form text. Building on this idea, some methods have proposed decoupled pipelines to enhance modularity and adaptability. The DCSEG framework [512] learns class-agnostic features through contrastive supervision from a 2D instance proposal network, clusters these features into coarse 3D masks, and then assigns semantic labels using a 2D open-vocabulary segmentation model. This design not only provides a plug-and-play interface for swapping different modules but also ensures multi-object instance segmentation without extra cost, leveraging rich 3D geometry for robust scene understanding. A key insight from such decoupled systems is the recognition that the semantic quality of the final segmentation is heavily dependent on the chosen 2D backbone, highlighting a trade-off between flexibility and fidelity.

To address the limitations of purely feature-embedding paradigms, recent work has proposed an “extrinsic” paradigm that decouples geometry from semantics at a more fundamental level, resonating with the decoupling strategies introduced in the previous subsection. The ExtrinSplat framework [505] argues that embedding high-dimensional features directly into Gaussians leads to geometry-semantic inconsistency, semantic bloat, and rigidity. Instead, it clusters Gaussians into multi-granularity, overlapping 3D object groups. A Vision-Language Model (VLM) then interprets these groups to generate lightweight textual hypotheses, creating an extrinsic index layer. This approach natively supports polysemy and dramatically reduces storage overhead and scene adaptation time, outperforming embedding-based frameworks on open-vocabulary object selection and semantic segmentation tasks.

For achieving precise instance-level discrimination, many methods adopt bottom-up aggregation strategies that merge smaller geometric or semantic units. The use of superpoints, which are geometrically and semantically coherent regions, has proven particularly effective. PanoGS [513] proposes language-guided graph cuts to group Gaussian primitives into super-primitives, which are then clustered into 3D-consistent instances using graph clustering with SAM-guided edge affinity. This synergy between geometric structure and language cues addresses the challenge of distinguishing instances in complex scenes. Expanding on this, training-free methods have been developed to construct superpoint graphs directly from Gaussian primitives, aligning with the efficiency-focused trends in feature representation. The THGS framework [514] partitions the scene into spatially compact and semantically coherent regions, forming view-consistent 3D entities. An efficient reprojection strategy lifts 2D semantic features onto these superpoints, avoiding costly multi-view iterative training. This representation naturally supports hierarchical understanding, enabling both coarse- and fine-grained open-vocabulary perception and achieving state-of-the-art segmentation performance with significantly faster semantic field reconstruction.

Beyond mere object-level segmentation, the field is increasingly addressing the challenge of part-level and hierarchical understanding, which directly sets the stage for the scene-level reasoning and graph-based querying discussed in the following subsection. This requires methods that can reason across multiple scales of granularity. The H2G framework [515] introduces a hyperbolic affinity field for hierarchical 3D grouping. By interpreting foundation-model affinities through Dasgupta’s objective for similarity-based hierarchical clustering, it derives semantically organized tree supervision. This supervision is distilled into a single Lorentz hyperbolic feature field, which is well-suited for representing tree-like branching structures, allowing for semantic hierarchical grouping grounded in 2D foundation-model knowledge. This approach moves beyond flat object-centric segmentation to recover scene groups from fine object parts to complete objects. Similarly, the

Search3D method [516] constructs hierarchical open-vocabulary 3D scene representations, enabling 3D search at multiple levels of granularity, including fine-grained object parts, entire objects, or regions described by attributes like materials. By shifting toward a more flexible search paradigm, it moves beyond explicit object-centric queries to support a wider range of user intents, bridging the gap toward the interactive querying paradigms of the next subsection.

The integration of large language models (LLMs) and multimodal models is a major trend driving progress in this area, particularly for referring expression grounding and complex query understanding, foreshadowing the reasoning capabilities explored in the subsequent subsection. A key limitation of methods relying solely on CLIP is their difficulty with compositional and functional queries. To address this, OpenTrack3D [517] replaces the CLIP-based classifier with a multimodal large language model (MLLM), significantly enhancing compositional reasoning for complex user queries. This is combined with a novel visual-spatial tracker that constructs cross-view consistent object proposals online from an RGB-D stream, making the pipeline applicable to mesh-free and unstructured environments. The broader impact of MLLMs is evident in works like Group3D [518], which integrates semantic constraints directly into the instance construction process for open-vocabulary 3D detection. By maintaining a scene-adaptive vocabulary derived from an MLLM and organizing it into semantic compatibility groups, the method ensures that 3D fragments are merged only when they satisfy both semantic compatibility and geometric consistency, mitigating over-merging and fragmentation errors.

In the context of open-ended settings, where predefined class names are absent, methods must be capable of discovering and naming objects autonomously. The OE3DIS method [519] formalizes this as a novel problem, eliminating the need for predefined class names during testing. By leveraging 2D multimodal large language models, it aims to generate both the geometric masks and their associated class names, evaluated with an open-ended score that assesses both semantic and geometric quality. This represents a shift towards truly autonomous scene understanding, laying the groundwork for the autonomous reasoning and embodied interaction discussed in the following subsection.

Finally, addressing the challenge of semantic inconsistency and granularity mismatch across views is crucial for robust segmentation, a concern that also permeates the feature distillation methods of the previous subsection. The CAGS framework [495] tackles the issue of cross-view granularity inconsistency caused by 2D segmentation models like SAM, where an object may be segmented as a single entity in one view and multiple parts in another. By constructing local graphs to propagate contextual features across Gaussians and employing mask-centric contrastive learning, CAGS smooths SAM-derived features and reduces fragmentation errors, enabling more cohesive object reasoning. Similarly, COS3D [520] proposes a collaborative prompt-segmentation framework that integrates complementary language and segmentation cues through a collaborative field comprising an instance field and a language field. Its adaptive language-to-instance prompt refinement promotes high-quality prompt-segmentation inference, demonstrating leading performance on standard benchmarks and showing potential for applications like hierarchical segmentation and robotics, thus providing the structured, entity-level understanding that serves as the foundation for the scene graph construction and reasoning techniques to follow.

8.3 Scene Understanding, Reasoning, and Interactive Querying

8.3 Scene Understanding, Reasoning, and Interactive Querying

Building upon the capabilities of open-vocabulary segmentation and semantic-aware editing, the field of 3D Gaussian Splatting is now advancing toward higher-level scene understanding, reasoning, and interactive querying. While the embedding of features and the segmentation of objects enable perception, true spatial intelligence requires the ability to infer relationships, answer complex queries,

and ground natural language instructions in the structured representation of a 3D environment. This subsection examines how 3DGS is being integrated with scene graphs, object attributes, and spatial relationships to facilitate high-level reasoning and interactive querying, spanning from language-guided retrieval to embodied navigation and manipulation.

A central theme in this domain is the construction of hierarchical 3D scene graphs (3DSGs) that organize the environment at multiple levels of abstraction, moving beyond object-centric maps to include rooms, floors, and abstract concepts. Frameworks like [521] leverage Large Language Models to annotate nodes across these hierarchies, creating rich metric-semantic representations. For large environments, efficient management of these graphs is essential, especially as their serialization can easily exceed the context windows of LLMs. To address this, [522] proposes a hierarchical graph structure with keyframe-selected multi-modal information, coupled with a retrieval-augmented generation (RAG) pipeline to extract relevant context, enabling the processing of complex queries without overwhelming the reasoning model. Relatedly, the work on [523] introduces an asynchronous dual-stream architecture that incrementally builds a RAG-ready knowledge base, decoupling geometric segmentation from semantic reasoning to efficiently handle complex embodied queries. This approach underscores the importance of scalable and updatable knowledge structures for practical navigation and interaction.

The integration of scene graphs with large language models forms the basis of powerful reasoning pipelines. For instance, [524] constructs a learnable representation of a 3D scene graph that explicitly incorporates semantic relationships and feeds it into an LLM for vision-language tasks, demonstrating that relational information improves grounding performance over purely geometric approaches. Similarly, [525] tackles the scalability issue by proposing a structured interface—specifically, the Cypher query language—as a tool for LLMs to retrieve task-relevant information from a graph database. This approach significantly outperforms naive serialization of the scene graph for instruction following and question answering, showcasing a scalable pattern for grounding language in complex environments.

To bridge the gap between abstract reasoning and physical perception, methods have been developed to ground natural language commands directly into the scene graph structure. [526] investigates the impact of adding open- or closed-vocabulary spatial relations to 3DSGs for grounding natural language commands. Their study reveals that explicit spatial relations improve the ability of LLMs to ground objects, highlighting the value of relational information in disambiguating targets. Extending this to real-world navigation, [527] constructs a multi-level semantic scene graph, termed an “Embodied Graph,” that integrates map data with object entities to serve as a retrievable knowledge base for semantic navigation. This system explicitly handles dynamic objects across varying spatial granularities, demonstrating the robustness of 3DSG-based reasoning in challenging outdoor environments. The importance of relational knowledge is further emphasized in [528], which learns to generate 3D semantic relations and demonstrates that modeling object relations improves search efficiency in real-world navigation tasks.

Addressing the inherent ambiguity of natural language is another key challenge. Approaches like [529] propose a modular system (BBQ) that constructs a 3D scene graph with metric and semantic edges, enabling the reasoning over complex queries that involve spatial relations, a task where simpler CLIP-based methods fail. Building on this foundation, [530] formalizes an Open-Vocabulary 3D Scene Graph (OVSG) framework for grounding a variety of entities based on free-form queries, facilitating contextual localization such as “the cup on the kitchen table.” For more intricate, multi-step reasoning, [531] (OVIGo-3DHSR) and [532] integrate hierarchical scene graphs with LLMs to handle queries involving spatial reference chains. This integration leverages inter- and intra-layer connections to enhance spatial context understanding and enable multistep reasoning for complex instructions.

Recent trends also highlight a move towards more dynamic, interactive, and physically grounded representations. For example, [533] presents a hybrid instance-aware 3D Gaussian dynamic scene graph system, integrating a probabilistic voxel grid with explicit Gaussians for robust instance fusion, and developing a “3D Gaussian Mind” that couples structural relations with spatial-semantic information for multimodal reasoning. The multi-modal aspect is also explored in [534], which argues that text-only relational edges lose critical visual evidence. Their M3DSG preserves visual cues by using dynamically assigned images as relational edges, and the system demonstrates state-of-the-art performance on challenging navigation benchmarks. Furthermore, the integration of physics is a growing area of interest. [535] unifies symbolic reasoning with structured 3D geometry to model kinematic and physical properties, inferring materials and articulations for downstream tasks like affordance prediction. Similarly, [536] and [537] demonstrate the construction of semantic-kinematic 3D scene graphs that encode actionable articulation information, enabling robots to not only see but also physically interact with their environment. These works collectively show a trajectory from static semantic mapping toward dynamic, physically-aware, and interactive 3D scene representations, positioning 3DGS as a central technology for next-generation embodied AI.

8.4 Semantic-Aware Editing and Manipulation

Semantic-aware editing and manipulation represent one of the most active and transformative application areas of 3D Gaussian Splatting (3DGS), leveraging the explicit, primitive-based nature of the representation to enable granular control over scene content that was previously difficult to achieve with implicit methods like NeRF. Unlike implicit representations, where editing requires navigating a continuous function, 3DGS allows for direct manipulation of individual Gaussians, facilitating a wide range of operations from simple recolorization to complex geometric deformations. This capability forms the practical foundation for the higher-level reasoning and interactive querying discussed in the previous subsection: while scene graphs and language models provide semantic understanding, editing tools translate that understanding into actionable, user-driven modifications of the environment. Research in this area is broadly characterized by text-guided editing, drag-based manipulation, stylization, and interactive selection tools, often working in concert to deliver a seamless and intuitive user experience.

A dominant paradigm for text-guided editing adopts a render-edit-optimize pipeline, where images are rendered from a 3DGS scene, edited using a 2D diffusion model, and then used to fine-tune the 3D representation. A critical challenge in this pipeline is ensuring multi-view consistency, as independent 2D edits can introduce conflicting information across views, leading to artifacts and mode collapse. To address this, frameworks like EditSplat propose a Multi-view Fusion Guidance (MFG) module to incorporate multi-view information directly into the diffusion process, alongside an Attention-Guided Trimming (AGT) strategy that leverages the explicit nature of 3DGS to prune irrelevant Gaussians, enhancing optimization efficiency and enabling precise local edits [538]. Similarly, GaussCtrl achieves multi-view consistency by using depth-conditioned editing, which enforces geometric consistency across views by leveraging the natural consistency of depth maps, and by employing attention-based latent code alignment to unify the appearance of edited images [539]. Another approach, VcEdit, incorporates a Cross-attention Consistency Module and an Editing Consistency Module to reduce inconsistencies in the guidance images themselves, thereby mitigating mode collapse during the subsequent 3D optimization [540]. For more localized control, TIGER proposes a bottom-up language aggregation strategy to create denser language-embedded Gaussians for accurate retrieval, and a Coherent Score Distillation (CSD) that combines 2D and multi-view diffusion models to produce consistent edits with finer details [541].

Beyond simple appearance changes, enabling geometric modifications such as addition, removal,

replacement, and repositioning is a key goal. GaussianEditor enhanced precision by introducing Gaussian semantic tracing to track the editing target, Hierarchical Gaussian splatting (HGS) for stable results, and editing strategies for efficient object removal and integration [542]. A more streamlined, fully 3D-based paradigm is proposed by 3DSceneEditor, which operates directly on Gaussians using a pre-trained instance segmentation model and zero-shot grounding with CLIP to align target objects with user prompts, enabling operations like object addition, repositioning, recoloring, replacing, and removal without the need for iterative 2D-to-3D projections [543]. To overcome the limitations of text prompts alone, some methods integrate generative models more deeply. For instance, SVGS uses a single-view editing strategy grounded in multi-view diffusion models, reconstructing 3D scenes from consistent views to balance editing consistency and efficiency [544]. For complex, multi-part scenes, custom frameworks have been developed for specific tasks, such as G²Editor for driving videos, which uses a 3D Gaussian representation of the object as a dense prior for pose control and a scene-level 3D bounding box layout to reconstruct occluded areas, supporting object repositioning, insertion, and deletion in a unified framework [545].

Another major strand of research focuses on drag-based editing, which provides a more intuitive and spatially precise alternative to text prompts for geometric deformation. These methods often rely on user-specified control points and masks to define the edit. 3DGS-Drag bridges the gap between deformation-based and 2D-editing-based methods by leveraging deformation guidance for consistent geometric modifications and diffusion guidance for content correction, enabling a wide range of edits including motion change, shape adjustment, and inpainting [546]. DragGaussian also enables drag-style manipulation on pre-trained 3D Gaussian object models, using diffusion models for multi-view consistent editing [547]. Building on the concept of as-rigid-as-possible (ARAP) deformation, ARAP-GS directly applies ARAP constraints to 3D Gaussians, preserving shape coherence and visual continuity during drag-driven geometric transformations, and incorporates a diffusion prior to enhance visual quality [548]. Similarly, DYG integrates an implicit triplane representation as a geometric scaffold to overcome the sparsity of 3DGS in editing regions and incorporates a drag-based Latent Diffusion Model to enable flexible, multi-view consistent editing [549]. To handle more aggressive and topology-changing edits, MVDrag3D employs a multi-view diffusion model as a generative prior, followed by a reconstruction model to create 3D Gaussians, and uses view-specific deformation networks to correct misalignments [550]. For object-level edits in images that can be lifted to 3D, ObjectMorpher lifts target instances into editable 3DGS using an image-to-3D generator, allowing fast manipulation via control points with ARAP constraints for non-rigid deformations [551].

In addition to text and drag, stylization and reference-based editing form an important category. ReGS adapts 3DGS for reference-based stylization by introducing a texture-guided control mechanism that adaptively adjusts local Gaussians to a new geometric arrangement, guided by texture clues and regularized by scene depth to preserve the original geometry [552]. For more generalized and creative edits, frameworks often incorporate 2D diffusion models to generate novel content, but must carefully manage the editing region to preserve the scene. The Mastering Regional 3DGS method leverages 2D diffusion editing to identify modification regions, followed by inverse rendering for 3D localization and a refined coarse 3DGS initialization to support iterative, view-consistent editing [553]. Similarly, GSEditPro introduces an attention-based progressive localization module to add semantic labels to each Gaussian during rendering, enabling precise localization of editing areas based on their relevance to the editing prompts [554]. To preserve non-target regions and reduce edit leakage, TransSplat formulates language-driven editing as a multi-view unbalanced semantic transport problem, explicitly characterizing the semantic relationship between edited 2D evidence and 3D Gaussians and using transport residuals to suppress erroneous edits [555].

Beyond static scenes, semantic-aware editing is increasingly applied to dynamic scenes and 4D

representations, extending the scope of manipulation from spatial structures to spatio-temporal coherence. SemDynReg uses SAM to extract semantic features and construct an object-ID map, guiding a semantics-aware regularization that imposes consistency constraints on the deformation parameters of Gaussians belonging to the same object, improving object-level deformation consistency in dynamic 3DGS rendering [556]. Mono4DEditor extends text-driven editing to 4D scenes reconstructed from monocular video by augmenting 3D Gaussians with quantized CLIP features, enabling point-level localization and targeted edits using diffusion-based video editing with flow and scribble guidance for temporal coherence [557]. For more complex and interactive control, 4DGS-Craft introduces a 4D-aware InstructPix2Pix model that incorporates 4D VGGT geometry features to ensure view and temporal consistency, along with a Gaussian selection mechanism to preserve non-edited regions and an LLM-based module for user intent understanding to decompose complex instructions into atomic operations [558]. Similarly, Catalyst4D transfers high-quality 3D edits to dynamic 4D Gaussian scenes using an Anchor-based Motion Guidance (AMG) module that establishes correspondences between original and edited Gaussians via optimal transport, and a Color Uncertainty-guided Appearance Refinement (CUAR) module to preserve temporal consistency [559].

Finally, the development of interactive and AI-assisted selection tools is crucial for practical Gaussian-level control, serving as the user-facing interface that bridges the gap between high-level editing intents and the underlying primitive-based representation. ArtisanGS presents an interactive suite of tools that combines a fast AI-driven method for propagating user-guided 2D selection masks to 3DGS with flexible manual selection and segmentation tools, enabling users to achieve virtually any binary segmentation of an unstructured 3DGS scene for downstream editing tasks [560]. The importance of accurate 3D masks is underscored by work on robust part-level editing. RoMaP proposes a robust 3D mask generation module using spherical harmonics to model view-dependent label variations and a regularized Score Distillation Sampling (SDS) loss with an anchor loss to confine modifications to the target region while preserving context [561]. Similarly, AG²aussian uses an anchor-graph structure to organize semantic features, enabling clean and accurate instance-level Gaussian selection for interactive click-based queries, object removal, and physics simulation [562]. Furthermore, TASE projects pretrained 2D semantic features into a truncation-aware embedding space, enabling text-driven edits with explicit control over how strongly edits adhere to the original scene content [563]. Look Before You Edit introduces a method for attention-guided camera placement and multi-view alignment, probing the diffusion model’s attention to find the optimal distance for editing cameras and using multi-view attention alignment to ensure the same edit is applied consistently across views [564]. These tools and techniques collectively position 3DGS as a highly versatile and controllable medium for semantic-aware editing, blurring the lines between professional content creation tools and intuitive, user-friendly interfaces, and providing the executable substrate upon which the scene understanding and reasoning capabilities described in the previous subsection can be deployed for real-world interaction.

9 Inverse Rendering, Relighting, and Advanced Appearance Modeling

9.1 Physically-Based Shading and Material Decomposition in Gaussian Splatting

Physically-based inverse rendering within the 3D Gaussian Splatting (3DGS) framework represents a significant paradigm shift from view synthesis to the disentanglement of intrinsic scene properties. The core objective is to jointly recover geometry, material attributes (such as albedo, metallic, and roughness), and illumination from multi-view images, enabling downstream applications like relighting and material editing. This subsection reviews the fundamental extensions that have

enabled this decomposition, focusing on shading pipelines, approximation strategies, and the integration of physically-based rendering objectives.

A primary challenge in adapting 3DGS for inverse rendering is that the standard rasterization pipeline is designed for forward shading, where the final color is computed per-primitive and composited via alpha-blending. This approach is insufficient for physically-based decomposition because it entangles lighting and material properties within the view-dependent spherical harmonic coefficients. To address this, a significant body of work has adopted deferred shading pipelines. In these frameworks, the rasterization pass renders geometric and material attributes—such as normals, albedo, and roughness—into intermediate G-buffers. A subsequent physically-based shading stage then computes the final pixel color using a rendering equation that explicitly models light transport. This separation is crucial as it allows for the independent optimization of material parameters and lighting while maintaining the real-time rendering capabilities of 3DGS. For instance, DeferredGS decouples texture and lighting by modeling illumination via a learnable environment map and defining attributes like texture parameters and normal direction directly on Gaussians, enabling more realistic relighting through deferred shading [565]. Similarly, Phys3DGS identifies problems incurred by hidden Gaussians in standard volume rendering and proposes adopting deferred rendering alongside a hybrid mesh-Gaussian representation to improve inverse rendering quality [566].

To make the rendering equation tractable for optimization, many methods employ the split-sum approximation for specular reflections. This technique approximates the integral of the Bidirectional Reflectance Distribution Function (BRDF) and the environment lighting into a pre-filtered environment map and a separable BRDF shading term, significantly simplifying the computation. Relightable 3D Gaussian represents the 3D scene as a set of relightable 3D Gaussian points, where each point is associated with normal direction, BRDF parameters, and incident lights, optimizing the scene through 3D Gaussian Splatting while decomposing BRDF and lighting via physically-based differentiable rendering with an innovative point-based ray-tracing approach for visibility baking [567]. Building on this, GS-IR extends 3DGS to estimate scene geometry, surface material, and environment illumination by proposing an efficient optimization scheme that incorporates depth-derivation-based regularization for normal estimation and a baking-based occlusion method to model indirect lighting [568]. Furthermore, the use of 2D Gaussian surfels, as opposed to volumetric 3D Gaussians, has been shown to improve surface alignment and material decomposition [569]. For example, MaterialClusterGS represents scene materials using a compact global palette of shared BRDF prototypes assigned via a continuous spatial material field, jointly optimizing the material field, palette prototypes, and environment lighting under a physically-based rendering objective, which recovers compact and spatially coherent attributes for editing and relighting [570].

However, the simplified shading models described above are inherently limited by their reliance on direct lighting and simplified visibility assumptions. They often fail to capture complex light transport phenomena such as inter-reflection, self-occlusion, and global illumination, which are crucial for achieving photorealistic relighting. Beyond the split-sum approximation, a body of work integrates Monte Carlo path tracing to model the full rendering equation, including inter-reflection and global illumination. This approach provides more physically accurate light transport at the cost of increased computational complexity. GI-GS calculates indirect lighting using efficient path tracing with deferred shading, rendering a G-buffer first to capture geometry and material properties, then performing physically-based rendering for direct lighting and using a lightweight path tracing method for indirect lighting, which achieves better novel view synthesis and relighting under any given lighting conditions [571]. Similarly, IRGS leverages differentiable 2D Gaussian ray tracing to compute incident radiance on the fly, applying the full rendering equation without simplification and demonstrating the capability to accurately model complex inter-reflection effects [572]. However, the

variance from Monte Carlo sampling poses a significant challenge for gradient-based optimization. To mitigate this, PTIR-GS proposes a splatting-free path-traced inverse rendering framework where forward light transport and backward gradient propagation are defined within a unified ray-tracing pipeline, ensuring consistency between the forward and backward passes and improving material inversion [573]. Another strategy, employed by GOGS, uses a two-stage framework based on 2D Gaussian surfels, where the first stage establishes robust surface reconstruction using split-sum approximation and geometric priors, and the second stage performs material decomposition by leveraging Monte Carlo importance sampling of the full rendering equation, modeling indirect illumination via differentiable 2D Gaussian ray tracing [574]. A more holistic approach to inter-reflection is presented in IRGS++, which introduces a unified robust and efficient Gaussian inverse rendering framework that employs differentiable 2D Gaussian ray tracing for visibility and indirect radiance queries during transport-aware optimization, making the physical core of Gaussian inverse rendering more grounded [575].

The parameterization of materials within Gaussian primitives is another area of active research, involving the assignment of spatially-varying BRDF properties. These properties are often fine-tuned per-primitive, which can be under-constrained. To address this, OMG introduces a novel approach that augments the opacity term by introducing a neural network that takes material properties as input, providing a physically correct activation function based on radiative transfer principles. This adds a dependency of opacity on material cross-section, facilitating constraint for material optimization and improving material modeling across different baselines [576]. In contrast, spatially-varying representations have been developed to allow per-Gaussian spatially varying parameters, addressing the limitation of constant material parameters in each Gaussian. SVG-IR proposes a Spatially-varying Gaussian (SVG) representation with a SVG splatting scheme akin to vertex/fragment shading, which, when combined with a physically-based indirect lighting model, significantly improves relighting quality by allowing for more detailed material and normal variation across a single primitive [577]. Furthermore, the use of explicit geometry guidance can enhance light transport modeling. GeoSplatting augments 3DGS by differentially constructing a surface-grounded 3DGS from an optimizable mesh, leveraging well-defined mesh normals and the opaque mesh surface to facilitate mesh-based ray tracing for efficient, occlusion-aware light transport calculations [578]. Finally, the need for robustness in sparse-view settings is addressed by GAINS, which introduces a two-stage inverse rendering framework that leverages foundation models as priors to stabilize geometry and material estimation, unifying these priors with physically-based representations in an optimization scheme to improve material parameter accuracy under sparse captures [579].

In summary, the field of physically-based shading and material decomposition within 3DGS is characterized by a progression from simplified shading models to more physically faithful light transport simulations. The evolution from forward to deferred shading pipelines has been pivotal in enabling the disentanglement of scene properties. Current state-of-the-art methods integrate advanced techniques like split-sum approximations, Monte Carlo ray tracing, and complex material parameterizations, all unified under differentiable objectives that allow for high-quality reconstruction, relighting, and material editing. The challenge remains to balance the computational cost of physically-based light transport with the real-time capabilities of 3DGS, a balance that continues to drive innovation in hybrid and adaptive rendering strategies. These fundamental decomposition techniques provide the essential substrate upon which the more sophisticated illumination modeling approaches—spanning HDR environment estimation, explicit emitter representation, and global illumination inference—are built, as explored in the following subsection.

9.2 Advanced Lighting Estimation, Global Illumination, and Indirect Transport Modeling

Advanced lighting estimation and global illumination modeling represent one of the most active and challenging frontiers in 3D Gaussian Splatting research. While the base 3DGS framework excels at novel view synthesis through view-dependent spherical harmonics, it fundamentally bakes illumination into the radiance field, preventing relighting, material editing, and physically consistent scene manipulation. Building upon the physically-based shading and material decomposition techniques discussed in the previous subsection—which established the disentanglement of geometry, materials, and lighting—this subsection surveys the sophisticated illumination modeling techniques that complete this decomposition by explicitly estimating and simulating the lighting itself. These techniques span HDR environment estimation, explicit emitter representations, spatially-varying lighting fields, and the critical challenge of indirect illumination through path tracing, radiance transfer fields, and inter-reflection modeling.

HDR Environment Map Estimation and Lighting Decomposition. A foundational step toward relightable 3DGS is the estimation of high-dynamic-range illumination that can be separated from scene materials. Early approaches in this direction leverage parametric lighting models to capture scene energy. [580] bridges the gap between parametric light models and realistic panorama generation by combining spherical Gaussian parameters with GAN-based LDR panorama extrapolation, yielding HDRI-ready lighting that supports editable light direction, intensity, and count. For street scenes, [581] proposes a hybrid representation containing an HDR sky dome for extreme solar intensity alongside a volumetric lighting model for spatially-varying appearance, enabling differentiable virtual object insertion. More recently, [582] introduces the concept of using HDR Gaussian Splats themselves as light source representations, marking the first time regular images serve directly as light sources in a 3D renderer through a two-stage process of diffusion-based dynamic range enhancement followed by 3D lighting modeling. In the domain of outdoor scenes, [583] employs a two-stage pipeline where monocular normal priors guide 2D Gaussian Splatting geometry reconstruction, followed by a hybrid lighting model combining spherical Gaussian functions for high-frequency sunlight with spherical harmonics for low-frequency skylight, achieving state-of-the-art relighting accuracy for unbounded outdoor environments. Similarly, [298] addresses the exposure inconsistency problem in autonomous driving data by jointly decomposing a view-invariant linear HDR radiance field, per-view exposure scales, and tone-mapping functions directly from LDR images, enforcing relative-exposure consistency and HDR-domain regularization based on the physical image-formation process.

Explicit Area Emitters and Spatially-Varying Lighting. A key limitation of environment-map-based approaches is their inability to model local light sources with physical spatial extent. [584] directly tackles this by explicitly modeling local area emitters within a relightable Gaussian Splatting representation, integrating multiple importance sampling with targeted regularization in a differentiable deferred rendering pipeline to accurately simulate local light transport, produce correct light attenuation and realistic shadows, and enhance downstream tasks including controlled relighting and virtual object insertion in scenes with complex local lighting. For indoor scenes with spatially varying illumination, [585] proposes a Volumetric Spherical Gaussian representation that parameterizes exitant radiance on a voxel grid, paired with a physics-based differentiable renderer that formulates energy-conserving image formation for joint training of albedo, normals, depth, and lighting without requiring ground-truth HDR data. In the context of large-scale indoor environments, the Texture-based Lighting (TBL) representation introduced in [586] combines 3D mesh with HDR textures to efficiently model direct and infinite-bounce indirect lighting of entire scenes, complemented by a hybrid lighting representation with precomputed irradiance to improve

optimization efficiency and reduce rendering noise.

Path Tracing and Global Illumination Integration. Modeling indirect illumination—light that reaches a surface after one or more bounces—remains the most challenging aspect of inverse rendering, as it requires evaluating visibility and secondary light transport. This challenge connects directly to the Monte Carlo path tracing and inter-reflection techniques introduced for material decomposition in the previous subsection, where methods like GI-GS and PTIR-GS were first discussed for their role in improving specular and indirect light estimation during decomposition. Here, we focus on their broader implications for illumination modeling in its own right. Several methods have pushed 3DGS toward physically grounded global illumination. [571] renders a G-buffer to capture geometry and material properties, performs PBR only for direct lighting, then computes indirect lighting through lightweight path tracing using the G-buffer and previous rendering results, effectively modeling indirect illumination under arbitrary lighting conditions. [573] goes further by defining a splatting-free path-traced inverse rendering framework where both forward light transport and backward gradient propagation operate within a unified ray-tracing pipeline, introducing a path-space equivalent interaction model for overlapping Gaussian primitives that makes Monte-Carlo path tracing unbiased for the induced light-transport integral while replaying pathwise gradients over the same ray-traced interactions. This pipeline consistency, together with the full rendering equation and ray-traced visibility, yields more plausible shadows, reflections, and relighting under global illumination. For dynamic scenes, [587] formulates a surface light transport model for 3D Gaussians and combines a fast compound stochastic ray-tracing algorithm with an optimized rasterizer, achieving over 40 fps with interactive material editing and dynamic multi-light setups while capturing mutual multi-bounce transport between Gaussians and meshes.

Inter-Reflection and Radiance Transfer Modeling. Inter-reflections among multiple objects introduce higher-order lighting effects that simple direct illumination models cannot capture. While the previous subsection reviewed inter-reflection methods such as IRGS and IRGS++ primarily from the perspective of decomposing materials from reflections, this subsection re-examines these same techniques through the lens of illumination estimation itself—where they serve as essential tools for recovering the incident radiance field that characterizes a scene’s lighting environment. [572] applies the full rendering equation without simplification, computing incident radiance on the fly through differentiable 2D Gaussian ray tracing, supported by an efficient optimization scheme for Monte Carlo sampling and a novel strategy for querying indirect radiance during relighting. Its successor, [575], extends this with metallic-aware material modeling, robust reflective initialization for glossy and specular surfaces, multiple importance sampling and denoising for finite-sample rendering stability, and mesh-based secondary-attribute queries to reduce relighting costs. The concept of radiance transfer fields is explored in [588], which combines forward rendering for radiance transfer with deferred rendering for reflections to separate high-frequency and low-frequency appearances, mitigating floating artifacts from spherical harmonic overfitting while refining BRDF and lighting decomposition. [589] takes a different approach, locating surface points through refined sphere tracing and explicitly tracing incoming indirect lights based on reflection, estimating each identified light through an efficient neural network and employing Leibniz’s integral rule to resolve non-differentiability at boundary lights. [590] introduces Monte Carlo sampling-based path tracing with neural radiance caching for indirect illumination, using spherical Gaussians for smooth environment illumination and a radiance consistency constraint between implicit neural radiance and path tracing results of unobserved rays to improve inter-reflection decomposition. For scenes with participating media, [591] estimates direct illumination via ray tracing and computes indirect illumination with spherical harmonics, avoiding lengthy indirect bounces while preventing energy loss, and [592] formalizes kernel-based volumetric primitives with closed-form transmittance and free-flight distance sampling within any off-the-shelf volumetric path tracer, supporting full path-tracing functionality

including scattering, relighting, and complex camera models. Multi-bounce path tracing with reservoir sampling is explored in [593], which extracts explicit triangular mesh geometry and employs Monte Carlo integration with reservoir sampling to handle noise in the estimation of self-shadowing and internal reflections. In the co-located light-camera setup, [594] integrates neural implicit surfaces with a neural radiance cache for global illumination approximation, introducing a dynamic radiance cache that adapts to sharp lighting discontinuities from near-field motion and a surface-angle-weighted radiometric loss to suppress specular artifacts.

Hybrid Rasterization-Ray Tracing Approaches. Given the complementary strengths of real-time rasterization and physically accurate ray tracing, a number of hybrid approaches have emerged. These approaches parallel the hybrid mesh-Gaussian strategies discussed in the previous subsection for material decomposition (e.g., GeoSplatting), but here the focus shifts toward leveraging this hybridity for efficient and accurate illumination estimation. [595] uses a neural field for primary rays while employing an explicit mesh reconstructed from the underlying field for secondary rays that produce cast shadows and higher-order lighting effects, enabling photorealistic relighting with specular and shadow effects on outdoor datasets. [578] augment 3DGS with explicit mesh guidance by differentially constructing a surface-grounded 3DGS, leveraging well-defined mesh normals and opaque surfaces to facilitate mesh-based ray tracing for occlusion-aware light transport. [596] combines pre-trained neural fields with a Neural Radiance Cache for indirect illumination, using differentiable marching tetrahedrons initialized from the SDF and single-bounce differentiable MC ray tracing to jointly optimize geometry, material, and light. [575] also adopts a hybrid strategy where mesh-based secondary-attribute queries reduce the cost of relighting under novel illumination while rasterization handles primary visibility. From a different angle, [597] proposes a generalizable 3D light transport embedding that approximates global illumination directly from 3D scene configurations without rasterized or path-traced cues, using a scalable transformer to model global point-to-point interactions and cross-attention for query aggregation at render time, with applicability extending to path guiding for glossy materials. These hybrid approaches demonstrate that the future of high-quality inverse rendering lies not in a single rendering paradigm but in the principled combination of complementary techniques, balancing physical accuracy with computational efficiency for practical deployment across indoor, outdoor, urban, and dynamic scenarios.

In summary, this subsection has traversed the landscape of advanced illumination modeling within 3DGS, from HDR environment estimation and explicit emitter representations to the intricate modeling of global illumination and inter-reflections. Together with the physically-based shading and material decomposition techniques established in the previous subsection, these illumination modeling approaches form the two pillars of inverse rendering in the 3DGS framework: the disentanglement of scene properties and the explicit recovery of the lighting that governs them. The specialized appearance modeling techniques for specular reflections, transparency, and subsurface scattering covered in the following subsection build upon this foundation, addressing the residual challenges in rendering complex real-world materials under the recovered illumination.

9.3 Complex Appearance Modeling: Specular Reflections, Transparency, and Subsurface Scattering

The standard 3D Gaussian Splatting (3DGS) framework excels at rendering diffuse and Lambertian surfaces but faces significant challenges when modeling complex optical phenomena such as specular reflections, transparency, and subsurface scattering. These phenomena are inherently view-dependent and require a nuanced understanding of light transport that goes beyond the simple alpha-blending of static Gaussian primitives. This subsection delves into specialized Gaussian-based techniques

developed to address these challenging appearances, categorizing them into three main areas: specular and anisotropic reflections, semi-transparent surfaces, and subsurface scattering.

A primary limitation of vanilla 3DGS in handling specular surfaces is its use of spherical harmonics (SH) to encode color. SH are low-frequency basis functions and are ill-suited for capturing the high-frequency, view-dependent variations characteristic of glossy reflections. To overcome this, researchers have explored enhancing the Gaussian primitives themselves. One line of work augments the standard Gaussian kernel with view-dependent opacity, allowing the Gaussian to become “visible” or “invisible” based on the viewing angle. This approach, exemplified by [598], explicitly models specular effects, thereby decoupling them from the diffuse components. Building on this, [599] introduces an additional symmetric matrix to modulate the opacity of each Gaussian based on the viewer’s perspective, enabling the model to suppress certain Gaussians for specific views and more accurately represent specular highlights without corrupting the scene’s overall integrity. These methods demonstrate that view-dependent opacity is a powerful and efficient mechanism for capturing reflective phenomena.

Another prominent strategy for modeling specular reflections involves extending the representation into higher dimensions. The concept of N-dimensional Gaussians (N-DG) inspired [600], which revisits the 6D spatial-angular representation. This method enhances both color and opacity representation by leveraging the additional directional information in 6D space, allowing for optimized Gaussian control and significantly improved modeling of view-dependent effects while maintaining the real-time rendering capabilities of the original 3DGS framework. Similarly, [601] proposes a hybrid model, Viewing Direction Gaussian Splatting (VDGS), which combines the explicit shape representation of Gaussians with a NeRF-based multi-layer perceptron (MLP) that takes Gaussian parameters and viewing direction as input to produce view-dependent changes in color and opacity. This hybrid approach better captures complex optical behaviors like shadows and reflections. In a different approach, [602] equips each 3D Gaussian with a universal latent feature descriptor, which is then fed into two parallel convolutional neural networks (CNNs) to decode specular and diffuse colors separately, before a view-dependent mask merges them. This allows for a more expressive and accurate decomposition of the scene’s appearance.

Further advancing beyond simple view dependence, some works aim for physically-based accuracy. The explicit modeling of reflection and transmission is tackled by frameworks like [603], which integrates a microfacet material model with ray tracing to jointly model specular reflection and transmittance using separate Gaussian primitives. This allows for the simultaneous reconstruction of distant reflections and objects behind semi-transparent surfaces. Similarly, [604] proposes a deferred shading method that leverages per-pixel reflection gradients to propagate nearly correct normal estimations across the surface of reflective objects, leading to high-quality specular effects while maintaining a rendering speed nearly identical to that of vanilla 3DGS. Anisotropic reflections, often seen on brushed metals, have also been a focus. While not strictly 3DGS, [605] demonstrates the power of Anisotropic Spherical Gaussian (ASG) distributions for modeling such appearances and has inspired similar techniques within the 3DGS community. This influence is seen in [606], which explicitly models surface normals and BRDF parameters, using ASG to approximate the normal distribution function under a microfacet model for efficient reconstruction of glossy surfaces.

Modeling transparency and semi-transparent surfaces presents another set of challenges. The order-dependent alpha-blending in vanilla 3DGS is a significant bottleneck, as it creates popping artifacts and fails to correctly represent the transmittance of light through multiple transparent layers. To address the issue of order-dependence, a key innovation is the use of moment-based order-independent transparency. The work [34] characterizes the density distribution along a camera ray using statistical moments, from which a continuous transmittance function can be reconstructed without the need for ray tracing or per-pixel sorting. This allows for physically accurate modeling of

light attenuation in complex translucent media, bridging the gap between rasterization and physical accuracy. Another framework, [71], tackles the complex interplay of reflection and transmission on semi-transparent specular surfaces by disentangling geometric occupancy from optical opacity. This factorization allows the representation to act as a surface for reflections and a volume for transmission, effectively handling phenomena that would otherwise cause blurry reflections or occluded backgrounds. The authors introduce a Specular-Aware Gradient Gating mechanism to prevent misleading gradients from specular regions from corrupting the transmission branch. For scenes containing planar transparent surfaces like glass, [607] combines 3D Gaussians with learnable reflection planes, modeling the transmitted and reflected components separately and blending them based on a Fresnel-based, view-dependent weighting scheme. Complex refraction, as seen in transparent objects with intricate shapes, is tackled by [608], which introduces transparent Gaussian primitives that enable specular refraction through a deferred refraction strategy and uses Gaussian light field probes (GaussProbe) to encode ambient light and nearby contents for efficient querying.

Finally, subsurface scattering (SSS) represents the most complex optical phenomena, where light penetrates a surface, scatters internally, and exits at a different point. This effect is crucial for realistic rendering of materials like skin, wax, and marble. Applying standard 3DGS directly to these materials often fails because its primitives are surface-based and cannot capture volumetric light transport. A promising direction is to augment surface Gaussians with an implicit volumetric representation for scattering. [609] proposes a framework that decomposes the scene into an explicit 3D Gaussian surface with a spatially varying BRDF and an implicit volumetric representation of the scattering component, optimized jointly via ray-traced differentiable rendering. A more physically grounded approach is presented in [610], which decomposes reflectance into diffuse, specular, shadow, and subsurface scattering components. It introduces a learnable dipole-based scattering module for subsurface transport, drawing from the classic dipole approximation in radiative transfer, and an occlusion-aware shadow formulation. This enables high-fidelity relighting and physically plausible decomposition. Building on this, the challenge of data acquisition for SSS is addressed by [611], which uses diffusion-based data augmentation and strong geometric priors like multi-view silhouette and depth consistency to reconstruct translucent objects from as few as ten images, highlighting a path toward practical deployment. The integration of physics and neural networks is also explored in [612], which, while not exclusively a 3DGS method, introduces a formulation that reconciles physically complete light transport with general-purpose neural optimization, estimating optical properties and the full light field as neural fields through a residual objective derived from the Radiative Transfer Equation. This philosophy of using neural fields for light transport is echoed in [613], which solves the rendering equation by minimizing the norm of its residual using neural networks to represent the radiance distribution. Together, these methods—encompassing view-dependent opacity, hybrid renderers, order-independent transmittance, and specialized scattering modules—represent a significant stride in expanding 3DGS from simple diffuse scenes to a wide array of complex, real-world appearances, complementing the physically-based illumination modeling discussed previously and laying the groundwork for the spectral extensions explored in the next section.

9.4 Spectral Extensions, Multispectral Rendering, and Cross-Modal Appearance Adaptations

The trichromatic RGB color space forms the foundation of most 3D Gaussian Splatting (3DGS) frameworks, yet it represents only a narrow window of the electromagnetic spectrum. Extending 3DGS beyond this conventional paradigm unlocks the ability to capture wavelength-dependent phenomena critical for scientific imaging, material analysis, remote sensing, and robust perception

under challenging conditions. This subsection surveys the emerging body of work that adapts 3DGS to multispectral and hyperspectral rendering, spanning new radiance representations, cross-modal scene reconstruction strategies, and data-efficient learning techniques. The progression from appearance modeling in the previous subsection to spectral extensions here is a natural one: while view-dependent effects and scattering phenomena operate within the visible spectrum, spectral methods generalize these concepts across wavelength bands, requiring analogous innovations in radiance representation, physical modeling, and data efficiency.

Multispectral and Hyperspectral Radiance Representations. The most direct approach to spectral extension involves augmenting each Gaussian primitive with a wavelength-aware color representation. In the multispectral domain, [614] proposes augmenting each Gaussian with spectral radiance encoded via per-band spherical harmonics, supervised under a dual-loss scheme that jointly optimizes RGB and multispectral signals. A notable finding of this work is that performing spectral-to-RGB conversion at the pixel level—rather than at the per-Gaussian level—retains richer spectral cues during optimization, leading to improved rendering quality on scenes with translucent materials and anisotropic reflections. Building on the observation that naive per-band optimization often produces poor reconstructions due to inconsistent geometry across spectral domains, [615] systematically evaluates three strategies: separate per-band reconstruction, split optimization (where RGB geometry is first optimized and then transferred to spectral bands), and joint optimization with an initial RGB-only warm-up phase. The authors find that the joint strategy significantly improves spectral reconstruction and even enhances RGB rendering quality through spectral cross-talk, suggesting that spectral information should be integrated directly into the spherical harmonics color components of each Gaussian.

For hyperspectral settings, where the number of spectral bands can be extremely large, [616] introduces wavelength-aware spherical harmonics combined with a Kullback-Leibler divergence-based spectral loss. This method addresses the challenge of reconstructing continuous spectral radiance fields from sparse hyperspectral observations and demonstrates state-of-the-art performance on real-world scenes from the Hyper-NeRF dataset. A complementary perspective is offered by [617], which diverges from per-channel optimization entirely. This work encodes multispectral information into a learned, compact per-splat feature embedding, decoded by a shallow multilayer perceptron to produce spectral color values. This neural color representation captures underlying spectral and spatial correlations across modalities without requiring cross-modal camera calibration, making it versatile across thermal, near-infrared, and other spectral domains. The authors demonstrate its practical utility in agricultural applications, where rendering vegetation indices such as the normalized difference vegetation index (NDVI) provides actionable insights into crop health.

Physically-Based Spectral Modeling and Spectral-Spatial Coupling. Beyond simple radiance encoding, several methods integrate physical models of spectral phenomena into the 3DGS optimization framework. [618] introduces a cross-spectral rendering framework that estimates reflectance and illumination per spectral band using an improved physically-based rendering approach for Gaussian splats. By jointly optimizing geometry, per-spectra reflectance, and lighting, this method achieves superior performance compared to both spectral NeRF baselines (e.g., XNeRF and SpectralNeRF) and non-spectral state-of-the-art methods, while also enabling spectral-aware scene editing such as style transfer and inpainting. For thermal-infrared modalities, [619] leverages two physics-based principles: Fourier’s law of heat conduction is integrated prior to alpha blending to model intensity interpolation caused by thermal conduction between neighboring Gaussians, and the Stefan-Boltzmann law combined with the inverse-square law formulates a depth-aware thermal radiation map that imposes geometric constraints. This approach models the Lambertian nature of thermal emission and reduces the number of Gaussians required for high-fidelity RGB-T reconstruction.

The challenge of spectral-spatial coupling is further addressed in [620], which, while not a 3DGS method itself, highlights the importance of frequency enhancement and spatial-spectral state-space modeling for infrared hyperspectral data. Its core insights—that state-space modeling captures global spectral dependencies while wavelet decomposition and Fourier gating recover structural contours and high-frequency details—are directly relevant to the design of spectral 3DGS kernels that must balance spectral fidelity with spatial sharpness. Similarly, [621] demonstrates that a single vision transformer backbone, conditioned on the active sensor modality, can effectively span NIR, SWIR, and LWIR bands. This suggests that spectral-adaptive conditioning mechanisms could be transferred to Gaussian-based representations to enable seamless cross-band rendering with a shared underlying scene representation.

Cross-Modal Scene Reconstruction and Modality Adaptation. A significant body of work addresses the challenge of reconstructing scenes when multiple modalities, such as visible and thermal cameras, observe the same geometry. [622] presents the first thermal 3DGS approach, which calibrates RGB and thermal cameras and learns multimodal 3D Gaussians with multimodal regularization constraints. The authors introduce smoothing constraints tailored to the physical characteristics of thermal data, preventing overfitting to any single modality while reducing model storage costs by 90%. A more recent framework, [623], proposes a Spectrum-Aware Adaptive Modulation mechanism that dynamically conditions shared latent features on thermal structural priors, guiding visible texture synthesis with reliable cross-modal geometric cues. Additionally, a Modality-Adaptive Geometric Decoupling scheme learns independent opacity offsets for the thermal branch, accommodating modality-specific geometric inconsistencies, while a hybrid rendering pipeline integrates explicit spherical harmonics with implicit neural decoding to preserve both semantic consistency and high-frequency details.

The issue of modality-specific geometry inconsistencies is further explored in [624], which proposes using only thermal infrared images combined with depth estimation to derive radiance fields, eliminating the reliance on visible light. This method, Thermal-to-Depth Gaussian Splatting (TDg), outperforms multiple single-modal Gaussian baselines while reducing training time by 55%, demonstrating that thermal-only reconstruction is viable for applications such as surveillance and industrial inspections. For scenarios where one or more modalities are scarce, [625] introduces SPoILeR, which learns mutual correlations between RGB and unconventional modalities (infrared, polarimetric, multispectral) during a multimodal pre-training phase. During fine-tuning, the model can predict accurate renderings of unconventional modalities using only RGB supervision, offering a data-efficient paradigm for cross-modal rendering.

Spectral Rendering Infrastructure and Datasets. Underpinning these advances are new representations for spectral rendering and datasets that facilitate evaluation. [626] proposes a compact latent representation enabling spectral rendering using standard RGB rendering operations. By encoding spectral images with a small number of RGB passes followed by a decoding step, and by introducing a learned non-negative linear encoder-decoder architecture that preserves scaling and addition while approximating multiplicativity under the Hadamard product, this method achieves significant color error reduction over RGB baselines at substantially lower computational cost than naive n -sample spectral rendering. At the data level, [627] presents a single-spectrograph system spanning the visible-near-infrared to short-wavelength infrared spectrum (450–1500 nm), reconstructing dense broadband hyperspectral reflectance alongside accurate 3D geometry. The authors model the image formation of broadband dispersed structured light and validate their approach on diverse real-world scenes, demonstrating accurate reconstruction of metameric materials and imaging through opaque layers.

Data-Efficient and Diffusion-Augmented Approaches. Given the scarcity of spectral training data, several methods employ generative models to augment reconstruction. [616] integrates

a diffusion-based denoiser into the 3DGS pipeline, leveraging generative priors to enhance reconstruction quality across the entire spectral range. For the related task of hyperspectral image generation, [628] introduces a latent diffusion model with multi-condition control, enhanced by a novel two-stage hyperspectral super-resolution framework that first applies RGB-band super-resolution and then uses a Rectangular Guided Attention Network for guided hyperspectral super-resolution. While not directly a 3DGS method, the underlying principles of spectral fidelity preservation and spatial diversity augmentation are directly transferable to spectral 3DGS training. In the mobile photography domain, [629] leverages the complementarity between RGB and low-resolution multispectral images to predict shading, reflectance, and material semantic priors, integrating these priors into an HDRNet for dynamic range enhancement. This work highlights how spectral information can guide appearance modeling even when spectral data is low-resolution.

Spectral Foundation Models and Cross-Domain Generalization. The development of spectral foundation models represents a promising direction for improving the generalization of spectral 3DGS. [630] introduces a spectral encoder with a self-attention-cross-attention mechanism that distills salient spectral information into camera-agnostic learned representations, demonstrating robustness to spectral heterogeneity across medical, autonomous driving, and satellite imaging domains. Similarly, [631] makes an initial attempt to build a generalized RGBT multispectral foundation model using a Cross-Modality Structural Sparsity metric and a GMM-based progressive masking strategy for self-supervised pre-training. These foundation models, while developed for 2D imagery, provide a pathway toward spectrally-aware feature embeddings that could be distilled into Gaussian primitives, enabling zero-shot or few-shot generalization across unseen spectral sensors. In summary, the spectral extension of 3DGS is rapidly evolving from per-band spherical harmonic extensions toward physics-informed, feature-based, and generative approaches that jointly model spectral radiance, material properties, and cross-modal correlations, laying the foundation for wavelength-aware real-time rendering across diverse application domains.

10 Domain-Specific Applications and System Integration

10.1 3DGS for SLAM and Robotic Navigation

Section 10.1 provides a comprehensive examination of 3D Gaussian Splatting (3DGS) applied to Simultaneous Localization and Mapping (SLAM), a domain where the explicit, differentiable nature of Gaussian primitives has catalyzed a paradigm shift from traditional sparse feature-based localization and dense implicit neural mapping toward real-time, photorealistic dense reconstruction. This subsection systematically analyzes the architectural diversity, algorithmic innovations, robustness mechanisms, and collaborative extensions that define the current landscape of 3DGS-SLAM, situating these developments within the broader context of robotic perception and establishing the foundational techniques that subsequently enable the large-scale urban and autonomous driving applications discussed in Section 10.2.

Sensor Modalities and SLAM Pipelines. The architectural design of a 3DGS-SLAM system is fundamentally dictated by its sensor configuration, with monocular, RGB-D, and multi-modal setups each presenting unique trade-offs between hardware accessibility, reconstruction fidelity, and tracking robustness. RGB-D pipelines have established the baseline for high-quality dense reconstruction: Gaussian-SLAM organizes scenes into independently optimized sub-maps from real-world single-camera RGBD video, achieving interactive-time photo-realistic rendering while scaling to larger environments through sub-map decomposition [632]. The RGBD GS-ICP SLAM system further optimizes efficiency by fusing Generalized Iterative Closest Point (G-ICP) with 3DGS, enabling a single Gaussian map to serve both tracking and mapping while exchanging covariances

and aligning scales to minimize redundant computation, ultimately achieving system speeds up to 107 FPS [633]. However, the depth sensor dependency of these approaches constrains their applicability on low-cost or power-limited platforms. This limitation has motivated substantial research into monocular pipelines. MonoGS++ eliminates the depth sensor requirement by leveraging online visual odometry for real-time sparse point cloud generation, and introduces dynamic Gaussian insertion with clarity-enhancing densification to achieve a 5.57x improvement in frames per second over prior monocular state-of-the-art [634]. KiloGS-SLAM tackles the coupled challenges of fragile long-term pose tracking and excessive memory overhead in kilometer-scale outdoor scenes, introducing a motion-adaptive hybrid tracking module with a condition-triggered three-tier solving pipeline and a lifecycle-managed Gaussian mapping strategy to scale to sequences exceeding 10,000 frames on a single GPU [635]. Beyond single-sensor approaches, multi-modal systems leverage complementary sensing modalities to enhance accuracy and robustness. MM3DGS SLAM incorporates inertial measurements into the 3DGS framework, achieving a 3x improvement in tracking accuracy and a 5% improvement in rendering quality by integrating IMU pre-integrated measurements directly into the loss functions [636]. Other works exploit learned priors to bootstrap monocular systems: GeoGS-SLAM and MyGO-Splat employ feed-forward reconstruction priors and foundation model depth estimates, with MyGO-Splat introducing a closed-loop design that analytically rasterizes Gaussians into pixel-wise depth and normals to actively supervise camera pose optimization, achieving performance comparable to RGB-D methods [243, 637]. Extending to multi-camera configurations, MCGS-SLAM presents the first purely RGB-based multi-camera SLAM system built on 3DGS, fusing dense RGB inputs from multiple viewpoints into a unified Gaussian map via multi-camera bundle adjustment, thereby enabling reconstruction of side-view regions that monocular setups inherently miss [247].

This progression from depth-dependent to monocular and multi-modal systems mirrors the sensor evolution observed in autonomous driving platforms, where heterogeneous camera rigs and varying exposure conditions necessitate increasingly flexible reconstruction frameworks—a theme that carries directly into the multi-camera handling strategies discussed in Section 10.2.

Real-Time Tracking, Mapping, and Memory Management. The stringent real-time requirements of SLAM necessitate careful co-design between tracking efficiency and mapping quality. A prevailing architectural pattern is decoupling, exemplified by GLAM-SLAM, which pairs a robust feature-based frontend for lightweight tracking with a structured, sparse anchor grid representation for scalable mapping [245]. This decoupling ensures that tracking remains computationally lightweight while the mapping module maintains scene coherence across extended sequences, as demonstrated on challenging benchmarks such as KITTI Odometry and Oxford RobotCar [245]. Memory management emerges as a critical bottleneck, since the number of Gaussian primitives can grow unboundedly during exploration. Pocket-SLAM introduces a rendering-area-aware pruning strategy that removes Gaussians based on their effective contribution to rendered images rather than relying solely on opacity or gradient magnitude, achieving over 60% memory reduction and a 2x FPS improvement in large-scale outdoor scenes [638]. Similarly, CaRtGS addresses computational misalignment by adaptively enhancing optimization iterations and refining densification to achieve high-fidelity rendering while maintaining a compact Gaussian representation [639]. At the system level, AGS proposes an algorithm-hardware co-design framework that exploits frame covisibility, using a coarse-then-fine-grained pose tracking method and a CODEC-assisted frame covisibility detection engine to achieve up to 17.12x speedup against high-end GPUs [640]. MGSO demonstrates that photometric SLAM can generate dense structured point clouds for Gaussian initialization, accelerating optimization and producing more efficient maps with fewer Gaussians while maintaining laptop-level performance [641].

Robustness to Dynamic Environments and Challenging Conditions. Early 3DGS-

SLAM systems inherited the static-world assumption from classical SLAM, which constitutes a major obstacle for real-world deployment in dynamic settings. A substantial body of research has emerged to address this limitation. DynaGSLAM goes beyond merely removing dynamic objects, achieving high-quality online rendering, tracking, and motion prediction of moving objects while jointly estimating ego-motion [642]. Dy3DGS-SLAM addresses monocular dynamic scenes by fusing optical flow masks and depth masks through a probabilistic model to derive a fused dynamic mask that constrains tracking and eliminates transient interference during mapping [401]. DAGS-SLAM introduces a spatiotemporal motion probability state per Gaussian and triggers semantic computation on demand via an uncertainty-aware scheduler, achieving a practical speed-accuracy tradeoff with reduced semantic invocations [404]. DL-SLAM computes dual-level dynamic probability maps that combine semantic and geometric information, enabling categorical pruning of dynamic Gaussians while allowing transiently static objects to contribute valuable geometric constraints for pose estimation [643]. WildGS-SLAM and UP-SLAM develop uncertainty-aware frameworks [644, 405], with UP-SLAM using a training-free uncertainty estimator that fuses multi-modal residuals to estimate per-pixel motion uncertainty and a probabilistic octree for adaptive Gaussian management [405]. For degraded imaging conditions, RoGER-SLAM introduces a Structure-Preserving Robust Fusion mechanism and an adaptive tracking objective to maintain trajectory accuracy and reconstruction quality under noise and low illumination [645]. These robustness strategies—particularly the handling of dynamic objects, exposure variations, and degraded visibility—lay crucial groundwork for the urban driving scenarios in Section 10.2, where moving traffic, changing illumination, and challenging weather conditions are inherent rather than exceptional.

Multi-Agent and Collaborative SLAM Architectures. The extension of 3DGS-SLAM to multi-robot systems introduces challenges in global consistency, communication efficiency, and data fusion. Surveys on collaborative SLAM with 3DGS systematically categorize approaches by centralized and distributed architectures, analyzing core components such as multi-agent consistency, communication-efficient Gaussian representation, and real-time scalability [646]. MAGiC-SLAM proposes a rigidly deformable 3D Gaussian-based scene representation to dramatically speed up the system, integrating loop closure for globally consistent map merging from multiple agents [647]. MAGS-SLAM presents the first RGB-only multi-agent 3DGS SLAM framework, integrating compact submap communication and occupancy-aware Gaussian fusion to enable coherent global reconstruction without active depth sensors [648]. Addressing communication bandwidth constraints, Compact Keyframe-Optimized Multi-Agent Gaussian Splatting SLAM incorporates a compaction step to remove redundant 3D Gaussians, achieving up to 85–95% reduction in transmitted data compared to state-of-the-art approaches [649]. For outdoor scenarios, CoMo3R-SLAM presents a collaborative monocular dense SLAM system that leverages learned feed-forward 3D reconstruction priors for cross-agent verification and global bundle adjustment, achieving globally consistent metric maps from monocular RGB alone [650]. These collaborative frameworks anticipate the multi-agent and fleet-level coordination requirements that emerge in autonomous driving contexts, where multiple vehicles must maintain consistent environmental representations.

Semantic Mapping and Scene Understanding. The integration of semantic information into 3DGS-SLAM has catalyzed a new class of spatial AI capabilities. SemGauss-SLAM was the first semantic SLAM system utilizing 3D Gaussian representation, incorporating semantic feature embedding and feature-level loss for precise semantic scene representation, alongside semantic-informed bundle adjustment for joint optimization [651]. GS3LAM models the scene as a Semantic Gaussian Field and jointly optimizes camera poses and the field via multimodal error constraints, introducing a Random Sampling-based Keyframe Mapping strategy to mitigate catastrophic forgetting [652]. Open-set semantics have been explored in OpenMonoGS-SLAM, the first monocular SLAM framework that unifies 3DGS with open-vocabulary understanding by

leveraging Visual Foundation Models such as SAM and CLIP [653]. FeatureSLAM integrates dense feature rasterization into novel-view synthesis, enabling entirely new downstream tasks via free-viewpoint, open-set segmentation [654]. LEGO-SLAM introduces a scene-adaptive autoencoder to distill high-dimensional language embeddings into a compact 16-dimensional feature space, reducing memory per Gaussian and enabling language-guided loop detection [655]. STAMICS integrates semantic information with 3D Gaussian representations, combining graph-based clustering for temporal semantic consistency with an open-vocabulary system to classify unseen objects [656]. These semantic capabilities not only improve mapping quality but also enable high-level reasoning and interaction essential for autonomous navigation and manipulation in complex environments, as highlighted by research on human-in-the-loop and LLM-enhanced object SLAM [657, 658]. The semantic segmentation and object-level understanding developed here find direct application in the autonomous driving systems of Section 10.2, where distinguishing and manipulating dynamic agents (vehicles, pedestrians) from static backgrounds is a prerequisite for safe closed-loop simulation and corner-case generation.

In summary, this subsection establishes the core algorithmic toolkit of 3DGS-SLAM—spanning sensor-adaptive architectures, real-time optimization strategies, dynamic robustness, collaborative consistency, and semantic understanding—that collectively provide the scalable, editable, and photorealistic scene representations on which the autonomous driving and urban reconstruction systems of Section 10.2 are built. The techniques surveyed here, from multi-camera fusion and exposure handling to dynamic scene decomposition and memory-efficient large-scale mapping, constitute the foundational capabilities that enable the transition from indoor and small-scale environments to the kilometer-scale, traffic-dense, sensor-heterogeneous driving scenarios that follow.

10.2 3DGS for Autonomous Driving and Urban Reconstruction

The application of 3D Gaussian Splatting (3DGS) to autonomous driving and large-scale urban reconstruction represents one of the most active and impactful research frontiers, fundamentally transforming how driving scenes are captured, simulated, and validated. The core appeal of 3DGS in this domain stems from its explicit, point-based representation that supports real-time rendering, photorealistic fidelity, and scene editability—properties that are indispensable for creating high-fidelity digital twins, closed-loop simulation environments, and synthetic data generators for safety-critical scenario testing. Building directly upon the sensor-adaptive architectures, dynamic robustness mechanisms, and memory-efficient mapping strategies established in the 3DGS-SLAM systems of Section 10.1, this subsection examines how these foundational techniques are scaled, adapted, and extended to meet the unique challenges of kilometer-scale, traffic-dense, and sensor-heterogeneous driving environments. The transition from SLAM to autonomous driving is characterized not only by a dramatic increase in spatial scale and dynamic complexity but also by a fundamental shift in emphasis: whereas SLAM prioritizes online tracking and incremental mapping for robot localization, autonomous driving applications demand offline photorealistic reconstruction, closed-loop simulation, and controllable scenario generation for perception validation. Nevertheless, the algorithmic toolkit developed in Section 10.1—including multi-camera fusion, handling of dynamic objects, and geometric regularization—reappears here in expanded and specialized forms, while the real-time rendering and interaction techniques explored in this section will subsequently find direct application to the constrained hardware and user-centric demands of Extended Reality in Section 10.3.

Sensor Simulation for Closed-Loop Testing and Photorealistic Digital Twins

A central application of 3DGS is the creation of photorealistic digital twins from real-world driving logs, which serve as the foundation for closed-loop simulation and perception model validation.

Unlike traditional game-engine-based simulators that suffer from a domain gap with real-world data, neural reconstruction methods like 3DGS offer a scalable solution for creating realistic virtual environments. This approach is exemplified by frameworks such as SceneCrafter, which leverages 3DGS to create an interactive and efficient simulator capable of generating realistic driving logs across diverse traffic scenarios while also enabling robust closed-loop evaluation of end-to-end models, thereby improving model generalization through synthetic data generation [659]. Similarly, the framework known as RealEngine reconstructs background scenes and foreground traffic participants separately from real-world data using 3DGS, allowing for flexible scene composition and photorealistic rendering across multiple sensor modalities, which supports non-reactive simulation, safety testing, and multi-agent interaction [660]. This decomposition of driving scenes into static backgrounds and dynamic agents mirrors the dynamic-object-handling strategies developed for 3DGS-SLAM in Section 10.1, but extends them to support independent control and re-composition of traffic participants for simulation purposes.

However, a significant challenge in this context is the “sim-to-real” gap introduced by the edited 3DGS-rendered videos, which often contain rendering artifacts, degraded foreground assets, and temporal flickering. To address this, RealityBridge proposes a structure-preserving and asset-aware framework that uses multimodal controls, including rendered videos, masks, and semantic maps, to repair 3DGS-specific artifacts and harmonize illumination, thereby enhancing the realism and temporal consistency of edited driving videos [661]. For large-scale digital twin integration, the DrivIng dataset provides a comprehensive multi-modal dataset with a complete geo-referenced digital twin of an 18 km route, leveraging 3DGS to enable a one-to-one transfer of real traffic into simulation while preserving agent interactions for realistic scenario testing [662]. Additionally, frameworks that synthesize novel trajectory views by leveraging world model priors, such as DriveDreamer4D, enhance 4D driving scene representation with improved spatiotemporal coherence for closed-loop simulation [663].

Handling Multi-Camera Setups and Varying Exposures

Real-world autonomous driving data is characterized by heterogeneous camera pipelines, varying exposures, and dynamic outdoor illumination. A critical limitation of conventional 3DGS is its implicit assumption of consistent exposure and tone mapping across views, which is violated in real driving data, leading to artifacts and inconsistent illumination. This challenge directly extends the multi-camera fusion and exposure-handling considerations introduced in the SLAM systems of Section 10.1, but in the driving context, the scale and heterogeneity of camera rigs—often spanning 6 to 12 cameras with differing exposure times and field-of-view characteristics—amplify the difficulty substantially. To address this, P2GS introduces a physically consistent Gaussian Splatting framework that jointly decomposes a view-invariant linear HDR radiance field, per-view exposure scales, and tone-mapping functions from LDR images without HDR supervision, ensuring photometric consistency and robust illumination across diverse scenes [298]. Furthermore, multi-camera setups require native support for diverse camera models. Instant NuRec presents a feed-forward neural reconstruction model that turns a short multi-view driving log into a simulatable 3DGS world in a single forward pass, with native support for non-pinhole camera models via 3DGUT, enabling rapid reconstruction and high-fidelity closed-loop simulation [664]. For modeling fine-grained variations between frames and camera viewpoints, ArmGS devises a multi-level appearance modeling scheme that optimizes transformation parameters at local Gaussian, global image, and dynamic actor levels, enabling composite Gaussian refinement that models both global appearance variations and local fine-grained changes of background and objects [665]. These techniques demonstrate how the multi-camera handling strategies of Section 10.1 are systematically extended to accommodate the hardware heterogeneity and photometric complexity inherent to production-grade autonomous driving platforms.

Geometric Consistency and Scalability for City-Scale Scenes

Scaling 3DGS to large-scale urban environments presents significant challenges in geometric consistency, memory efficiency, and computational scalability. While the SLAM systems of Section 10.1 typically operate at the scale of indoor rooms or bounded outdoor environments, autonomous driving reconstruction targets city-scale areas spanning kilometers, requiring fundamentally different strategies for spatial partitioning, memory management, and geometric regularization. The UrbanGS framework addresses these issues by introducing a depth-consistent D-normal regularization module that integrates monocular normal constraints with external depth supervision, ensuring comprehensive updates of all geometric parameters and multi-view depth alignment. Additionally, a spatially adaptive Gaussian pruning strategy and a unified partitioning scheme are designed to reduce redundancy and eliminate boundary artifacts, achieving superior performance in rendering quality, geometric accuracy, and memory efficiency for city-scale applications [225]. The memory-efficient pruning strategies developed here echo the rendering-area-aware pruning of Pocket-SLAM from Section 10.1, but are adapted to the much larger scale and more heterogeneous scene content of urban environments. For highway-specific scenarios, LiHi-GS proposes a LiDAR-supervised Gaussian splatting method that improves scene reconstruction through LiDAR supervision and supports LiDAR rendering, addressing the challenging highway scenes characterized by sparse sensor views and monotone backgrounds [234].

Scene Editing, Data Augmentation, and Safety-Critical Scenario Generation

A key advantage of 3DGS is its explicit representation, which facilitates scene editing and the generation of diverse synthetic data for safety-critical scenario creation. This capability builds on the semantic mapping and scene understanding foundations established in Section 10.1, but extends them toward controllable manipulation and synthetic data generation—a requirement unique to autonomous driving validation. DecoupleGS presents a framework that decomposes scenes into a static background and manipulable dynamic agents, using an object-centric canonical representation to enable high-fidelity background and controllable dynamic agents for end-to-end autonomous driving testing [666]. For photorealistic object insertion, R3D2 leverages a lightweight diffusion model to generate plausible rendering effects, such as shadows and consistent lighting, enabling realistic insertion of complete 3D assets into existing 3DGS scenes, which supports text-to-3D asset insertion and cross-scene object transfer [667]. Moreover, MultiEditor introduces a dual-branch latent diffusion framework that uses 3DGS as a structural and appearance prior to jointly edit images and LiDAR point clouds, thereby enhancing the perception accuracy of models on underrepresented vehicle classes [668]. DriveWeaver offers a point-conditioned video inpainting framework that generates temporally consistent vehicles for insertion, explicitly extracting 3D Gaussian representations to ensure real-time rendering and lighting consistency between foreground and background [669]. The CARLA-GS framework further unifies visual representation, semantic reasoning, and physics-based execution, using an editable Gaussian scene and a multi-agent LLM to generate physically feasible corner-case trajectories that are re-projected into the scene for photorealistic corner-case generation [670]. This integration of LLM-driven reasoning with editable 3DGS scenes anticipates the LLM-XR synergies that will be explored in Section 10.3, where language models similarly power intelligent content creation and interactive world manipulation.

Integration with Traffic Simulators and Feed-Forward Reconstruction

Integrating 3DGS with traffic simulators enables comprehensive safety validation. SynthDrive proposes a scalable real2sim2real system that automates asset mining and generation from real data, addressing the scalability limitations of both CG-based and learning-based approaches for generic objects [671]. To alleviate the per-scene optimization cost of traditional 3DGS, feed-forward reconstruction methods have been developed. ReconDrive leverages a 3D foundation model with hybrid Gaussian prediction heads and a static-dynamic 4D composition strategy to rapidly generate

high-fidelity 4D Gaussian representations for driving scenes, achieving performance competitive with per-scene optimization while being orders of magnitude faster [672]. The feed-forward paradigm introduced here—which prioritizes inference speed and generalization over per-scene optimization—represents a conceptual departure from the online optimization focus of the SLAM systems in Section 10.1, and anticipates the efficiency-first approaches that will become dominant in the XR applications of Section 10.3, where real-time interactivity on constrained hardware is paramount. Finally, to enable robust reconstruction without expensive annotations, S3Gaussian presents a self-supervised method that decomposes dynamic and static elements from 4D consistency using a spatial-temporal field network, demonstrating the ability to reconstruct street scenes without requiring tracked 3D vehicle bounding boxes, which broadens the applicability of 3DGS in-the-wild scenarios [673]. For further improving the fidelity of reconstructions under novel viewpoints, generative enhancers like GenRe distill diffusion priors into 3D Gaussian representations to generalize reliably to challenging unseen viewpoints [674]. Additionally, frameworks such as DecoupleGS emphasize asset compression and map-guided geometric registration to achieve balanced fidelity-efficiency trade-offs for large-scale evaluation [666].

In summary, this subsection has traced the evolution of 3DGS from the online, resource-constrained SLAM systems of Section 10.1 to the large-scale, editable, and simulation-oriented frameworks that define the autonomous driving and urban reconstruction landscape. The key technical transitions—from online tracking to offline reconstruction, from single-vehicle perception to multi-camera heterogeneous sensor fusion, and from passive mapping to active scene manipulation for safety-critical testing—collectively establish the representational power and system-level maturity of 3DGS as a core technology for next-generation autonomous driving platforms. These capabilities, particularly the ability to render photorealistic scenes in real-time, edit and recompose scene elements, and generate controllable synthetic data, lay the essential groundwork for the transition to Extended Reality applications in Section 10.3, where similar requirements for real-time rendering, scene editability, and user-driven interaction are reinterpreted within the context of immersive experiences on constrained hardware.

10.3 3DGS for Virtual, Augmented, and Mixed Reality (XR)

The transition from the structured, sensor-rich environments of autonomous driving and urban reconstruction to the more fluid and interactive domain of Extended Reality (XR) is marked by a fundamental shift in priorities. While the previous subsection emphasized geometric accuracy and scalability for large-scale city modeling, the integration of 3D Gaussian Splatting (3DGS) with Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) focuses on real-time interactivity, hardware constraints, and user-centric experiences. Moreover, as we will see in the subsequent subsection, the principles developed here—such as efficient rendering on constrained hardware and the integration of scene understanding—find direct parallels in the demands of medical imaging and remote sensing. In XR, the convergence of 3DGS with immersive technologies represents a paradigm shift in how digital content is created, rendered, and interacted with. 3DGS offers an explicit, point-based scene representation that delivers photorealistic novel view synthesis at real-time frame rates, directly addressing the stringent performance requirements of XR headsets and the demand for interactivity that prior implicit neural rendering methods could not satisfy [675]. This subsection explores the multifaceted integration of 3DGS into XR systems, focusing on real-time rendering on constrained hardware, hardware-software co-design, physics-aware asset creation, user-guided editing, telepresence, and the synergistic coupling with large language models (LLMs).

A primary challenge in deploying 3DGS within XR is achieving high-fidelity rendering on

resource-constrained, standalone headsets that have limited compute, memory, and power budgets—a constraint that is less prominent in the server-side processing typical of autonomous driving but becomes a dominant factor in consumer XR devices. The computational demands of rendering large-scale 3DGS models often exceed the capabilities of consumer-grade devices, highlighting a critical need for system-level optimization [676]. To address this, an end-to-end framework introduces a cache-centric pipeline that eliminates redundant computations, an efficiency-aware scheduler for elastic multi-GPU rendering, and optimized CUDA kernels. This synergy between representation and system design achieves up to a 5.35x performance improvement and supports 2K binocular rendering at over 120 FPS, demonstrating a scalable path for real-time neural rendering in immersive environments [676]. Beyond software optimizations, the architectural landscape of XR workloads—which are increasingly characterized by heterogeneous operator types and complex dataflow structures—demands a shift from conventional CNN-centric accelerators [677]. This insight underscores the need for phase-aware scheduling and elastic resource allocation in next-generation XR System-on-Chips (SoCs).

Hardware-software co-design is essential to unlock the full potential of 3DGS in XR. A notable example is an architecture-algorithm co-design framework that introduces axis-shared rasterization and order-independent transmittance. By precomputing common terms along X- and Y-axes, the design reduces multiply-and-accumulate operations by up to 38%, while a lightweight multilayer perceptron (MLP) approximates transmittance to remove the need for explicit sorting, enabling efficient alpha blending and achieving a 1.33 to 1.88x speedup over state-of-the-art accelerators [111]. Furthermore, the feasibility of native MR compositing on devices like the Meta Quest 3 has been explored, with benchmarking showing that the Snapdragon XR2 Gen 2 can support lightweight native MR content creation at 720p30, albeit with thermal limits restricting operation to 5–10 minutes. This study underscores the tight thermal and power headroom that must be managed in XR hardware design [678].

The creation of interactive and dynamic XR content is another frontier where 3DGS excels. While the previous subsection focused on static and dynamic scene reconstruction for simulation, XR elevates the requirement to real-time physical interaction. Traditional techniques for content creation are engineering-intensive and require extensive expertise. Systems like VR-GS represent a leap forward by integrating physical dynamics awareness into an interactive 3DGS system within VR. By constructing a two-level embedding strategy alongside deformable body simulations, VR-GS ensures real-time execution with highly realistic dynamic responses, enabling users to interact with and manipulate virtual objects in a physically plausible manner [679]. This capability is further enhanced by LLMs, which can predict reasonable physical parameters from static Gaussian assets in seconds, facilitating the rapid creation of physics-based dynamic assets without manual annotation. This integration of language models with 3DGS enables non-experts to author highly realistic, interactive VR scenes, as demonstrated in the LIVE-GS system [680].

Building upon the physical interactivity of Gaussian assets, user-guided scene editing and manipulation are crucial for spatial computing. The explicit nature of 3DGS primitives allows for intuitive and precise editing, which can be further controlled through semantic understanding. For instance, EditSplat introduces a text-driven 3D scene editing framework that integrates Multi-view Fusion Guidance to ensure multi-view consistency and Attention-Guided Trimming to selectively prune Gaussians, enabling semantically rich and localized edits [538]. Additionally, the rise of generative AI and LLMs is transforming XR authoring. New interfaces are emerging that allow users to create outdoor scenes through natural language interaction, combining offline scene understanding and fast 3D asset generation, as exemplified by the ImagineAR system [681]. Such systems translate natural language prompts into spatial intent and executable constraints for generative 3D design [682], moving towards continuous co-creation where part-aware representations enable

component-level control [683].

Another key application where 3DGS uniquely bridges the gap between the physical and virtual worlds is telepresence and teleoperation, building on the sensor fusion techniques used in autonomous driving but applied here to human-centric communication. Traditional teleoperation systems often rely on screen-based interfaces that lack depth cues, hindering task performance. Multi-view VR telepresence systems that fuse geometry from multiple cameras to produce GPU-accelerated point-cloud rendering on standalone VR hardware have shown improved usability, and integrating a localized RGB stream can further enhance performance [684]. In the domain of human telepresence, frameworks like Mon3tr integrate 3DGS-based parametric human modeling into telepresence, dividing the process into a one-time offline reconstruction and a monocular online inference phase. This amortized computation strategy, where a single RGB camera drives a 3DGS avatar, achieves photorealistic rendering at ~60 FPS on a Meta Quest 3 with an end-to-end latency of ~80 ms, while reducing bandwidth by over 1000x compared to point-cloud streaming [685]. This represents a major step toward accessible and realistic holographic communication.

Finally, the integration of 3DGS with large language models (LLMs) is enabling intelligent interactions and context-aware experiences, a trend that anticipates the multi-modal system integration challenges discussed in the next subsection. LLMs can power the real-time creation and modification of interactive XR experiences, with frameworks like LLMR demonstrating 4x improvements in error rate over standard GPT-4 by using scene understanding and task planning techniques [686]. Moreover, large multimodal models (LMMs) are being used to build voice-first assistants in MR that can understand physical environments and generate content, facilitating a more natural human-AI collaboration [687]. The combination of LLMs’ reasoning power with 3DGS’s visual fidelity promises to make XR environments not only more immersive but also more intelligent and responsive to user intent, pushing the boundaries of what is achievable in interactive media and virtual worlds [688].

10.4 3DGS for Medical Imaging, Remote Sensing, and Cross-Domain System Integration

The versatility of 3D Gaussian Splatting (3DGS) has catalyzed its adoption across a diverse range of domain-specific applications, moving well beyond the conventional boundaries of computer vision and graphics. This subsection surveys the significant strides made in integrating 3DGS within two particularly demanding fields: medical imaging and remote sensing. We also examine the system-level integration challenges and opportunities that arise when deploying 3DGS in these real-world pipelines, highlighting how the unique properties of Gaussian primitives are leveraged to address domain-specific constraints. While the previous subsection focused on the real-time interactivity and user-centric demands of XR, the applications discussed here shift the emphasis toward domain-specific constraints such as anatomical accuracy, multi-temporal consistency, and scalability to geospatial extents. Nevertheless, the principles of efficient rendering on constrained hardware and the integration of scene understanding developed in the XR context find direct parallels in the clinical and remote sensing pipelines explored below.

3DGS in Medical Imaging and Surgical Applications

The medical domain presents a unique set of challenges for 3D reconstruction, characterized by the need for high fidelity, anatomical accuracy, and the ability to handle complex, often dynamic, biological tissues. Traditional methods, while robust, are often computationally expensive and can lead to structural discontinuities, especially when working with sparse slice data [689]. 3DGS has emerged as a powerful solution to these challenges, offering a blend of rendering efficiency and geometric fidelity that is well-suited for clinical workflows. Just as the XR domain demanded real-

time performance on resource-constrained headsets, medical applications require rapid reconstruction for intraoperative decision-making, yet they impose the additional burden of strict anatomical correctness.

A primary application area is the reconstruction of surgical scenes for intraoperative navigation and simulation. Endoscopic procedures, for instance, suffer from photometric inconsistencies caused by non-Lambertian surfaces and dynamic motion due to patient breathing [690]. Here, 3DGS-based SLAM systems have been developed to provide robust real-time tracking and mapping, mirroring the real-time demands of XR telepresence systems discussed previously, but with the added complexity of deformable tissue and specular reflections. For example, EndoFlow-SLAM integrates optical flow loss as a geometric constraint and a depth regularization strategy to mitigate photometric inconsistencies, demonstrating superior performance in both static and dynamic surgical scenes [690]. Similarly, to address the limitations of Structure-from-Motion in feature-sparse endoscopic environments, SurGSplat proposes a paradigm that progressively refines 3DGS through geometric constraints, enabling detailed reconstruction of critical structures like vasculature for enhanced intraoperative decision-making [691].

Beyond navigation, 3DGS is being used to create immersive visualization platforms, directly building upon the XR foundations established in the previous subsection. Multi-Layer Gaussian Splatting introduces a layered representation for CT scans, enabling interactive exploration of different anatomical structures in virtual reality at frame rates suitable for immersive headsets—a task previously prohibitive with computationally intensive path tracing [692]. This approach retains a degree of interactivity, such as selective activation and clipping of layers, which is crucial for medical education and pre-operative planning [692]. Furthermore, the interplay between 3D reconstruction and advanced imaging modalities is being explored. For instance, PhysioSynth, a modality-specific simulator, can generate synthetic CT and MR images based on anatomical priors, which can be used to train and validate downstream reconstruction models in a data-efficient manner [693]. This synergy between simulation and reconstruction is vital for overcoming data scarcity and privacy concerns in medical AI, echoing the broader trend of using in-silico experiments to scale up model training [694].

The integration of 3DGS with other medical data types and deep learning models is a growing trend, paralleling the multi-modal fusion strategies seen in both XR and remote sensing. The success of AI models like nnU-Net and its variants in tasks such as brain tissue segmentation and lung registration underscores the importance of robust 3D representations, and 3DGS offers a complementary, rendering-centric approach [695]. The explicit nature of Gaussian primitives also facilitates the incorporation of geometric priors, which can be critical in complex tasks like surgical phase recognition [696]. More advanced frameworks, such as MedGS, enhance 3DGS with physically based relightable models, separating light effects from tissue properties to generate photorealistic reconstructions with better generalization across novel views and enabling realistic tissue modifications—a significant step towards practical clinical simulation tools [697].

3DGS in Remote Sensing and Satellite Photogrammetry

Remote sensing applications, particularly those using satellite or aerial imagery, present a distinct set of challenges, including large-scale scenes, multi-temporal data, radiometric inconsistencies, and complex geometric distortions. While the XR and medical domains prioritize real-time interactivity and anatomical fidelity, respectively, remote sensing emphasizes scalability, temporal robustness, and geometric accuracy across vast spatial extents. 3DGS is increasingly recognized as a valuable tool for overcoming these hurdles, especially for large-scale urban and environmental monitoring [698, 113].

One of the primary difficulties in applying 3DGS to satellite imagery is the incompatibility with the Rational Polynomial Coefficient (RPC) camera model, a challenge unique to, yet analogous

with, the sensor-specific constraints encountered in endoscopic imaging. SkySplat directly addresses this by integrating the RPC model into a generalizable 3DGS pipeline, enabling rapid and accurate digital surface model (DSM) reconstruction from sparse, multi-temporal satellite images [699]. This work demonstrates not only an $86\times$ speedup over per-scene optimization methods but also superior accuracy, highlighting the potential of 3DGS for large-scale geospatial applications [699]. Similarly, SwiftGS introduces a meta-learned system that predicts geometry-radiation-decoupled Gaussian primitives in a single forward pass, replacing expensive per-scene fitting with an episodic training approach. This allows for immediate, zero-shot surface recovery from multi-date satellite imagery, which is critical for rapid disaster response and environmental monitoring [700]. The framework’s design, which couples a differentiable physics graph with spatial gating, ensures both geometric accuracy and view-consistent rendering, overcoming the traditional cost of per-scene optimization [700].

Scalability to large urban and aerial scenes remains a key research focus, echoing the constraints of large-scale reconstruction discussed in the context of autonomous driving. Approaches like HUG and LoBE-GS introduce novel hierarchical and load-balanced frameworks to manage the massive memory consumption and slow training times associated with city-scale data [698, 113]. LoBE-GS, for instance, utilizes a load-balanced KD-tree scene partitioning scheme and visibility cropping to achieve up to $2\times$ faster end-to-end training times than state-of-the-art baselines, enabling reconstructions previously infeasible with vanilla 3DGS [113]. These advances are complemented by distributed computing frameworks like Radiant and LandMarkSystem, which partition regions based on heterogeneous environment information and leverage distributed parallel computing to handle large-scale scenes [701, 702]. Furthermore, research in this area often builds upon the foundation of classic stereo and multi-view satellite photogrammetry, and the integration of these established pipelines with modern neural rendering is an active area of exploration [703, 704].

Cross-Domain System Integration and Challenges

The integration of 3DGS into real-world pipelines across these domains reveals several common threads and challenges that also connect back to the XR applications discussed in the previous subsection. A significant challenge is the robustness of the reconstruction pipeline to imperfect or heterogeneous inputs. For surgery, this means handling tool occlusions, bleeding, and specular reflections [705]. For remote sensing, it involves dealing with illumination changes, sensor heterogeneity, and transient objects [699, 700]. These robustness requirements parallel the need for XR systems to handle dynamic user movements and varied environmental lighting conditions.

System-level integration often requires fusing 3DGS with other data modalities, such as LiDAR point clouds, thermal infrared images, or multi-spectral data [93, 623, 706]. For example, ThermoSplat demonstrates a cross-modal framework that integrates RGB and thermal infrared data for robust environmental perception, a capability crucial for both autonomous systems and defense applications [623]. The fundamental trade-off between high-fidelity visuals from neural representations like 3DGS and the robust geometric accuracy of traditional surface meshes is a critical consideration. For functional tasks like line-of-sight analysis and physics simulations, a hybrid approach that combines a traditional mesh scaffold for geometric integrity with a neural representation for visual detail is often the most practical solution [93].

In the medical domain, integrating 3DGS with other analytical tools, such as large language models for interactive querying or vision-language models for automated reporting, remains a significant frontier [707, 708]. This trend toward language-driven interaction mirrors the LLM-integrated XR systems discussed in the previous subsection, suggesting a convergent trajectory toward more intelligent and context-aware 3D applications. The ability to segment and understand the reconstructed scene is also paramount. While datasets like 3DLAND provide 3D lesion annotations for medical CT volumes, the seamless integration of dense 3D scene reconstruction with object-level

or semantic understanding is still an open research problem [709]. Overall, the deployment of 3DGS in these critical domains hinges on overcoming challenges related to computational scalability, data efficiency, multimodal fusion, and the development of robust evaluation protocols that reflect real-world operational conditions [37]. These challenges, while manifesting differently across XR, medicine, and remote sensing, collectively define the research agenda for the next generation of 3DGS-based systems, setting the stage for the cross-cutting methodological discussions that follow in the subsequent subsection.

11 Security, Quality Assessment, and Benchmarks

11.1 Threat Modeling and Security Analysis of 3DGS Systems

The rapid adoption of 3D Gaussian Splatting (3DGS) in safety-critical domains such as autonomous driving, augmented reality, and robotic navigation has made the security of its reconstruction pipeline a pressing concern. Unlike traditional deep learning models operating on flat 2D inputs, 3DGS involves a multi-stage pipeline that transforms a collection of posed images into an explicit, editable 3D representation. This multi-stage nature creates a unique and expanded attack surface, where malicious actors can intervene at various points—from the input data to the intermediate geometric representations to the final optimized Gaussian parameters. A formal threat-modeling framework is therefore essential to systematically analyze adversarial capabilities, identify vulnerabilities, and develop robust countermeasures.

The 3DGS reconstruction pipeline can be broadly segmented into three primary stages: data ingestion, geometric and appearance optimization, and model deployment. At the data ingestion stage, the adversary can poison the training images by injecting malicious textures, altering camera poses, or embedding backdoor triggers into the visual content [710]. During the optimization stage, the attacker might interfere with the densification and pruning processes, manipulate the gradient signals, or implant malicious weights into the Gaussian parameters. Finally, at the deployment stage, the rendered output can be attacked through adversarial patches applied to the physical environment or through model-level perturbations that trigger undesired behaviors in downstream perception systems [711]. It is crucial to note that the multi-stage nature of this pipeline produces heterogeneous intermediate representations—including multi-view images, point clouds, training dynamics, and Gaussian statistics—each of which may carry distinct forensic signals that reveal the presence of an attack. Indeed, recent work has emphasized that the detectability of poisoning is inherently stage-dependent, as an attack introduced at one stage may produce observable signatures only at later stages of the pipeline [712].

Within this framework, we can define a taxonomy of adversarial capabilities based on the attacker’s knowledge and access. A white-box attacker has full access to the model architecture, training data, and optimization pipeline, enabling them to execute highly sophisticated attacks such as gradient-guided backdoor injection or direct parameter manipulation [713]. Conversely, a black-box attacker operates with limited information, often relying on query access to the rendered outputs or the assumption of a specific model architecture [714]. This distinction is critical, as it shapes the feasibility and stealthiness of different attack vectors. For instance, white-box attacks can optimize for the joint objective of attack efficacy and perceptual realism, as demonstrated by the GaussTrap framework, which implants viewpoint-consistent poisoned renderings that yield high-quality output in non-target views while causing targeted scene confusion under specific conditions [710]. Gray-box settings, where the attacker possesses partial knowledge, are also relevant, particularly in scenarios involving model extraction or transferable attacks [715].

The attack landscape for 3DGS can be categorized into several distinct types. First, backdoor

injection attacks aim to embed hidden behaviors into the trained 3DGS model, causing it to render malicious content or misrepresent the scene when a specific trigger is present. These attacks often involve poisoning a small subset of training data with carefully crafted triggers that remain imperceptible to human inspection [710]. The triggers can be spatial (e.g., a specific pattern in the scene), spectral, or even semantic in nature, exploiting the model’s ability to learn and activate on distributed representations [716]. Second, computation cost attacks, or denial-of-service (DoS) attacks, seek to degrade the system’s performance by injecting samples that cause excessive densification or expensive gradient computations, leading to resource exhaustion and rendering delays. These attacks exploit the adaptive nature of the 3DGS optimization process, where a single malicious image can trigger the creation of an extremely large number of Gaussian primitives [717]. Third, adversarial patch poisoning involves the placement of physical or digital patches that deceive the neural components of the 3DGS pipeline, such as feed-forward reconstruction networks, into producing incorrect geometric or appearance representations [711]. This can compromise the entire scene reconstruction, leading to environmental misperception in downstream applications.

A significant challenge in defending 3DGS systems is the stage-dependent nature of detectability. Forensic signals that are indicative of an attack may be highly visible at one stage but completely absent at another. For instance, an illusory object injection attack might be challenging to detect from the final rendered images if the renderer maintains high fidelity, but the training dynamics—such as a sudden spike in the gradient norm or an unusual distribution of Gaussian opacities—might reveal the malicious intervention [712]. Similarly, computation cost attacks might leave minimal visual trace in the output but are likely to produce anomalous Gaussian parameter statistics, such as an unusually high primitive count in a specific region [717]. This underscores the need for multi-stage detection pipelines that monitor not only the final output but also the intermediate representations and training behaviors. Furthermore, the effectiveness of detection is not uniform across attack types; certain attacks, such as adversarial weight perturbations, may be better detected by analyzing the Gaussian covariance matrices, while others, like data poisoning, may be more apparent in the input image frequencies or the multi-view consistency of the reconstructed geometry [712].

Defense strategies for 3DGS systems must therefore adopt a defense-in-depth approach. At the input stage, detection mechanisms can filter adversarial patches and calibration errors [717]. During optimization, robust training techniques, such as adversarial training or gradient regularization, can reduce the impact of poisoned samples [718]. At the model level, post-hoc purification methods can be applied to remove backdoor triggers or to unlearn malicious associations [719, 720]. Importantly, these defensive measures must be understood in relation to the broader ecosystem of 3DGS asset protection. While the threat model defined here addresses attacks that corrupt the *integrity* and *availability* of a 3DGS model during its lifecycle, the *confidentiality* and *ownership* of the asset itself are defended by the active and passive copyright protection mechanisms discussed in the next subsection. Notably, the principles of threat modeling and the cat-and-mouse dynamic between attackers and defenders are mirrored in the copyright domain, where watermarking schemes are pitted against increasingly sophisticated purification and removal attacks. Notably, the concept of proactive defense—where the defender actively engages with the training process to amplify the forensic signals of an attack—offers a promising direction, as exemplified by prototype frameworks in other domains that promote “proactive” poison detection [721]. In the specific context of 3DGS, this could involve monitoring the distribution of Gaussian scales, the entropy of spherical harmonic coefficients, or the temporal consistency of rendered views to expose subtle anomalies introduced by an adversary [712].

In summary, a comprehensive threat model for 3DGS systems must account for the diverse attack surfaces introduced by its multi-stage pipeline, the variety of adversarial capabilities, and the heterogeneous nature of stage-dependent forensic signals. By establishing this taxonomy, we

can better guide the development of robust and reliable 3DGS systems that are resilient to both current and emerging security threats. This foundation of security analysis, which focuses on the integrity of the reconstruction process, naturally leads to the complementary question of protecting the *final 3DGS asset* itself, a topic that encompasses the copyright protection mechanisms explored in the following subsection. Future research must focus on standardized benchmarks that enable the systematic evaluation of detection and defense methods under unified protocols, ensuring that progress in this critical area is both measurable and reproducible [712, 717].

11.2 Active and Passive Copyright Protection for 3DGS Assets

The explosive growth of 3D Gaussian Splatting (3DGS) as a de facto standard for real-time, high-fidelity 3D scene representation has created a pressing need for robust intellectual property (IP) protection. While the preceding subsection established a threat model focused on the *integrity* and *availability* of 3DGS models—addressing adversarial attacks that corrupt the reconstruction process—a complementary security concern arises once a legitimate 3DGS asset has been created and deployed: the protection of its *confidentiality* and *ownership*. Unlike implicit representations like NeRF, the explicit and editable nature of Gaussian primitives—each with its own position, covariance, opacity, and appearance parameters—makes unauthorized copying, redistribution, and malicious manipulation of 3D assets significantly easier. Addressing this dual threat requires a two-pronged approach: passive protection mechanisms, primarily in the form of watermarking for ownership tracing, and active protection strategies that deter unauthorized editing. This subsection surveys the landscape of these protection mechanisms, analyzes their underlying principles, and discusses the attacks that threaten their efficacy, thereby establishing a direct bridge to the quality assessment challenges discussed in the following subsection, where the perceptual impact of protective measures must be carefully balanced against their security benefits.

Passive Protection: Watermarking for Ownership Tracing

Passive copyright protection for 3DGS is predominantly achieved through digital watermarking, which embeds imperceptible ownership information (e.g., a bitstream or a logo) into the model. Once embedded, this information can be reliably extracted to assert ownership if the asset is misused. Early efforts in this domain adapted existing concepts to the unique structure of 3DGS. For instance, [722] introduced an uncertainty-based method that constrains the perturbation of Gaussian parameters to embed watermarks invisibly, ensuring that message decoding is reliable from both the 3D model and rendered 2D images, even after various distortions. Similarly, [723] proposed a universal watermarking strategy that utilizes a pre-trained watermark decoder and treats the 3DGS generation module as an encoder, incorporating novel 3D distortion layers to bolster robustness against real-world point cloud transformations.

A significant challenge in 3DGS watermarking is the efficiency of embedding messages. The generalizability and message capacity of watermarking schemes have been key areas of innovation. For instance, [724] presents a generalizable framework that enables protection of Splatter Image-based 3DGS models through a single forward pass, overcoming the need for expensive per-message fine-tuning. To further enhance capacity, [725] introduces a bit-compression framework that encodes multiple message bits per token, breaking the traditional limits of bit-to-token methods and achieving high message recovery accuracy for larger payloads like 128-bit messages. To improve the invisibility and robustness trade-off, [726] presents a representation-native framework that separates carrier selection from quality preservation, using a Trio-Experts module and a Safety and Budget Aware Gate to allocate Gaussians effectively and achieve both high watermark persistence and explainability.

A further dimension of passive protection is steganography, which aims to hide the very existence of the secret communication. In the 3DGS context, [727] proposes an end-to-end, key-secured 3D

steganography framework that jointly optimizes the Gaussian model and a key-secured decoder. This framework highlights that Gaussian features contribute unequally to secret hiding and allows for multi-secret hiding while preventing unauthorized access, with a novel 3D-Sinkhorn distance introduced to quantify steganographic imperceptibility.

Active Protection: Edit Deterrence and Unified Frameworks

Passive watermarking, however, does little to prevent an adversary from maliciously editing a 3DGS asset. This has led to the development of active protection mechanisms. One approach involves “lifting” adversarial perturbations from 2D to 3D. For instance, [728] proposes the first editing safeguard for 3DGS, which prevents instruction-driven editing across arbitrary views by lifting strictly bounded 2D adversarial perturbations into the 3D Gaussian representation, effectively disrupting diffusion-based editing pipelines.

Recognizing that copyright protection requires both tracing and deterrence, recent research has moved toward unified frameworks. [729] presents the first unified protection framework that jointly optimizes a scene-wide watermarking objective with an adversarial objective for edit deterrence. This framework not only embeds a recoverable watermark but also uses adversarial techniques to divert the editing trajectory, preventing unauthorized modifications while preserving rendering fidelity. This approach effectively combines the strengths of passive and active protection, achieving a balance between bit accuracy, edit deterrence, and rendering quality, as demonstrated on Mip-NeRF 360 and Instruct-NeRF2NeRF scenes. This represents a significant step forward, suggesting that practical IP protection for 3DGS assets is best served by integrating ownership tracing and unauthorized editing deterrence into a single, coherent optimization process, rather than treating them as separate problems.

Threat Model and Attacks: Watermark Purification and Removal

Despite the advances in watermarking, the security of 3DGS watermarking systems has come under scrutiny. A critical question is whether these watermarks can withstand dedicated removal attacks, a concept analogous to adversarial attacks in image processing. [730] systematically explores the vulnerabilities of current 3DGS watermarking frameworks. It demonstrates that conventional 2D watermark removal techniques do not generalize effectively to the 3DGS scenario due to the specialized rendering pipeline and unique attributes of each Gaussian primitive. This insight led to the proposal of GSPure, the first watermark purification framework specifically for 3DGS. GSPure exploits view-dependent rendering contributions and geometrically accurate feature clustering to isolate and remove watermark-related Gaussian primitives while preserving scene integrity, reducing watermark PSNR by up to 16.34 dB with minimal degradation to original scene fidelity. This is a potent example of a purification attack, highlighting a serious vulnerability in current passive protection schemes.

Furthermore, the broader field of watermark security highlights that attacks can be universal and black-box. For instance, [731] introduces the Group-based Multi-objective Evolutionary Attack (GMEA), a first-of-its-kind universal black-box attack designed to challenge both 1D and 2D watermarks embedded within 3DGS. GMEA formulates the attack as a large-scale multi-objective optimization problem, balancing watermark removal with visual quality, thus demonstrating that effective watermark removal can be achieved without any internal knowledge of the watermarking scheme. The existence of such powerful attacks underscores the need for more robust and threat-aware watermarking solutions. The interconnectedness of these issues is also emphasized by the broader research agenda presented in [732], which argues that effective progress requires articulated security objectives and realistic threat models. This position paper advocates for scenario-driven formulations that clearly define adversarial capabilities and evaluate methods under explicit security assumptions, a principle that is critically important for advancing the robustness and practical utility of 3DGS copyright protection.

The cat-and-mouse dynamic between protection and attack mechanisms described here directly parallels the threat modeling discussion in the preceding subsection, where the detectability of security violations was shown to be stage-dependent. Similarly, the effectiveness of a watermarking scheme must be evaluated not only by its embedding capacity and robustness but also by its impact on perceptual quality—a consideration that connects naturally to the quality assessment methodologies reviewed in the following subsection. Indeed, the watermarking schemes discussed above often introduce subtle perturbations to Gaussian parameters, and the assessment of whether these perturbations remain perceptually invisible is, at its core, a quality assessment problem. The development of 3DGS-specific quality metrics, as surveyed in the next subsection, will consequently play a pivotal role in enabling the fair and robust evaluation of future copyright protection mechanisms, ensuring that security measures do not come at the cost of user-perceived fidelity. In summary, the field is moving from simple embedding techniques toward a sophisticated ecosystem where advanced unified protection frameworks contend with increasingly powerful purification and removal attacks, all while being constrained by the perceptual quality requirements that govern the deployment of 3DGS technology in practical applications.

11.3 Quality Assessment, Benchmarks, and Evaluation Protocols

The rapid proliferation of 3D Gaussian Splatting (3DGS) as a dominant representation for real-time novel view synthesis has created a critical demand for robust and standardized quality assessment methodologies. Unlike traditional image or video quality assessment, the evaluation of 3DGS-rendered content is complicated by a unique set of factors, including the interactive nature of view synthesis, the explicit but complex primitive-based representation, and the wide variety of distortion types introduced by optimization, compression, and rendering processes. This subsection reviews the current landscape of quality assessment for 3DGS, spanning subjective databases, objective prediction models, and the evaluation protocols that govern their comparability, while also recognizing the growing intersection between quality evaluation and the security mechanisms discussed in the preceding subsections.

A foundational challenge in 3DGS quality assessment is the lack of large-scale, purpose-built subjective datasets. Early efforts in the broader field of 3D graphics quality assessment, such as studies on textured meshes, established methodologies using paired-comparison protocols and crowdsourced subjective experiments to capture perceptual quality under geometry and texture distortions [733, 734]. Building upon these principles, the community has recently introduced several datasets specifically tailored to 3DGS. For instance, 3DGS-QA was introduced as the first subjective quality assessment dataset for 3DGS, comprising 225 degraded reconstructions across 15 object types to systematically study the perceptual impact of factors like viewpoint sparsity, limited training iterations, point downsampling, and noise [735]. Similarly, to address the specific distortions introduced by compression, 3DGS-IEval-15K provides a large-scale dataset with 15,200 images rendered from 10 real-world scenes using 6 representative compressed 3DGS algorithms, evaluated by 60 viewers [736]. Extending this to video sequences, 3DGS-VBench offers a video quality assessment benchmark with 660 compressed 3DGS models and corresponding videos from 11 scenes, while GScomp-QA focuses on isolating compression-induced distortions from synthesis artifacts across 9 state-of-the-art compression solutions [737, 738]. These datasets collectively provide a foundation for training and validating quality metrics specifically for 3DGS content.

Given the explicit nature of 3DGS, a notable trend in objective quality assessment is the development of no-reference (NR) models that operate directly on Gaussian primitives rather than rendered images. This approach is motivated by the inefficiency of rendering multiple views solely for quality evaluation. For example, the quality prediction model accompanying the 3DGS-QA

dataset directly extracts spatial and photometric cues from the Gaussian representation to estimate perceived quality in a structure-aware manner, eliminating the need for image rendering [735]. Similarly, Aes3D introduces a lightweight model, Aes3DGSNet, that predicts scene-level aesthetic scores directly from 3DGS representations, capturing high-level aesthetic cues like composition and harmony without rendering multi-view images [739]. SpatialQ goes a step further by proposing a multimodal quality assessment framework that augments a VGGT-based encoder with a quality head and geometric cues to learn structure-aware quality representations, utilizing a Qwen-based MLLM for grounded multimodal reasoning [740]. These primitive-based and multimodal approaches signify a departure from traditional image-based metrics, aiming for more efficient and 3D-aware quality prediction.

Beyond 3DGS-specific methods, the broader field of 3D-generated content (3DGC) quality assessment offers relevant insights and benchmarks. Datasets like 3DGCQA, built using 7 representative Text-to-3D generation methods, highlight common distortion categories in generated assets and expose the limitations of existing assessment algorithms [741]. Hi3DEval proposes a hierarchical evaluation framework that combines object-level and part-level assessments, explicitly evaluating material realism attributes such as albedo and metallicness, using a large-scale dataset and a 3D-aware automated scoring system [742]. Furthermore, platforms like 3D Arena collect large-scale human preferences through pairwise comparisons for generative 3D models, revealing that presentation formats (e.g., Gaussian splat vs. mesh) and texturing significantly influence human preferences [743]. These works underscore the importance of perceptual and utility-based evaluation in addition to pixel-level fidelity. The lessons learned from point cloud quality assessment, where no-reference metrics often rely on projected or extracted features [744, 745], also inform the design of 3DGS quality models, though the continuous and semi-transparent nature of Gaussians requires novel adaptations.

A significant portion of quality assessment research for 3DGS focuses on evaluation protocols, particularly for compression and rendering degradations. The evaluation of compressed 3DGS models typically involves perceptual rate-distortion analysis, where the trade-off between storage size and visual quality is a primary concern [738, 737]. However, studies have shown that many classical objective quality metrics, such as those evaluated on the GScomp-QA dataset, do not fully capture GS-specific distortions [738]. This necessitates the development of specialized metrics or the adaptation of existing ones. For instance, a study on high-fidelity compressed images emphasizes the need for metrics sensitive to just-noticeable differences (JND), a concept that is also highly relevant when assessing high-quality 3DGS renderings [746]. Moreover, the impact of benchmark design itself cannot be overstated; as observed in some studies, the choice of dataset splits can significantly affect the comparability of results across different methods [747]. Therefore, standardized evaluation protocols—such as those provided by benchmark toolkits like Splatwizard, which automate the calculation of image quality metrics, chamfer distances, frame rates, and resource consumption—are essential for fair and reproducible comparisons [748].

Finally, the assessment of 3DGS quality also intersects with the security and robustness concerns detailed in the preceding subsections. While the primary focus of quality assessment is on perceptual fidelity, the integrity of the models themselves is increasingly important. The threat of adversarial attacks and backdoor poisoning can severely degrade the visual quality of a 3DGS model, and watermarking techniques for copyright protection often introduce imperceptible perturbations that should not adversely affect perceived quality. Consequently, future quality assessment methodologies must consider not only native rendering quality but also the resilience of models to malicious modifications and the fidelity preservation under protective measures. This dual concern mirrors the tension between protection and usability discussed in the context of active and passive copyright protection, where the balance between watermark robustness and rendering fidelity is a central

design challenge. In summary, while significant strides have been made in establishing subjective datasets and objective prediction models for 3DGS, the field still lacks a unified, standardized framework that addresses the unique properties of this representation. Future work should focus on developing more comprehensive benchmarks that encompass diverse distortion types, evaluation protocols that are agnostic to dataset splits, and metrics that are sensitive to both perceptual quality and the subtle artifacts introduced by compression, security mechanisms, and real-world deployment conditions. Such efforts will lay the groundwork for the empirical security evaluations discussed in the following subsection, ensuring that 3DGS technology can be deployed with confidence in both quality and safety.

11.4 Adversarial Robustness, Poisoning Defenses, and Security Benchmarks

The empirical evaluation of security and robustness in 3D Gaussian Splatting (3DGS) systems is a nascent but critical area of study that naturally extends the quality assessment landscape discussed in the preceding subsection. While quality assessment focuses on perceptual fidelity under benign distortions, a parallel and equally pressing concern is the integrity of 3DGS models under adversarial manipulation. The explicit, primitive-based nature of 3DGS introduces a unique attack surface that is distinct from both traditional deep learning classifiers and implicit neural representations like NeRF. This subsection synthesizes findings from the broader adversarial machine learning literature to establish a framework for understanding adversarial robustness, poisoning defenses, and security benchmarks applicable to 3DGS pipelines. The core challenge lies in adapting established concepts—such as data poisoning, backdoor injection, and adversarial training—to a domain where the “model” is a collection of optimizable geometric primitives and the “inference” is a differentiable rasterization process. Just as quality benchmarks like 3DGS-QA and GScomp-QA provide standardized evaluations of rendering quality, the security community must develop analogous protocols to quantify attack success and defense robustness, a theme that will be further elaborated in the following subsection.

A foundational step for securing 3DGS is adapting the taxonomy of attacks and defenses from the wider field. A key concern is the threat of data poisoning and backdoor injection, where an adversary manipulates the training images used to optimize the Gaussians—a vector that directly parallels the distortion types examined in quality assessment, but with malicious intent. The literature on backdoor attacks provides a comprehensive overview of the attack surfaces across the machine learning pipeline, which can be mapped directly to 3DGS. These surfaces include data collection, where poisoned images are introduced, and outsourcing, where a pre-optimized 3DGS model might be distributed [749]. The goal of a backdoor attack on a 3DGS scene could be to cause specific artifacts to appear in the rendered output when a trigger is present in the input view, or to subtly corrupt the geometric reconstruction in a way that causes downstream failures in navigation or manipulation—distortions that would be invisible to conventional quality metrics yet catastrophic in practice. The stealthy nature of these attacks, where the model performs nominally on benign inputs but maliciously on triggered ones, is a primary threat [749].

To defend against such threats, several strategies from the classical domain can be re-contextualized for 3DGS, mirroring the way quality assessment methods have been adapted from traditional 3D graphics. Poisoned sample detection (PSD) is a first line of defense, aiming to filter out malicious training images before they influence Gaussian optimization, much as quality metrics filter out distorted renders. The effectiveness of these detection methods has been shown to be correlated with the strength of the backdoor effect; methods like Sharpness-Aware Minimization (SAM) can be used to train a separate classifier to amplify the backdoor effect, thereby making poisoned samples easier to detect [750]. This suggests that a pre-processing step to identify and remove poisoned

frames from a training set for 3DGS is feasible. Another promising direction is the use of machine unlearning techniques. Instead of retraining from scratch on a cleaned dataset, one could train a 3DGS model and then “unlearn” the influence of suspected poisoned samples. This is analogous to methods for purifying backdoored models using a small clean dataset, which derive an upper bound on backdoor risk and use adversarial training to unlearn the malicious behavior [751]. However, applying unlearning to a continuous, non-parametric representation like 3DGS is non-trivial, as the influence of a single training image is distributed across many Gaussian primitives. This remains an open challenge.

Beyond data-centric defenses, the optimization process itself can be hardened. Backdoor attacks often rely on the model learning a “shortcut” that connects the trigger to the target behavior. Research has shown that model training can be made more robust by decoupling the learning process into distinct stages, such as supervised learning, active unlearning, and semi-supervised fine-tuning, to actively prevent the model from fitting the poisoned samples [752]. Translating these training strategies to the 3DGS optimization loop—particularly the adaptive density control and densification phases—could help mitigate the risk of the model structurally encoding an embedded backdoor. Furthermore, a critical lesson from the broader field is the fragility of defenses under adaptive attacks. A defense that is effective against a naive attack can often be bypassed by an adversary who is aware of the defense mechanism [753]. For instance, defenses that rely on the assumption that poisoned and clean samples are separable in the latent space of a classifier can be broken by adaptive attacks that deliberately make the latent representations of poisoned samples more diverse and similar to clean ones. In 3DGS, this means that any defense mechanism must be evaluated against adversaries who can specifically target its assumptions.

The evaluation of these security properties in 3DGS must also be rigorous and standardized, echoing the call for unified benchmarks in quality assessment. The existing body of work on backdoor learning has highlighted the dangers of inconsistent evaluation setups and flawed threat models, leading to potentially overstated effectiveness of both attacks and defenses [754, 755]. A major concern is the “quantization blindspot,” where backdoor defenses that work perfectly on full-precision models completely fail when the model is quantized for deployment [756]. This is highly relevant to 3DGS, where model compression and quantization are essential for real-time performance on mobile and edge devices—the same compression scenarios analyzed in quality databases like 3DGS-IEval-15K and GScomp-QA, but now viewed through a security lens. Security evaluations must therefore be conducted at the same precision and with the same rendering pipeline as the final deployment target. To address this, the community should develop comprehensive benchmarks for 3DGS that include standardized protocols for evaluating both attack success and defense robustness, similar to how quality benchmarks standardize perceptual fidelity assessments. Such benchmarks would need to define clear threat models (e.g., white-box, black-box), specify the attack’s goals (e.g., inducing rendering artifacts, degrading geometric accuracy), and provide metrics for both security (e.g., attack success rate, defense detection rate) and utility (e.g., PSNR, SSIM of clean renders) [757]. These benchmarks are crucial for facilitating fair comparisons and reproducible research, avoiding the pitfalls of evaluating defenses in isolation without considering the computational overhead and their behavior under benign conditions [758].

Finally, the broader robustness of 3DGS to adversarial perturbations should be considered, just as quality assessment considers a spectrum of distortion types. Unlike a classifier, where an adversarial example is a single input image, an adversarial example for a 3DGS system could be a crafted input view that causes a catastrophic failure in the rendering process or leads to a significant error in a downstream perception task. The concept of “adversarial hypervolume,” which assesses robustness across a range of perturbation intensities rather than a single one, is a useful model for evaluating 3DGS systems [759]. It moves beyond the brittle, single-point evaluation of robustness

and acknowledges that attacks can vary in strength. Therefore, a comprehensive security evaluation of a 3DGS system must consider not only whether it can be poisoned or backdoored but also how gracefully it degrades under a spectrum of adversarial perturbations to its inputs, both during training and at inference time. This requires a shift towards threat-aware evaluation protocols that rigorously define attacker capabilities and system vulnerabilities, ensuring that 3DGS technology can be deployed with confidence in security-sensitive applications. Such protocols will complement the quality assessment frameworks discussed earlier and provide the empirical foundation for the theoretical security considerations to be explored in the following subsection.

12 Future Directions, Open Challenges, and Conclusion

12.1 Open Challenges and Current Limitations of 3DGS

Despite the remarkable advances and transformative potential of 3D Gaussian Splatting (3DGS) across numerous domains, the field continues to grapple with several fundamental limitations that impede its widespread adoption and constrain its practical utility. These challenges, which have been recurring throughout this survey, span the core pillars of robustness, scalability, generalization, and efficiency, and represent critical barriers that must be addressed for 3DGS to mature into a foundational technology for real-time photorealistic 3D reconstruction. Understanding these limitations is essential, as they directly motivate the active research directions—hybrid representations, generative pipelines, and optimization refinements—that are explored in the subsequent discussion on future frontiers.

One of the most significant open challenges is the **robustness of 3DGS to extreme and challenging lighting conditions**. The standard 3DGS pipeline inherently assumes consistent exposure and tone mapping across all input views, an assumption that is routinely violated in real-world data. Heterogeneous camera pipelines, dynamic outdoor illumination, and diverse camera exposure settings can bake photometric inconsistencies directly into the radiance field, leading to visible artifacts and a breakdown in the fidelity of the reconstruction [298]. In particularly challenging scenarios, such as low-light environments, the acquired images are plagued by severe noise, low dynamic range, and unreliable Structure-from-Motion (SfM) initialization, which collectively destabilize the optimization process [760]. The difficulty is compounded by the fact that the spherical harmonic functions used for color representation are often ill-suited for these degraded linear color spaces, and the limited SfM points fail to capture sufficient scene detail, especially in distant regions [761]. While methods have been proposed to address these issues—through physical priors, per-view adjustments, or illumination-invariant representations—achieving robust and photorealistic rendering under such diverse and unconstrained illumination remains an active area of research [293, 762]. These efforts underscore the need for more physically grounded appearance models, a theme that resonates with the hybrid and regularization strategies discussed later.

Another critical limitation is the **scalability of 3DGS to unbounded and large-scale scenes**, particularly in terms of memory footprint and computational cost. The explicit, primitive-based nature of 3DGS, while advantageous for rendering speed, leads to significant storage demands and excessive GPU memory consumption, making it impractical for large-scale applications on mobile or head-mounted devices [451, 54]. As scenes grow to city-scale, the sheer number of Gaussian primitives required for high-fidelity reconstruction explodes, leading to prohibitively long training times and intractable memory requirements [113, 698]. Furthermore, attempting to render city-scale 3DGS scenes in real-time on resource-constrained mobile devices poses significant challenges due to computationally intensive stages such as level-of-detail search and rasterization [763]. While divide-and-conquer strategies, such as block-based or hierarchical partitioning, have been proposed

to mitigate these issues, they often introduce new bottlenecks like severe load imbalance, inefficient use of the coarse stage, or degradation in rendering quality, particularly in areas with high visual complexity and at high resolutions [113, 73]. The omnipresent visibility of panoramic images further invalidates existing partitioning strategies that rely on local camera frustums, causing block-wise optimization to degenerate into global training [109]. This scalability bottleneck is precisely what motivates the exploration of more compact and adaptive representations, including the hybrid mesh-Gaussian and level-of-detail approaches highlighted in the future directions.

The **generalization of 3DGS across unseen scenes and domains** represents a third major challenge. The standard 3DGS pipeline is inherently per-scene, requiring expensive optimization that does not generalize to novel environments. Feed-forward methods that attempt to infer Gaussian parameters in a single pass face fundamental bottlenecks, such as being memory-prohibitive at high resolutions and struggling to fuse dense multi-view observations consistently [236]. The robustness of these models on unseen scenes is also limited by the “sparsity shield” of current benchmarks, which rewards hallucination over physical fidelity and fails to capture the complexities of real-world data [764]. Moreover, the performance of 3DGS models often degrades sharply when deployed in real-world scenarios due to unpredictable noise, dynamic motion, and sensor imperfections—conditions that are largely absent from sanitized training data [765]. Even in the domain of semantic understanding, generalization remains a hurdle, as many Language Gaussian Splatting methods are evaluated only on rendered 2D views close to training viewpoints, limiting insights into holistic 3D understanding [766]. This limitation directly motivates the emerging shift toward generative and feed-forward pipelines, which aim to replace per-scene optimization with learned priors that can generalize across scenes, as elaborated in the following subsection.

Finally, the **real-time performance constraints on mobile and resource-constrained platforms** continue to be a significant open problem. While 3DGS enables real-time rendering on high-end workstations, its training and rendering demands often exceed the capabilities of mobile devices and embedded systems. Training on-device requires negotiating a fundamental tension between training efficiency, memory compactness, and modeling quality, and often remains impractical under minute-scale time budgets and limited peak memory [115]. Even for rendering, the computational cost of rasterization can be prohibitive on low-power GPUs, necessitating software and hardware co-design, aggressive compression, level-of-detail systems, or remote streaming solutions to maintain interactive frame rates [114, 415, 483]. The need for predictable resource consumption is paramount for production-level deployment, yet heuristic Gaussian growth processes often lead to unpredictable memory and computational needs [764]. Addressing these constraints requires not only algorithmic innovations—such as more efficient densification strategies and compact representations—but also system-level co-design, a theme that carries forward into the optimization and robustness research discussed in the next section.

In summary, while 3DGS has achieved remarkable success, the field is confronted with significant open challenges related to lighting robustness, scalability, generalization, and resource efficiency. These limitations are not isolated; they are deeply interconnected and collectively define the research agenda for the field. They highlight the need for continued innovation in physical modeling, compact and adaptive representations, and efficient system-level design—precisely the directions that the emerging work on hybrid representations, feed-forward generative models, and advanced optimization strategies aims to address, as detailed in the remainder of this survey.

12.2 Emerging Research Directions in Representation and Optimization

The field of 3D Gaussian Splatting is rapidly evolving, and several promising research directions are emerging to address its current limitations and unlock its full potential. As highlighted in

the preceding discussion, the core challenges of lighting robustness, scalability, generalization, and resource efficiency are deeply interconnected, and the avenues explored below—spanning hybrid models, generative and feed-forward pipelines, and optimization refinements—directly target these bottlenecks. Together, they lay the algorithmic groundwork that will later be integrated into the unified, scalable, and deployable systems discussed in the following subsection.

Hybrid Representations and Unified Frameworks. A significant trend that directly addresses the scalability and geometric fidelity limitations is the departure from pure Gaussian-only models towards hybrid representations that combine the strengths of 3DGS with other complementary data structures. The integration of explicit meshes is a key focus, aiming to provide structural fidelity and geometric integrity that pure point-based methods often lack, thereby reducing the number of Gaussians needed for large-scale scenes. For instance, an end-to-end mesh-Gaussian representation binds 3D Gaussians to mesh faces, enabling a direct information pathway for jointly learning explicit geometry and renderable appearance [90]. Similarly, hybrid approaches that combine 3DGS with textured meshes have been proposed to handle texture-rich flat areas more efficiently, reserving Gaussians for intricate geometries [47]. This strategy of leveraging a mesh scaffold for large-scale structure and Gaussians for fine details is also echoed in other works [93, 226]. Furthermore, integrating implicit fields, such as Signed Distance Functions (SDFs), is a powerful direction to enforce global surface continuity and improve mesh extraction, as demonstrated by models that squeeze geometry regularization into a compact MLP within a unified framework [209]. This move towards unified, single-model frameworks that jointly optimize for both high-quality rendering and accurate surface reconstruction is a critical area of exploration [209]. The success of these hybrids suggests that the future of 3D scene representation lies not in a single primitive type, but in the synergistic combination of multiple, where each component handles the tasks it is best suited for—a principle that directly informs the system-level design discussions on compact and adaptive representations later in this survey.

Generative Modeling and Feed-Forward Reconstruction. Another major frontier that directly confronts the generalization limitation is the shift from per-scene optimization to generalizable, feed-forward reconstruction, which aims to infer a 3D representation from a single forward pass. This paradigm is crucial for scalability and real-time applications, as it replaces the expensive per-scene fitting with learned priors. Current research is actively rethinking the core prediction framework, moving away from simple pixel-aligned predictions to more robust voxel-aligned or geometry-guided formulations to improve multi-view consistency and handle occlusions [767, 768]. The use of powerful transformer architectures as the backbone for these feed-forward models is becoming increasingly common, enabling the prediction of semantically rich and geometrically coherent primitives [769, 672]. A key challenge being addressed is the development of efficient and stable training paradigms for these feed-forward models. For iterative refinement, methods like those using self-supervised densification policies and geometric regularization are being developed to improve the stability of iterative feed-forward pipelines [236]. Another promising area is the use of episodic meta-training to learn transferable priors, allowing the model to reconstruct scenes in a single forward pass by predicting geometry and radiance together [700]. Furthermore, the integration of diffusion models into the reconstruction process is a powerful tool to enhance the quality of predicted primitives or fill in unobserved regions, with some works using that as a robust refinement mechanism [770, 771]. The development of these feed-forward and generative methods is crucial for moving 3DGS from an offline tool to an on-the-fly, interactive system [772], and it provides the scalable, generalizable foundation upon which the streaming and edge-cloud architectures of the next subsection will depend.

Optimization, Densification, and Robustness. Key to the success and efficiency of 3DGS is its optimization strategy, and research here is focused on making training faster, more stable, and

less memory-intensive—directly tackling the efficiency and robustness challenges outlined earlier. Improving the efficiency of 3DGS is a critical goal, with innovations in the core prediction and optimization pipeline. For instance, sparse query-based models are being explored to reduce memory usage and inference time by allocating capacity only where needed [773]. The “densification” step, where new Gaussian primitives are added, is a critical point of failure that is being addressed through various new strategies, such as Hessian-assisted sampling to guide where new primitives are generated for super-resolution [774], or explicit identification of missing regions through visibility-aware analysis to recover geometry [775]. Some methods are looking to Gaussian processes for principled densification [776]. Stability remains a concern, especially in challenging settings like the low-light and sparse-view scenarios previously discussed, with techniques like alternating densification phases reducing artifacts in sparse-input scenarios [179]. Finally, the issue of geometric fidelity is being tackled by denoising the optimization process through depth and density-guided strategies [264], or by using energy-based models for soft geometric guidance when priors are incomplete [194]. These refinements are essential for making 3DGS reliable and efficient enough for widespread adoption, providing the predictable resource consumption and robust performance that the hardware-software co-design and federated learning frameworks in the following subsection will require.

12.3 Toward Unified, Scalable, and Deployable Systems

12.3 Toward Unified, Scalable, and Deployable Systems

Building upon the optimization strategies and generative advancements discussed in the preceding section, the trajectory of 3D Gaussian Splatting (3DGS) from a high-fidelity novel view synthesis technique to a foundational technology for interactive applications hinges on its ability to operate within unified, scalable, and deployable system architectures. While core algorithmic advances have propelled rendering quality and speed, the practical integration of 3DGS into real-world products—spanning mobile devices, autonomous vehicles, and immersive XR experiences—demands a holistic rethinking of the entire pipeline, from data acquisition and semantic understanding to model compression, transmission, and hardware co-design.

Unified Multimodal Representations and Semantic Integration. A key pillar of deployable systems is the seamless fusion of 3D geometry with high-level semantic, linguistic, and multi-sensor information. As discussed in Section 8, embedding semantic features into Gaussian primitives (e.g., via CLIP or SAM distillation) enables interactive scene understanding, open-vocabulary segmentation, and language-guided editing. However, unifying these capabilities within a single, coherent 3DGS framework requires addressing cross-modal representation gaps. The “semantic gap” between low-level photometric radiance and high-level concepts necessitates architectures that can jointly optimize for geometric fidelity and semantic consistency. Drawing inspiration from multimodal foundation models that process text, images, and audio in a shared token space [777], future 3DGS systems could adopt a unified tokenization strategy where Gaussian primitives, semantic features, and textual queries are embedded in a common representational space. This would enable more direct interaction with large language models for scene reasoning and task-oriented communication [778], rather than relying on post-hoc feature aggregation layers.

Moreover, scaling 3DGS to support diverse sensor modalities, including LiDAR, radar, and depth sensors, is critical for applications like autonomous driving and robotic navigation. Sensor data provides crucial geometric priors that can regularize Gaussian optimization and improve metric accuracy in textureless or reflective regions (Section 5.4). A unified system would need to perform online multi-modal fusion, aligning and integrating these heterogeneous data streams in real-time to construct a coherent scene representation. The principle of semantic communication, where raw sensor data is reduced to goal-oriented messages rather than raw bitstreams, is particularly relevant

here. For instance, in collaborative robotics, transmitting only task-relevant semantic features—such as object locations and identities extracted from 3DGS maps—rather than full image streams, can significantly reduce bandwidth requirements and enable more efficient edge-cloud collaboration [779].

System-Level Compression, Streaming, and Scalability. A fundamental obstacle to ubiquitous 3DGS deployment is its substantial memory footprint (Section 7). While intra-model compression techniques like vector quantization and codebook-based methods offer high compression ratios, a scalable, deployable system must integrate these with standardized streaming protocols. This requires moving beyond per-scene optimizations toward rate-distortion-optimized storage and transmission that explicitly account for human visual perception and semantic importance. For networked and remote rendering scenarios, such as XR, the system architecture must be designed to deliver a satisfying Quality of Experience (QoE) under varying network conditions. The industry standardization efforts by MPEG to manage media enablers and hardware resources underscore the necessity for standardized frameworks to lower the barrier for complex XR experiences [780] and to reconcile the game-engine and multimedia streaming paradigms [781]. A key challenge is the dynamic allocation of computation between the edge and the cloud. Existing edge-cloud collaborative architectures often use fixed-size representations [782], which are not adaptive to fluctuating bandwidth. Future 3DGS streaming systems should therefore employ progressive or scalable encodings, allowing the server to transmit a coarse base representation followed by refinements, as network conditions permit. This approach can be informed by adaptive streaming strategies that adjust both resolution and frame rate based on scene motion and bandwidth constraints to optimize perceptual quality [488].

Furthermore, scalability to city-scale and unbounded scenes requires out-of-core and distributed training and rendering frameworks that can manage billions of primitives (Section 7.3). Beyond raw computation, this demands a hierarchical memory system capable of paging Gaussian data from storage to GPU memory on demand, similar to virtual texture mapping in traditional graphics. To achieve this, the Gaussian representation must be rebuilt around Level-of-Detail (LOD) hierarchies, where coarse representations at distance are replaced by finer ones as the user approaches. This LOD structure is also crucial for efficient streaming in multi-user environments, as it naturally supports progressive data download and discard.

Hardware-Software Co-Design for Real-Time Edge Deployment. The ultimate goal of deployable 3DGS systems is real-time, photorealistic interaction on resource-constrained devices like smartphones and embedded platforms. This cannot be achieved by algorithmic efficiency alone; it requires a tight coupling between algorithm design and the underlying hardware architecture. The challenge is analogous to that faced by on-device large language models, where the computational and energy budgets far exceed the capabilities of mobile CPUs and GPUs, leading to the development of specialized acceleration strategies [783]. For 3DGS, the dense rasterization, sorting, and alpha-blending pipeline must be mapped onto hardware accelerators, such as NPUs and Tensor Cores. Research on binary networks for segmentation highlights the potential of extreme quantization combined with hardware-specific instructions to achieve significant speedups and energy savings [784]. Similarly, for 3DGS, implementing rasterization with low-precision arithmetic and custom data paths could unlock real-time performance on mobile GPUs.

The performance bottleneck is not limited to rendering; the training and optimization stages are also extremely computationally intensive. Therefore, on-device fine-tuning and adaptation are challenging but necessary for personalized experiences. Federated learning frameworks, which evaluate the costs of local fine-tuning on real smartphones and edge boards, provide a valuable blueprint for sustainable, privacy-preserving model updates [785]. A unified 3DGS platform must similarly support the ability to incrementally update and refine a scene model on-device, sending

only model “deltas” back to a central server. This parallels the development of efficient federated learning methods for edge LLMs, which balance communication, memory, and energy footprints [785].

Moreover, the system software stack must be designed to maximize the utilization of heterogeneous hardware. As evidenced by VLM deployment studies [786], current frameworks often over-utilize the CPU while leaving NPUs and GPUs idle due to inefficient kernel implementations or memory transfer bottlenecks. Future 3DGS deployments must avoid such pitfalls through rigorous profiling and hardware-aware scheduling. This includes dynamic partitioning of tasks, where the edge handles primary visibility and simple lighting, while the cloud computes complex global illumination or updates dense semantic maps, as demonstrated by distributed rendering systems [787]. A successful architecture will be one that can intelligently schedule workloads across CPU, GPU, and NPU, balancing latency, energy, and bandwidth constraints. This necessitates a co-design process where the mathematical formulation of the Gaussian primitive, the rasterization algorithm, and the instruction set architecture are developed in tandem. Custom on-device training frameworks and federated learning offer a pathway to ensure that device-side personalization and continuous improvement are achievable without compromising the system’s privacy and latency guarantees [785], making 3DGS a truly deployable ubiquitous technology. Together, these system-level considerations complement the algorithmic foundations established earlier and set the stage for the broader outlook on the field’s future trajectory presented in the concluding section.

12.4 Future Outlook and Concluding Remarks

As we conclude this comprehensive survey, the trajectory of 3D Gaussian Splatting (3DGS) reveals a field that has evolved from a breakthrough representation into a versatile and foundational tool for 3D vision and graphics [5]. Building directly upon the system-level considerations of unified representations, scalable architectures, and hardware-software co-design outlined in the previous section, we now synthesize the broader implications of these advances and chart the future directions that will define the next phase of 3DGS research and deployment. The rapid progress since its introduction underscores a paradigm shift in how we approach photorealistic scene reconstruction, moving from computationally intensive implicit neural representations toward explicit, GPU-friendly primitives that balance visual fidelity, rendering speed, and editability [37]. The convergence of technical advances across representation design, optimization strategies, geometric fidelity, robustness, and domain-specific applications positions 3DGS not merely as an alternative to NeRF but as a transformative technology bridging neural rendering, computer graphics, and embodied intelligence [4].

Looking forward, the unified multimodal representations discussed in the previous section serve as a springboard for one of the most promising research directions: the deep integration of semantics, language, sensor data, and physical properties directly into the Gaussian-based scene model. The extension of 3DGS with semantic feature distillation, open-vocabulary segmentation, and language-embedded queries has already enabled richer scene understanding and interactive editing [8]. As large language models become increasingly integrated into 3D pipelines, systems such as LIVE-GS demonstrate how LLM-powered analysis of physical properties can transform static Gaussian assets into dynamic, interactive VR experiences with minimal manual effort [680]. Similarly, the fusion of 3DGS with scene graphs and hierarchical representations paves the way for high-level reasoning systems that enable embodied navigation and manipulation in complex environments [788]. The emergence of physics-aware generation and simulation frameworks further highlights how 3DGS can

go beyond visual realism to incorporate physical plausibility, enabling applications in robotics and autonomous systems that require adherence to real-world laws [789].

The scalability and system-level integration challenges identified in the previous section—memory footprint, streaming protocols, and hardware constraints—remain critical open problems that will shape future adoption. Current methods for large-scale urban and aerial scene reconstruction, such as hierarchical block-based approaches and streamlined pipelines, have made significant strides in reducing computational overhead while maintaining quality [698, 790]. Yet the field still confronts fundamental trade-offs between memory footprint, training time, and rendering quality, particularly in unbounded environments and resource-constrained platforms [37]. Out-of-core training frameworks, distributed optimization, and hardware-software co-design strategies—including custom accelerators and compression-aware pipelines—will be essential to realize billion-scale reconstruction at interactive rates [702]. Compression techniques, including pruning, quantization, entropy coding, and factorized representations, have demonstrated that high compression ratios are achievable while preserving visual quality [791]. The development of standardized compression and streaming protocols, as advocated in our discussion of system-level deployment, will be crucial for enabling real-time collaborative experiences across devices, from mobile edge platforms to cloud-based rendering farms [5].

Another emerging frontier that directly extends the algorithmic foundations laid earlier in this survey is the convergence of generative modeling and feed-forward reconstruction with 3DGS. Traditional per-scene optimization approaches, while high-quality, are inherently limited by their computational cost and lack of generalization—a limitation that the scalable and unified systems discussed previously must overcome. Recent advances in feed-forward frameworks that directly regress Gaussian parameters from multi-view images have demonstrated remarkable speed improvements while maintaining competitive fidelity [792, 672]. The integration of diffusion models with 3DGS for both scene generation and refinement is opening new possibilities—from text-to-3D synthesis using video diffusion self-distillation [793] to multi-modal diffusion frameworks that synthesize RGB, depth, and semantic outputs for urban reconstruction [241]. These generative approaches have the potential to break the dependence on dense captured data, enabling reconstruction from sparse, degraded, or even entirely synthetic inputs, and they align with a broader shift in the field from visual realism toward interaction readiness in which generated content must support physical simulation, articulation, and task execution [794, 795].

Dynamic and 4D Gaussian representations represent another vital research axis with substantial applications in digital humans, autonomous driving simulation, and embodied AI, complementing the static scene understanding enabled by unified multimodal systems. Advances in canonical deformation fields, explicit 4D primitives, and motion decomposition strategies have expanded the expressiveness of 3DGS to temporally varying scenes [796]. Physics-based dynamic simulation through differentiable material point methods and 4DGS enables instance-level editing and realistic object-object interactions, supporting the generation of challenging corner cases for autonomous driving validation [797]. The fusion of 3DGS with SLAM systems continues to push the boundaries of online, real-time dense mapping in dynamic and large-scale environments, with methods such as global Sim(3) optimization and voxelized representations enabling robust camera tracking and efficient loop closure [798, 799]. These systems are foundational for robotics, augmented reality, and autonomous navigation, where the ability to reconstruct and update scenes in real time from monocular or multi-modal streams is paramount [800].

Despite the remarkable progress, several fundamental challenges persist that cut across the algorithmic and system-level themes of this survey. Robustness to extreme lighting, adverse weather, sparse views, and imperfect sensor inputs remains an active area of research, with methods leveraging physical priors, multi-modal fusion, and generative enhancement to close the gap between

controlled conditions and in-the-wild settings [801, 802]. The simulation-to-real gap, particularly acute in autonomous driving, is being addressed through hybrid approaches that combine neural reconstruction with physics-based rendering, enabling interactive frame rates and multi-modal sensor simulation [803]. Benchmarks such as the Extrapolated Urban View Synthesis benchmark highlight the need for more robust NVS methods under large viewpoint deviations, emphasizing that current approaches often overfit to training views [804].

Security, privacy, and intellectual property protection for 3DGS assets are also emerging as critical considerations as Gaussian representations become commercially valuable, paralleling the deployment concerns raised in our discussion of scalable systems. The explicit parametric structure of 3DGS introduces unique vulnerabilities to adversarial attacks and backdoor poisoning, as well as novel opportunities for watermarking and ownership tracing [791]. Standardized threat modeling, robust defense mechanisms, and comprehensive quality assessment benchmarks will be necessary to establish trust in 3DGS systems across industrial deployments [5].

In summary, 3DGS has fundamentally transformed the landscape of real-time photorealistic 3D reconstruction and rendering. Its explicit, editable, and GPU-friendly nature positions it as a cornerstone technology for a broad spectrum of applications, ranging from virtual and augmented reality to robotics, autonomous driving, and digital twins. The future of 3DGS lies in continued advances toward unified multimodal representations, scalable and efficient systems, physics-aware interactivity, and robust generalization—building directly upon the system-level foundations established in the preceding section. As the field converges with generative AI, foundation models, and embodied intelligence, we anticipate that 3DGS will evolve from a scene representation technique into a comprehensive infrastructure for spatial intelligence—enabling machines not only to perceive and render the world but also to understand, reason, and interact within it. The journey from concept to capability is well underway, and the full impact of 3D Gaussian Splatting on interactive media, autonomous systems, and digital society is only beginning to unfold.

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